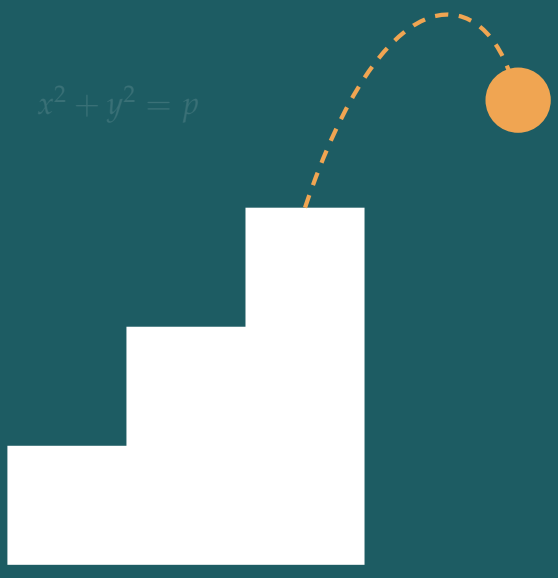
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1995 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1995 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1995 Worked Solutions

Worked solutions to all 42 questions of the 1995 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

Contents

STEP I

Q1 Polynomial Inequalities Pure . . . 3
Q2 Integration by Substitution Pure . . . 7
Q3 Telescoping Series Pure . . . 10
Q4 De Moivre's Theorem Pure . . . 13
Q5 Binomial Series and Integration Pure . . . 16
Q6 Bernoulli Differential Equations Pure . . . 19
Q7 Vector Geometry of Triangles Pure . . . 22
Q8 Linear Differential Equations Pure . . . 25
Q9 Projectile Envelope Mechanics . . . 28
Q10 Elastic Strings and Bouncing Mechanics . . . 31
Q11 Friction Between Cylinders Mechanics . . . 35
Q12 Arrangements and Counting Probability and Statistics . . . 39
Q13 Poisson Thinning Probability and Statistics . . . 42
Q14 Estimating a Proportion Probability and Statistics . . . 45

STEP II

Q1 Differentiating Geometric Series Pure . . . 49
Q2 Counting Colourings Recursively Pure . . . 52
Q3 Logical Deduction and Counting Pure . . . 55
Q4 Wallis Reduction Formula Pure . . . 58
Q5 Optimisation and Concavity Pure . . . 61
Q6 Roots of Unity Pure . . . 64
Q7 Heron's Formula Pure . . . 68
Q8 Logistic Differential Equation Pure . . . 72
Q9 Friction and Equilibrium Mechanics . . . 76
Q10 Collisions and Restitution Mechanics . . . 79
Q11 Linear Air Resistance Mechanics . . . 82
Q12 Sequential Duels Probability and Statistics . . . 85
Q13 Poisson Distribution Optimisation Probability and Statistics . . . 89

Q14 Hypothesis Testing Probability and Statistics . . . 93

STEP III

Q1 Simultaneous Linear Equations Pure . . . 97
Q2 Reduction Formulae Pure . . . 101
Q3 Damped Harmonic Motion Pure . . . 104
Q4 Trigonometric Series Summation Pure . . . 107
Q5 Maclaurin Series Pure . . . 110
Q6 Loci and Inversion Pure . . . 113
Q7 Groups and Isomorphisms Pure . . . 117
Q8 Planes and Spheres Pure . . . 121
Q9 Compound Pendulum Mechanics . . . 125
Q10 Projectiles on Inclined Planes Mechanics . . . 129
Q11 Relative Velocity Mechanics . . . 132
Q12 Radially Symmetric Densities Probability and Statistics . . . 136
Q13 Noisy Binary Channel Probability and Statistics . . . 140
Q14 Moment Generating Functions Probability and Statistics . . . 144

WORKED SOLUTION

STEP I 1995, Question 1

Pure • Polynomial Inequalities

STEP I 1995, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q1

The key idea

Part (i) is a cubic that factorises by grouping, and once it is a product of three linear factors its sign can be read off from a number line. Part (ii) is the same cubic in disguise: every term has total degree 3, so dividing by y^3 turns it into the cubic of part (i) in the variable x/y. The same factorisation then gives three lines through the origin, and part (iii) is a sign chart drawn around the origin instead of along a line.

Solution

Part (i)

Grouping the terms in pairs,

$$\begin{aligned} x^3 - 4x^2 - x + 4 &= x^2(x-4) - (x-4) \\ &= (x-4)(x^2-1) \\ &= (x+1)(x-1)(x-4). \end{aligned}$$

The roots are -1, 1 and 4. Each factor is linear and changes sign only at its own root, so the product changes sign at each of the three roots. For large positive x all three factors are positive, so the sign pattern, reading from the right, is

x	$x < -1$	$-1 < x < 1$	$1 < x < 4$	$x > 4$
sign	$-$	$+$	$-$	$+$

Including the roots, where the cubic is zero, the inequality holds exactly when

$$-1 \leq x \leq 1 \quad \text{or} \quad x \geq 4.$$

Part (ii)

Write $f(x,y) = x^3 - 4x^2y - xy^2 + 4y^3$. It is part (i)'s cubic with each term made up to degree 3 by powers of y, and the same grouping works:

$$\begin{aligned} f(x,y) &= x^2(x-4y) - y^2(x-4y) \\ &= (x-4y)(x-y)(x+y). \end{aligned}$$

A product is zero exactly when one of its factors is zero, so $f(x,y) = 0$ on the three lines

$$x = 4y \text{ (i.e. } y = \tfrac{1}{4}x), \qquad y = x, \qquad y = -x,$$

all of which pass through the origin.

? **Ask yourself** Why should the answer to (ii) have anything to do with (i)?

For $y \neq 0$, put $t = x/y$. Then $f(x,y) = y^3\,(t^3 - 4t^2 - t + 4)$, which is y^3 times the cubic of part (i). That cubic vanishes at $t = -1, 1, 4$, that is, on $x = -y$, $x = y$ and $x = 4y$. (When $y = 0$, $f = x^3$ vanishes only at the origin, which is already on all three lines.)

Part (iii)

The three lines cut the plane into six sectors meeting at the origin. Measured anticlockwise from the positive x-axis, the six half-lines point in the directions

$$\arctan\tfrac{1}{4} \approx 14^\circ, \quad 45^\circ, \quad 135^\circ, \quad 194^\circ, \quad 225^\circ, \quad 315^\circ.$$

Inside a sector no factor of f is zero, so f has a constant sign there. Crossing one of the lines changes the sign of exactly one factor (the three lines are distinct, and away from the origin we cross only one at a time), so the sign of f alternates from one sector to the next.

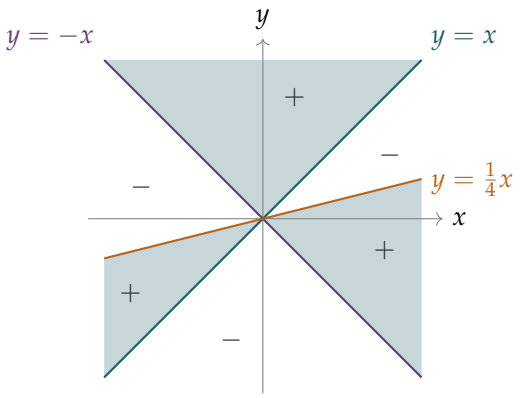
To fix the signs, take $(1,0)$: $f(1,0) = 1 > 0$, and $(1,0)$ lies in the sector between 315° and 14°. Going round, the signs are therefore

sector (degrees)	$-45, 14$	$14, 45$	$45, 135$	$135, 194$	$194, 225$	$225, 315$
sign of f	$+$	$-$	$+$	$-$	$+$	$-$

As a spot check, $f(0,1) = 4 > 0$ in the third column and $f(-3,-1) = 8 > 0$ in the fifth.

So $f(x,y) \geq 0$ on the three shaded sectors below, together with all three lines (where $f = 0$):

- the sector containing the positive x-axis, between $y = -x$ (below) and $y = \frac{1}{4}x$ (above), for $x > 0$;
- the sector containing the positive y-axis, between $y = x$ and $y = -x$, for $y > 0$;
- the thin sector in the third quadrant between $y = x$ and $y = \frac{1}{4}x$.



Ask yourself Why is the picture unchanged by a half-turn about the origin, with the shading swapped?

Every term of f has odd degree 3, so $f(-x,-y) = -f(x,y)$. Rotating a point through $180°$ about the origin flips the sign of f, so each shaded sector sits opposite an unshaded one. That is a quick check on any sketch: shaded and unshaded sectors must come in opposite pairs, and they do.

Checks

- On the line $y = 1$, $f(x,1)$ is exactly the cubic of part (i), so the shaded part of $y = 1$ should be $-1 \le x \le 1$ and $x \ge 4$. In the sketch, $y = 1$ is shaded between $y = -x$ and $y = x$ (from $x = -1$ to $x = 1$), and it re-enters a shaded sector where it crosses $y = \frac{1}{4}x$, at $x = 4$.
- On the x-axis $f = x^3$, which is non-negative exactly for $x \ge 0$: the positive x-axis is shaded and the negative one is not.
- Numerically, f was evaluated at 2×10^5 random points, and its sign agreed with the six-sector chart every time; for instance $f(1,0) = 1$, $f(0,-1) = -4$, $f(-3,-1) = 8$.

Common slips

- Dividing by y^3 and keeping the inequality sign for $y < 0$. Since $y^3 < 0$ there, the condition becomes $t^3 - 4t^2 - t + 4 \le 0$, which is why the lower half-plane looks different from the upper.
- Shading using part (i)'s answer for $y > 0$ only and then forgetting $y < 0$ altogether.
- Leaving out the lines themselves: the inequality is non-strict, so they belong to the region.

Beyond STEP

A polynomial whose terms all have the same degree is called *homogeneous*. Its zero set is a union of lines through the origin, and its sign depends only on direction. Homogeneous polynomials are the starting point of projective geometry, where a point is a whole line through the origin, and they appear again when you classify conics and quadratic forms in a linear algebra course. Strang's *Introduction to Linear Algebra* covers quadratic forms, and Siklos's *Advanced Problems in Mathematics* has more STEP questions in this style.

WORKED SOLUTION

STEP I 1995, Question 2

Pure • Integration by Substitution

STEP I 1995, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q2

The key idea

In part (i) neither integral is easy on its own, but they pair up beautifully: adding them gives a fraction that cancels to 1, and subtracting them gives a numerator that is exactly the derivative of the denominator. Two easy integrals give two simultaneous equations for S and T. In part (ii) the substitution $x = \sin^2 t$ is chosen because it turns $\frac{1}{x} - 1$ into a perfect square, $\cot^2 t$, so the square root disappears; after that it is double-angle formulae all the way.

Solution

Part (i)

Adding the integrands,

$$S + T = \int \frac{\cos x + \sin x}{\cos x + \sin x}\,\mathrm{d}x = \int 1\,\mathrm{d}x = x + c_1.$$

Subtracting them,

$$S - T = \int \frac{\cos x - \sin x}{\cos x + \sin x}\,\mathrm{d}x.$$

The numerator is the derivative of the denominator, since $\frac{\mathrm{d}}{\mathrm{d}x}(\cos x + \sin x) = -\sin x + \cos x$. So this is of the form $\int \frac{f'(x)}{f(x)}\,\mathrm{d}x$, and

$$S - T = \ln|\cos x + \sin x| + c_2.$$

Adding and subtracting these two equations and halving,

$$S = \tfrac{1}{2}(x + \ln|\cos x + \sin x|) + C,$$
$$T = \tfrac{1}{2}(x - \ln|\cos x + \sin x|) + C',$$

where C and C' are arbitrary constants. These hold on any interval on which $\cos x + \sin x \neq 0$, where the integrands are defined.

Part (ii)

Let I be the required integral. With $x = \sin^2 t$ we have $\mathrm{d}x = 2\sin t\cos t\,\mathrm{d}t$. For the limits, $x = \frac{1}{4}$ gives $\sin t = \frac{1}{2}$, so $t = \frac{\pi}{6}$, and $x = \frac{1}{2}$ gives $\sin t = \frac{1}{\sqrt{2}}$, so $t = \frac{\pi}{4}$. (We take t in $[\frac{\pi}{6}, \frac{\pi}{4}]$; as t runs over this interval, x increases from $\frac{1}{4}$ to $\frac{1}{2}$.)

Next, the square root:

$$\frac{1}{x} - 1 = \frac{1-\sin^2 t}{\sin^2 t} = \frac{\cos^2 t}{\sin^2 t}, \qquad \text{so} \qquad \sqrt{\frac{1}{x}-1} = \frac{\cos t}{\sin t}.$$

? **Ask yourself** Why is the square root $\cos t/\sin t$ and not $|\cos t/\sin t|$?

In general $\sqrt{\cot^2 t} = |\cot t|$. But for $\frac{\pi}{6} \le t \le \frac{\pi}{4}$ both $\sin t$ and $\cos t$ are positive, so $\cot t > 0$ and the modulus can be dropped. This is why we chose t in the first quadrant.

Substituting, the $\sin t$ cancels:

$$I = \int_{\pi/6}^{\pi/4} (1 - 4\sin^2 t)\,\frac{\cos t}{\sin t}\,2\sin t\cos t\,\mathrm{d}t = \int_{\pi/6}^{\pi/4} 2\cos^2 t\,(1-4\sin^2 t)\,\mathrm{d}t.$$

Now we use double angles to make this easy to integrate. Since $2\cos^2 t = 1 + \cos 2t$ and $8\sin^2 t\cos^2 t = 2\sin^2 2t = 1 - \cos 4t$,

$$2\cos^2 t - 8\sin^2 t\cos^2 t = (1+\cos 2t) - (1-\cos 4t) = \cos 2t + \cos 4t.$$

Therefore

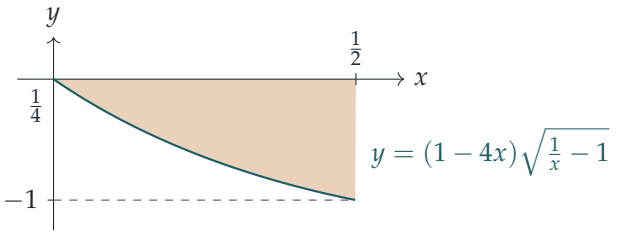
$$\begin{aligned} I &= \int_{\pi/6}^{\pi/4} (\cos 2t + \cos 4t)\,\mathrm{d}t = \left[\tfrac{1}{2}\sin 2t + \tfrac{1}{4}\sin 4t\right]_{\pi/6}^{\pi/4} \\ &= \left(\tfrac{1}{2}\sin\tfrac{\pi}{2} + \tfrac{1}{4}\sin\pi\right) - \left(\tfrac{1}{2}\sin\tfrac{\pi}{3} + \tfrac{1}{4}\sin\tfrac{2\pi}{3}\right) \\ &= \tfrac{1}{2} - \left(\tfrac{\sqrt{3}}{4} + \tfrac{\sqrt{3}}{8}\right), \end{aligned}$$

so

$$\int_{1/4}^{1/2} (1-4x)\sqrt{\frac{1}{x}-1}\,\mathrm{d}x = \frac{1}{2} - \frac{3\sqrt{3}}{8} = \frac{4-3\sqrt{3}}{8}.$$

Checks

- Sign: on $(\frac{1}{4}, \frac{1}{2}]$ we have $1 - 4x < 0$ and the square root is positive, so the integrand is negative and I must be negative. Indeed $4 - 3\sqrt{3} \approx -1.196$, so $I \approx -0.1495$. The graph below shows the integrand.
- Simpson's rule applied directly to the original integral gives $-0.149519\ldots$, matching $\frac{1}{2} - \frac{3\sqrt{3}}{8}$ to ten decimal places.
- For part (i), differentiate: $\frac{\mathrm{d}}{\mathrm{d}x}\,\frac{1}{2}\big(x + \ln(\cos x + \sin x)\big) = \frac{1}{2}\left(1 + \frac{\cos x - \sin x}{\cos x + \sin x}\right) = \frac{\cos x}{\cos x + \sin x}$. Over $[0, 1]$, for example, both S and T agree numerically with quadrature (0.66168 and 0.33832).



Common slips

- Forgetting to change the limits to $t = \frac{\pi}{6}$ and $t = \frac{\pi}{4}$, or substituting back to x incorrectly.
- Dropping the factor $2\sin t\cos t$ from $\mathrm{d}x$, or writing $\mathrm{d}x = 2\sin t\,\mathrm{d}t$.
- Giving a positive answer: the integrand is negative throughout, so a positive value signals a sign error.

Beyond STEP

The trick in part (i), pairing an awkward integral with a "partner" so that the sum and difference are both easy, appears again and again, for instance with $\int \mathrm{e}^x \cos x\,\mathrm{d}x$ and $\int \mathrm{e}^x \sin x\,\mathrm{d}x$, and in its complex form by integrating $\mathrm{e}^{(1+\mathrm{i})x}$. Part (ii) is a small example of a general principle: a well-chosen substitution can turn an algebraic integrand into a trigonometric polynomial, which is the starting point of Fourier series. For many more integrals of this flavour, with careful discussion, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1995, Question 3

Pure • Telescoping Series

STEP I 1995, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q3

The key idea

When consecutive terms of a sum share most of their parts, writing each term as a difference $\mathrm{f}(r) - \mathrm{f}(r-1)$ makes almost everything cancel: the sum *telescopes*. The work is in choosing f so that the difference is the term you want; here the question chooses it for us. For the alternating series, the trick is to write it as *all* the cubes minus twice the even ones, so that part (ii) does the work twice.

Solution

Part (i)

Write the sum out in full:

$$\sum_{r=1}^{n} \big\{\mathrm{f}(r) - \mathrm{f}(r-1)\big\} = \big\{\mathrm{f}(1) - \mathrm{f}(0)\big\} + \big\{\mathrm{f}(2) - \mathrm{f}(1)\big\} + \cdots + \big\{\mathrm{f}(n) - \mathrm{f}(n-1)\big\}.$$

Each of $\mathrm{f}(1), \mathrm{f}(2), \ldots, \mathrm{f}(n-1)$ appears once with a plus sign and once with a minus sign, so they all cancel. Only $-\mathrm{f}(0)$ from the first bracket and $\mathrm{f}(n)$ from the last survive, giving $\mathrm{f}(n) - \mathrm{f}(0)$, as required.

If you'd like something more formal than "everything cancels", use induction on n. For $n = 1$ the sum is $\mathrm{f}(1) - \mathrm{f}(0)$. If the result holds for $n = k$, then adding the next term gives

$$\mathrm{f}(k) - \mathrm{f}(0) + \mathrm{f}(k+1) - \mathrm{f}(k) = \mathrm{f}(k+1) - \mathrm{f}(0),$$

which is the result for $n = k+1$. So it holds for all $n \geq 1$.

Part (ii)

With $f(r) = r^2(r+1)^2$ we have $f(r-1) = (r-1)^2r^2$. Both contain the factor r^2, so take it out and use a difference of two squares:

$$\begin{aligned} f(r) - f(r-1) &= r^2\big[(r+1)^2 - (r-1)^2\big] \\ &= r^2\big[(r+1) + (r-1)\big]\big[(r+1) - (r-1)\big] \\ &= r^2 \cdot 2r \cdot 2 = 4r^3. \end{aligned}$$

Now sum from $r = 1$ to n and apply part (i). Since $f(0) = 0$,

$$4\sum_{r=1}^{n} r^3 = f(n) - f(0) = n^2(n+1)^2, \qquad \text{so} \qquad \boxed{\sum_{r=1}^{n} r^3 = \tfrac{1}{4}n^2(n+1)^2.}$$

Part (iii)

The series has $2n+1$ terms: odd cubes with a plus sign and even cubes with a minus sign, ending on the odd cube $(2n+1)^3$.

Ask yourself How can we deal with the alternating signs using only the formula for $\sum r^3$?

Start from the sum of *all* the cubes up to $(2n+1)^3$. That has every even cube with a plus sign, where we want a minus sign, so subtract each even cube twice:

$$1^3 - 2^3 + 3^3 - \cdots + (2n+1)^3 = \sum_{k=1}^{2n+1} k^3 - 2\sum_{m=1}^{n} (2m)^3.$$

The even cubes are $(2m)^3 = 8m^3$ for $m = 1, \ldots, n$, the largest being $(2n)^3$. Using part (ii) twice, once with $2n+1$ in place of n,

$$\begin{aligned} \sum_{k=1}^{2n+1} k^3 - 16\sum_{m=1}^{n} m^3 &= \tfrac{1}{4}(2n+1)^2(2n+2)^2 - 16 \cdot \tfrac{1}{4}n^2(n+1)^2 \\ &= (2n+1)^2(n+1)^2 - 4n^2(n+1)^2 \\ &= (n+1)^2\big[(2n+1)^2 - (2n)^2\big] \\ &= (n+1)^2(4n+1), \end{aligned}$$

where in the second line we used $(2n+2)^2 = 4(n+1)^2$, and in the last the difference of two squares $(2n+1)^2 - (2n)^2 = (2n+1+2n) \cdot 1$. So

$$\boxed{1^3 - 2^3 + 3^3 - 4^3 + \cdots + (2n+1)^3 = (n+1)^2(4n+1).}$$

Checks

- For $n = 0$ the series is just $1^3 = 1$, and $(0+1)^2(0+1) = 1$.

- For $n = 1$: $1 - 8 + 27 = 20$, and $2^2 \times 5 = 20$.
- For $n = 2$: $1 - 8 + 27 - 64 + 125 = 81$, and $3^2 \times 9 = 81$.
- The answer should grow like $(2n+1)^3/2$, roughly half the last term, since the terms pair up as $(2m+1)^3 - (2m)^3 \approx 12m^2$. Indeed $(n+1)^2(4n+1) \approx 4n^3$, matching $\sum 12m^2 \approx 4n^3$.

We also checked parts (ii) and (iii) by direct summation for every n from 0 to 199.

Common slips

- Subtracting the even cubes only once, which gives $\sum k^3 - \sum (2m)^3$, the sum of the *odd* cubes, not the alternating sum.
- Taking the even cubes up to $(2n+2)^3$ or using n in place of $2n+1$ in the formula for all the cubes: count the terms carefully.

Beyond STEP

Part (i) is the discrete analogue of the fundamental theorem of calculus: the difference $\mathrm{f}(r) - \mathrm{f}(r-1)$ plays the role of the derivative, and summing it recovers f, just as integrating F' gives $F(b) - F(a)$. This "calculus of finite differences" lets you sum any polynomial by finding a suitable f, and the same telescoping idea underlies many convergence proofs for series. Siklos, *Advanced Problems in Mathematics*, has more practice with telescoping and series of this kind.

WORKED SOLUTION

STEP I 1995, Question 4

Pure • De Moivre's Theorem

STEP I 1995, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q4

The key idea

De Moivre's theorem turns $\sin 5\theta$ into the imaginary part of $(\cos\theta + \mathrm{i}\sin\theta)^5$, and the binomial theorem expands that power into products of $\cos\theta$ and $\sin\theta$. Every term of the imaginary part has an odd power of $\sin\theta$, so we can take out one factor and turn the rest into cosines with $\sin^2\theta = 1 - \cos^2\theta$. Then the angle $\frac{1}{5}\pi$ is special because $\sin 5\theta = 0$ there, which leaves a quadratic in $\cos^2\theta$.

Solution

The identity

Write $c = \cos\theta$ and $s = \sin\theta$. By de Moivre's theorem,

$$\cos 5\theta + \mathrm{i}\sin 5\theta = (c + \mathrm{i}s)^5.$$

By the binomial theorem, with coefficients $1, 5, 10, 10, 5, 1$,

$$(c + \mathrm{i}s)^5 = c^5 + 5c^4(\mathrm{i}s) + 10c^3(\mathrm{i}s)^2 + 10c^2(\mathrm{i}s)^3 + 5c(\mathrm{i}s)^4 + (\mathrm{i}s)^5.$$

Since $\mathrm{i}^2 = -1$, $\mathrm{i}^3 = -\mathrm{i}$, $\mathrm{i}^4 = 1$ and $\mathrm{i}^5 = \mathrm{i}$, the imaginary terms are those with odd powers of $\mathrm{i}s$:

$$\mathrm{Im}(c + \mathrm{i}s)^5 = 5c^4 s - 10c^2 s^3 + s^5.$$

Both c and s are real, so equating imaginary parts gives

$$\begin{aligned}
\sin 5\theta &= s\left(5c^4 - 10c^2 s^2 + s^4\right) \\
&= s\left(5c^4 - 10c^2(1 - c^2) + (1 - c^2)^2\right) \\
&= s\left(5c^4 - 10c^2 + 10c^4 + 1 - 2c^2 + c^4\right) \\
&= s\left(16c^4 - 12c^2 + 1\right),
\end{aligned}$$

using $s^2 = 1 - c^2$ in the second line. That is,

$$\sin 5\theta = \sin\theta\left(16\cos^4\theta - 12\cos^2\theta + 1\right),$$

as required.

Evaluating $\cos^2\frac{1}{5}\pi$

Put $\theta = \frac{1}{5}\pi$. Then $\sin 5\theta = \sin\pi = 0$, while $\sin\frac{1}{5}\pi \neq 0$ (as $0 < \frac{1}{5}\pi < \pi$). So we can divide the identity by $\sin\theta$, and $x = \cos^2\frac{1}{5}\pi$ satisfies

$$16x^2 - 12x + 1 = 0, \qquad \text{so} \qquad x = \frac{12 \pm \sqrt{144 - 64}}{32} = \frac{12 \pm 4\sqrt{5}}{32} = \frac{3 \pm \sqrt{5}}{8}.$$

? **Ask yourself** The quadratic has two roots. Which one is $\cos^2\frac{1}{5}\pi$?

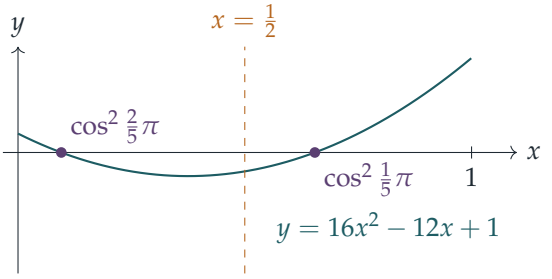
We need one more fact about $\cos\frac{1}{5}\pi$ to pick the root. Since $0 < \frac{1}{5}\pi < \frac{1}{4}\pi$ and cos is decreasing on $[0, \frac{1}{2}\pi]$, we have $\cos\frac{1}{5}\pi > \cos\frac{1}{4}\pi = \frac{1}{\sqrt{2}}$, so

$$\cos^2\tfrac{1}{5}\pi > \tfrac{1}{2}.$$

Now $\sqrt{5} < 3$ gives $\frac{3-\sqrt{5}}{8} < \frac{3}{8} < \frac{1}{2}$, while $\sqrt{5} > 1$ gives $\frac{3+\sqrt{5}}{8} > \frac{1}{2}$. Exactly one root is bigger than $\frac{1}{2}$, so

$$\cos^2\tfrac{1}{5}\pi = \frac{3+\sqrt{5}}{8}.$$

(The other root is not spurious: $\theta = \frac{2}{5}\pi$ also has $\sin 5\theta = 0$ and $\sin\theta \neq 0$, and $\cos^2\frac{2}{5}\pi = \frac{3-\sqrt{5}}{8}$.)



Deducing $\cos\frac{1}{5}\pi$

Since $0 < \frac{1}{5}\pi < \frac{1}{2}\pi$, $\cos\frac{1}{5}\pi > 0$, so we take the positive square root:

$$\cos\tfrac{1}{5}\pi = \sqrt{\frac{3+\sqrt{5}}{8}} = \sqrt{\frac{6+2\sqrt{5}}{16}} = \frac{\sqrt{6+2\sqrt{5}}}{4}.$$

Finally, $(1+\sqrt{5})^2 = 1+2\sqrt{5}+5 = 6+2\sqrt{5}$ and $1+\sqrt{5} > 0$, so $\sqrt{6+2\sqrt{5}} = 1+\sqrt{5}$ and

$$\cos\tfrac{1}{5}\pi = \tfrac{1}{4}\left(1+\sqrt{5}\right),$$

as required.

Checks

- At $\theta = \frac{1}{2}\pi$ the identity gives $\sin\frac{5}{2}\pi = 1\cdot(0-0+1) = 1$, which is right. Small θ: both sides are about 5θ, since the bracket tends to $16-12+1=5$.
- Numerically, $\cos 36^\circ \approx 0.809017$ and $\frac{1}{4}(1+\sqrt{5}) \approx 0.809017$; also $\cos^2 36^\circ \approx 0.654508 = \frac{3+\sqrt{5}}{8}$. We checked the identity at 10^4 random angles as well.
- $\frac{1}{4}(1+\sqrt{5})$ is half the golden ratio φ, the ratio of diagonal to side in a regular pentagon, as an angle of 36° suggests.

Common slips

- Forgetting the signs from powers of i: the s^3 term carries $\mathrm{i}^3 = -\mathrm{i}$, giving $-10c^2s^3$.
- Choosing the root of the quadratic without a reason. You need an inequality such as $\cos^2\frac{1}{5}\pi > \frac{1}{2}$ to rule out $\frac{3-\sqrt{5}}{8}$.
- Dividing by $\sin\theta$ without noting that $\sin\frac{1}{5}\pi \neq 0$.

Beyond STEP

The same method writes $\sin n\theta / \sin\theta$ and $\cos n\theta$ as polynomials in $\cos\theta$, the Chebyshev polynomials, which turn up throughout numerical analysis. The numbers $\mathrm{e}^{2\pi\mathrm{i}k/5}$ are the fifth roots of unity, and the fact that $\cos\frac{1}{5}\pi$ involves only square roots is why a regular pentagon can be constructed with ruler and compasses; Gauss's theorem on which regular polygons are constructible is a highlight of undergraduate Galois theory. For the geometry behind de Moivre's theorem, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP I 1995, Question 5

Pure • Binomial Series and Integration

STEP I 1995, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q5

The key idea

Differentiating a polynomial term by term kills the awkward $1/r$ factors, and what is left is almost a binomial expansion: the coefficients $(-1)^{r+1}\binom{n}{r}$ are exactly those of $-(1-x)^n$ with its constant term removed. Once f' is known, the fundamental theorem of calculus (with $\mathrm{f}(0)=0$) gives f as an integral, and a substitution $y = 1-t$ turns it into the required form. The integrand there is a geometric series, which integrates at once.

Solution

The derivative

Write

$$\mathrm{f}(x) = \sum_{r=1}^{n} (-1)^{r+1} \binom{n}{r} \frac{x^r}{r}.$$

This is a polynomial, so we may differentiate term by term:

$$\mathrm{f}'(x) = \sum_{r=1}^{n} (-1)^{r+1} \binom{n}{r} x^{r-1}.$$

For $x \neq 0$, multiply by x and use $(-1)^{r+1}x^r = -(-x)^r$:

$$\begin{aligned} x\,\mathrm{f}'(x) &= -\sum_{r=1}^{n} \binom{n}{r} (-x)^r \\ &= -\Big(\sum_{r=0}^{n} \binom{n}{r} (-x)^r - 1\Big) \\ &= 1 - (1-x)^n, \end{aligned}$$

by the binomial theorem. Dividing by x,

$$\mathrm{f}'(x) = \frac{1-(1-x)^n}{x} \qquad (x \neq 0),$$

as required. (At $x = 0$ the series gives $\mathrm{f}'(0) = n$, which is also the limit of the right-hand side as $x \to 0$, since $1 - (1-x)^n = nx + O(x^2)$.)

The integral formula

Every term of f contains a factor x, so $\mathrm{f}(0) = 0$. By the fundamental theorem of calculus,

$$\mathrm{f}(x) = \mathrm{f}(0) + \int_0^x \mathrm{f}'(t)\,\mathrm{d}t = \int_0^x \frac{1-(1-t)^n}{t}\,\mathrm{d}t.$$

Ask yourself Is it safe to integrate through $t = 0$, where the formula has t in the denominator?

Yes. The integrand is really the polynomial $\mathrm{f}'(t)$; the fraction is just a compact way of writing it for $t \neq 0$, and its value at $t = 0$ is the limit n. So there is no singularity, only a removable gap in the formula.

Now substitute $y = 1 - t$, so $\mathrm{d}t = -\mathrm{d}y$; when $t = 0$, $y = 1$, and when $t = x$, $y = 1 - x$. Then

$$\begin{aligned}\mathrm{f}(x) &= \int_1^{1-x} \frac{1-y^n}{1-y}\,(-\mathrm{d}y) \\ &= \int_{1-x}^1 \frac{1-y^n}{1-y}\,\mathrm{d}y,\end{aligned}$$

reversing the limits to absorb the minus sign. This is the required result, and it holds for every real x.

The value of $\mathrm{f}(1)$

Put $x = 1$, so the lower limit is 0. For $y \neq 1$ the integrand is the sum of a geometric series,

$$\frac{1-y^n}{1-y} = 1 + y + y^2 + \cdots + y^{n-1},$$

and (as before) the value at $y = 1$ is the limit n, so the integrand is this polynomial on the whole of $[0, 1]$. Hence

$$\begin{aligned}\mathrm{f}(1) &= \int_0^1 (1 + y + \cdots + y^{n-1})\,\mathrm{d}y \\ &= \Big[y + \frac{y^2}{2} + \cdots + \frac{y^n}{n}\Big]_0^1 \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n},\end{aligned}$$

as required.

Notice what this says: putting $x = 1$ in the original series gives the striking identity

$$\sum_{r=1}^{n} \frac{(-1)^{r+1}}{r}\binom{n}{r} = 1 + \frac{1}{2} + \cdots + \frac{1}{n},$$

which is far from obvious directly.

Checks

- $n = 1$: $\mathrm{f}(x) = x$, and $\big(1 - (1 - x)\big)/x = 1 = \mathrm{f}'(x)$; also $\mathrm{f}(1) = 1$.
- $n = 4$: $\mathrm{f}(1) = 4 - \frac{6}{2} + \frac{4}{3} - \frac{1}{4} = \frac{25}{12} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$.
- We checked the derivative formula exactly at 300 random rational points, compared the integral (by Simpson's rule) with the series for random $x \in [-1.5, 1.5]$ and $n \le 15$ (relative agreement better than 10^{-11}), and confirmed the identity for $\mathrm{f}(1)$ exactly for every $n \le 40$.

Common slips

- Forgetting the $r = 0$ term when matching with the binomial expansion of $(1 - x)^n$: the sum from $r = 1$ is $(1 - x)^n - 1$, not $(1 - x)^n$.
- Writing $\mathrm{f}(x) = \int_0^x \mathrm{f}'(t)\,\mathrm{d}t$ without noting that $\mathrm{f}(0) = 0$.
- Getting the limits the wrong way round after the substitution $y = 1 - t$: the sign of $\mathrm{d}t = -\mathrm{d}y$ is what flips them back.

Beyond STEP

The numbers $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ are the harmonic numbers; they grow like $\ln n$, and $H_n - \ln n$ tends to Euler's constant $\gamma \approx 0.5772$. The trick used here, differentiating to remove awkward denominators and then integrating back, is the same one that gives the power series for $\ln(1 + x)$ and $\tan^{-1} x$, and its justification for infinite series (term-by-term differentiation of power series) is a core topic of first-year analysis. Abbott's *Understanding Analysis* treats this carefully.

WORKED SOLUTION

STEP I 1995, Question 6

Pure • Bernoulli Differential Equations

STEP I 1995, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q6

The key idea

Neither equation is linear in y, but each is linear in a suitable power of $1/y$. The left-hand side contains y'/y^2, which is (up to sign) the derivative of $1/y$; so writing $u = 1/y$ turns the equation into a first-order linear equation for u, which an integrating factor finishes off. In part (ii) the term y'/y^3 suggests looking at $1/y^2$ instead.

Solution

Part (i)

Put $u = 1/y$. By the chain rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x},$$

so the first term of the equation is exactly $-\mathrm{d}u/\mathrm{d}x$, and the second is u. The equation becomes

$$-\frac{\mathrm{d}u}{\mathrm{d}x} + u = \mathrm{e}^{2x}, \qquad \text{that is,} \qquad \frac{\mathrm{d}u}{\mathrm{d}x} - u = -\mathrm{e}^{2x}.$$

This is linear in u. The integrating factor is $\mathrm{e}^{\int -1\,\mathrm{d}x} = \mathrm{e}^{-x}$, and multiplying through gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(u\,\mathrm{e}^{-x}\right) = \mathrm{e}^{-x}\frac{\mathrm{d}u}{\mathrm{d}x} - \mathrm{e}^{-x}u = -\mathrm{e}^{x}.$$

Integrating, with A an arbitrary constant,

$$u\,\mathrm{e}^{-x} = A - \mathrm{e}^{x}, \qquad \text{so} \qquad u = A\mathrm{e}^{x} - \mathrm{e}^{2x}.$$

Since $y = 1/u$,

$$y = \frac{1}{A\mathrm{e}^{x} - \mathrm{e}^{2x}},$$

which is the required general solution. (The original equation involves $1/y$, so $y \neq 0$ throughout and the substitution loses no solutions; the formula is valid on any interval where $A\mathrm{e}^{x} \neq \mathrm{e}^{2x}$, i.e. avoiding $x = \ln A$ when $A > 0$.)

Part (ii)

Ask yourself Which substitution plays the role of $u = 1/y$ here?

We want the derivative of the new variable to produce y'/y^3. Since

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{y^2}\right) = -\frac{2}{y^3}\frac{\mathrm{d}y}{\mathrm{d}x},$$

the right choice is $v = 1/y^2$. Then the first term is $-\frac{1}{2}\,\mathrm{d}v/\mathrm{d}x$ and the second is v, so

$$-\frac{1}{2}\frac{\mathrm{d}v}{\mathrm{d}x} + v = \mathrm{e}^{2x}, \qquad \text{that is,} \qquad \frac{\mathrm{d}v}{\mathrm{d}x} - 2v = -2\mathrm{e}^{2x}.$$

Now the integrating factor is e^{-2x}:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(v\,\mathrm{e}^{-2x}\right) = -2\mathrm{e}^{2x}\cdot\mathrm{e}^{-2x} = -2.$$

Integrating, with B an arbitrary constant,

$$v\,\mathrm{e}^{-2x} = B - 2x, \qquad \text{so} \qquad \frac{1}{y^2} = (B-2x)\,\mathrm{e}^{2x}.$$

Hence

$$y^2 = \frac{\mathrm{e}^{-2x}}{B-2x}, \qquad y = \pm\frac{\mathrm{e}^{-x}}{\sqrt{B-2x}}.$$

Because $1/y^2 > 0$, we need $B - 2x > 0$: the solution exists only for $x < \frac{1}{2}B$, and the sign is fixed by whichever sign y has initially (it can't change sign, since y is never zero). Unlike part (i), the answer isn't a difference of two exponentials: here e^{2x} on the right exactly cancels the integrating factor e^{-2x}, leaving a constant to integrate, which is why the linear term $-2x$ appears.

Checks

- Substitute back in (ii): with $1/y^2 = (B-2x)\mathrm{e}^{2x}$, differentiating gives $-2y'/y^3 = -2\mathrm{e}^{2x} + 2(B-2x)\mathrm{e}^{2x}$, so $y'/y^3 = \mathrm{e}^{2x} - 1/y^2$, which is the equation.
- Numerically: taking $y(0) = 1$ in (ii) gives $B = 1$, so the formula predicts $y(0.4) = \mathrm{e}^{-0.4}/\sqrt{0.2} \approx 1.49888$. Stepping the differential equation from $x = 0$ numerically gives the same value to ten decimal places. Both general solutions also satisfy their equations at many random points and constants.

Common slips

- Forgetting the minus sign (or the factor 2 in part (ii)) in $\mathrm{d}u/\mathrm{d}x$, which wrecks the linear equation.
- In part (ii), writing only the positive square root, or not noticing that the solution requires $x < \frac{1}{2}B$.

Beyond STEP

Both equations are examples of Bernoulli equations, $y' + P(x)y = Q(x)y^n$: dividing by y^n and substituting $u = y^{1-n}$ always produces a linear equation. Here part (i) is $n = 2$ and part (ii) is $n = 3$. The fact that solutions can "blow up" at a finite x (as in $x \to \frac{1}{2}B$) is a typical feature of nonlinear equations, studied in a first course on ODEs; see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP I 1995, Question 7

Pure • Vector Geometry of Triangles

STEP I 1995, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q7

The key idea

Putting the origin at the point equidistant from A, B and C means the position vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ all have the same length. That one fact does all the work, because for vectors of equal length $(\mathbf{b}+\mathbf{c})\cdot(\mathbf{c}-\mathbf{b}) = |\mathbf{c}|^2 - |\mathbf{b}|^2 = 0$. Once we find that Q has the beautifully simple position vector $\mathbf{a}+\mathbf{b}+\mathbf{c}$, both parts become a line or two of algebra.

Solution

Choosing the origin

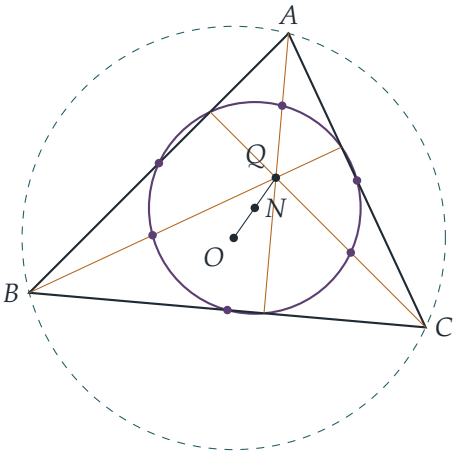
The points equidistant from A and B form the perpendicular bisector of AB, a straight line perpendicular to AB; similarly for B and C. Since A, B, C are not collinear, AB and BC are not parallel, so neither are their perpendicular bisectors. These two lines therefore meet in exactly one point, which is equidistant from A and B and from B and C, and so from all three. (It is the circumcentre of the triangle.)

Take this point as O, write $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ for the position vectors of A, B, C, and let R be the common distance, so

$$|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = R.$$

The centroid has position vector $\mathbf{g} = \frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$. Since $\overrightarrow{GQ} = 2\overrightarrow{OG}$ means $\mathbf{q}-\mathbf{g} = 2\mathbf{g}$,

$$\mathbf{q} = 3\mathbf{g} = \mathbf{a}+\mathbf{b}+\mathbf{c}, \qquad \mathbf{n} = \tfrac{1}{2}\mathbf{q} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}).$$



The figure shows the circle through A, B, C centred at O (dashed), the altitudes meeting at Q, and the circle of radius $\frac{1}{2}R$ centred at N through the six midpoints of part (ii).

Part (i)

We have $\overrightarrow{AQ} = \mathbf{q} - \mathbf{a} = \mathbf{b} + \mathbf{c}$ and $\overrightarrow{BC} = \mathbf{c} - \mathbf{b}$, so

$$\overrightarrow{AQ} \cdot \overrightarrow{BC} = (\mathbf{c} + \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b}) = |\mathbf{c}|^2 - |\mathbf{b}|^2 = R^2 - R^2 = 0,$$

using $\mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{b}$ to cancel the cross terms. Hence $\overrightarrow{AQ}$ is perpendicular to $\overrightarrow{BC}$.

? **Ask yourself** Does $\overrightarrow{AQ} \cdot \overrightarrow{BC} = 0$ really put Q on the altitude from A? What if $\overrightarrow{AQ}$ is the zero vector?

If $Q \neq A$, the line AQ passes through A and is perpendicular to BC, so it *is* the altitude from A, and Q lies on it. If $Q = A$ (which happens exactly when $\mathbf{b} + \mathbf{c} = \mathbf{0}$, i.e. BC is a diameter and the angle at A is a right angle), then Q lies on the altitude from A trivially. Either way, Q is on the altitude from A.

Nothing in the argument singled out A: the expression $\mathbf{a} + \mathbf{b} + \mathbf{c}$ is symmetric, so in the same way

$$\overrightarrow{BQ} \cdot \overrightarrow{CA} = (\mathbf{c} + \mathbf{a}) \cdot (\mathbf{a} - \mathbf{c}) = |\mathbf{a}|^2 - |\mathbf{c}|^2 = 0,$$
$$\overrightarrow{CQ} \cdot \overrightarrow{AB} = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} - \mathbf{a}) = |\mathbf{b}|^2 - |\mathbf{a}|^2 = 0,$$

and Q also lies on the altitudes from B and from C. So all three altitudes pass through the single point Q: they are concurrent. (Q is the orthocentre.)

Part (ii)

The midpoint of AQ has position vector $\frac{1}{2}(\mathbf{a} + \mathbf{q}) = \mathbf{a} + \frac{1}{2}(\mathbf{b} + \mathbf{c})$, so its displacement from N is

$$\mathbf{a} + \tfrac{1}{2}(\mathbf{b} + \mathbf{c}) - \tfrac{1}{2}(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \tfrac{1}{2}\mathbf{a}.$$

The midpoint of BC has position vector $\frac{1}{2}(\mathbf{b}+\mathbf{c})$, so its displacement from N is

$$\tfrac{1}{2}(\mathbf{b}+\mathbf{c}) - \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}) = -\tfrac{1}{2}\mathbf{a}.$$

Both of these have length $\frac{1}{2}|\mathbf{a}| = \frac{1}{2}R$.

By the same symmetry, the midpoints of BQ and CA are displaced from N by $\pm\frac{1}{2}\mathbf{b}$, and the midpoints of CQ and AB by $\pm\frac{1}{2}\mathbf{c}$. Since $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = R$, all six midpoints are at distance $\frac{1}{2}R$ from N, which is what we had to show. In fact we have shown a little more: the midpoint of AQ and the midpoint of BC are at opposite ends of a diameter of this circle, and likewise for the other two pairs.

Checks

- An equilateral triangle: then $\mathbf{a}+\mathbf{b}+\mathbf{c} = \mathbf{0}$, so O, G, Q and N all coincide, and the six midpoints should all be at distance $\frac{1}{2}R$ from the centre. Indeed the midpoints of the sides are at distance $R\cos 60^\circ = \frac{1}{2}R$, and the midpoints of AQ etc. are halfway along radii.
- A right angle at A: then $Q = A$, as the argument above predicts, since the two legs are themselves altitudes through A.
- Numerically, we took 10^5 random triangles, computed the circumcentre directly, and confirmed $Q = \mathbf{a}+\mathbf{b}+\mathbf{c}$, that all three dot products vanish, and that all six midpoints are at distance $\frac{1}{2}R$ from N, with largest relative error about 10^{-13}.

Common slips

- Reading $\overrightarrow{GQ} = 2\overrightarrow{OG}$ as $\mathbf{q} = 2\mathbf{g}$. It says $\mathbf{q}-\mathbf{g} = 2\mathbf{g}$, so $\mathbf{q} = 3\mathbf{g}$.
- Stopping after one perpendicularity. Concurrency needs Q on all three altitudes; say explicitly that the same calculation works for B and C.
- Forgetting to justify the origin: the reason it exists is that the perpendicular bisectors of two non-parallel sides must meet.

Beyond STEP

Since $\mathbf{q} = 3\mathbf{g}$ and $\mathbf{n} = \frac{3}{2}\mathbf{g}$, the points O, G, N, Q all lie on one line, the *Euler line*, with $OG : GQ = 1 : 2$. The circle of part (ii) is the *nine-point circle*: it also passes through the three feet of the altitudes (visible in the figure), because each foot sees a diameter such as the one from the midpoint of AQ to the midpoint of BC at a right angle. Choosing a clever origin so that a symmetric expression does the work is a recurring trick in vector geometry and in linear algebra; Strang's *Introduction to Linear Algebra* develops the vector and dot-product tools further, and Siklos's *Advanced Problems in Mathematics* has more STEP-style geometry in this spirit.

WORKED SOLUTION

STEP I 1995, Question 8

Pure • Linear Differential Equations

STEP I 1995, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q8

The key idea

The unknowns are three functions, but at each fixed x they are just three numbers $f(x)$, $g(x)$, $h(x)$. Substituting the three given solutions into $(*)$ gives three linear equations in these three numbers, which we solve as simultaneous equations. For the second paragraph, the left-hand side of $(*)$ is *linear* in y, so the result for a combination of x, 1 and x^{-1} follows from what each piece produces separately.

Solution

Finding f, g and h

Write Ly for the left-hand side $y'' + f(x)y' + g(x)y$, so $(*)$ says $Ly = h(x)$. We substitute each solution in turn.

- $y = 1$: here $y' = y'' = 0$, so $g(x) = h(x)$.
- $y = x$: here $y' = 1$, $y'' = 0$, so $f(x) + x\,g(x) = h(x)$.
- $y = x^{-1}$: here $y' = -x^{-2}$, $y'' = 2x^{-3}$, so

$$\frac{2}{x^3} - \frac{f(x)}{x^2} + \frac{g(x)}{x} = h(x).$$

Using $h = g$ in the second equation gives $f(x) = (1-x)\,g(x)$. Putting both of these into the third equation and multiplying through by x^3 (allowed, as $x > 0$):

$$\begin{aligned} 2 - x(1-x)\,g(x) + x^2\,g(x) &= x^3\,g(x) \\ 2 = g(x)\left(x^3 - 2x^2 + x\right) &= g(x)\,x(1-x)^2. \end{aligned}$$

Since $0 < x < 1$, the factor $x(1-x)^2$ is non-zero, so we can divide by it. Therefore

$$g(x) = h(x) = \frac{2}{x(1-x)^2}, \qquad f(x) = (1-x)\,g(x) = \frac{2}{x(1-x)}.$$

At each x we solved the three equations step by step with no free choices, so these functions are the only ones that work. The restriction $0 < x < 1$ keeps us away from $x = 0$ and $x = 1$, where they blow up.

So the equation is

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{2}{x(1-x)}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2}{x(1-x)^2}y = \frac{2}{x(1-x)^2}.$$

When is $ax + b + c/x$ a solution?

The operator L is linear: $L(\alpha u + \beta v) = \alpha\, Lu + \beta\, Lv$ for constants α, β, because differentiation is linear and f, g multiply y' and y only. From the first paragraph,

$$L(x) = L(1) = L(x^{-1}) = \mathrm{h}(x).$$

Hence, for $y = ax + b + c/x$,

$$Ly = a\,L(x) + b\,L(1) + c\,L(x^{-1}) = (a + b + c)\,\mathrm{h}(x).$$

Ask yourself Why can we conclude $a + b + c = 1$, rather than just "$a + b + c = 1$ at some x"?

We need $(a + b + c)\,\mathrm{h}(x) = \mathrm{h}(x)$ for all $0 < x < 1$, that is, $(a + b + c - 1)\,\mathrm{h}(x) = 0$. But $\mathrm{h}(x) = 2/\big(x(1-x)^2\big) > 0$ on this interval, so we can divide by it. So $y = ax + b + c/x$ is a solution of $(*)$ if and only if

$$a + b + c = 1.$$

Checks

- Taking $(a, b, c) = (1, 0, 0)$, $(0, 1, 0)$ or $(0, 0, 1)$ recovers the three given solutions, and each has $a + b + c = 1$.
- Directly: with $x = \frac{1}{2}$ we get $\mathrm{f} = 8$, $\mathrm{g} = \mathrm{h} = 16$. For $y = 1/x$, $y = 2$, $y' = -4$, $y'' = 16$, and $16 - 32 + 32 = 16$, as required.
- The computer checked all three solutions exactly at 49 rational points in $(0, 1)$, and confirmed $Ly = (a + b + c)\mathrm{h}$ for 2000 random (a, b, c, x).

Common slips

- Writing $y'' = -2x^{-3}$ for $y = x^{-1}$: two minus signs multiply to give $+2x^{-3}$.
- Answering $a + b + c = 0$. That is the condition for $ax + b + c/x$ to solve the *homogeneous* equation $Ly = 0$; here the right-hand side is h, not 0.

Beyond STEP

This is the structure of solutions of a linear differential equation: the difference of any two solutions of $Ly = \mathrm{h}$ solves $Ly = 0$. Here $x - 1$ and $x^{-1} - 1$ solve the homogeneous equation, and for a second-order equation two independent solutions give all of them. So $1 + \alpha(x - 1) + \beta(x^{-1} - 1)$ is the general solution, which is exactly the family with $a + b + c = 1$. The Wronskian makes "independent" precise. Boyce and DiPrima's *Elementary Differential Equations* covers all of this.

WORKED SOLUTION

STEP I 1995, Question 9

Mechanics • Projectile Envelope

STEP I 1995, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q9

The key idea

Fix a point (x, y) and ask: is there an angle of projection whose trajectory passes through it? The trajectory equation, written in terms of $\tan\theta$ rather than θ, is a *quadratic* in $\tan\theta$. A point can be reached exactly when that quadratic has a real root, so the whole question comes down to a discriminant.

Solution

The model

We treat the particle as a point mass moving under gravity alone: no air resistance, g constant in size and direction, and nothing in the way of the path. The particle then moves in the vertical plane containing its initial velocity, and since only the horizontal *distance* from O matters, by symmetry we may take $x \geq 0$, with the particle projected towards increasing x at angle θ above the horizontal, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. (Straight up, $\theta = \frac{\pi}{2}$, is dealt with at the end.)

The trajectory

Write $u = \sqrt{2gh}$, so $u^2 = 2gh$. With constant acceleration g downwards, after time t

$$x = ut\cos\theta, \qquad y = ut\sin\theta - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(u\cos\theta)$ and using $\sec^2\theta = 1 + \tan^2\theta$,

$$y = x\tan\theta - \frac{gx^2}{2u^2}\left(1 + \tan^2\theta\right) = x\tan\theta - \frac{x^2}{4h}\left(1 + \tan^2\theta\right),$$

since $2u^2 = 4gh$. Writing $T = \tan\theta$ and multiplying through by $4h$, the trajectory passes through (x, y) if and only if

$$x^2T^2 - 4hx\,T + \left(x^2 + 4hy\right) = 0. \qquad (*)$$

As θ runs over $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $T = \tan\theta$ takes every real value exactly once, so choosing an angle is the same as choosing a real number T.

Ask yourself Why write the trajectory in terms of $\tan\theta$ instead of θ?

In terms of θ the equation mixes sin, cos and $\sec^2$, which is awkward. With $\sec^2\theta = 1 + \tan^2\theta$ everything is a polynomial in the one variable T, and "is there an angle?" becomes "does this quadratic have a real root?", a question we can answer at once.

Which points can be reached

Take $x > 0$. Then $(*)$ is a genuine quadratic in T, and it has a real root if and only if its discriminant is non-negative:

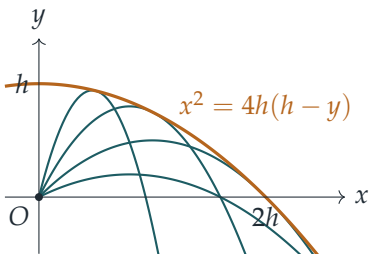
$$\begin{aligned}(4hx)^2 - 4x^2\left(x^2 + 4hy\right) \geq 0 &\iff 4x^2\left(4h^2 - x^2 - 4hy\right) \geq 0 \\ &\iff x^2 \leq 4h^2 - 4hy = 4h(h-y),\end{aligned}$$

where we divided by $4x^2 > 0$. So if (x, y) lies *above* the parabola, meaning $x^2 > 4h(h-y)$, or equivalently $y > h - \dfrac{x^2}{4h}$, then $(*)$ has no real root and no angle of projection sends the particle through (x, y).

It remains to deal with $x = 0$, the vertical line through O. Points on it with $y \neq 0$ are reached only by projecting vertically (any other angle gives $x > 0$ for all $t > 0$). Projected straight up with speed u, the particle rises to height $u^2/(2g) = h$ and no further, which is exactly the point where the parabola meets $x = 0$. So on this line too nothing above the parabola is reached.

Hence, whatever the angle of projection, the particle can never reach a point above the parabola $x^2 = 4h(h-y)$. ■

The bound is sharp: when $x^2 = 4h(h-y)$ the discriminant is zero and there is exactly one angle, $\tan\theta = 2h/x$, whose path just touches the parabola. The parabola is the *envelope* of all the trajectories, often called the parabola of safety.



Four trajectories (teal) with speed $\sqrt{2gh}$, each touching the boundary parabola (amber) once.

Simplifying assumptions

- The particle is a point mass (no size, no spin).
- Air resistance (and any other force except gravity) is neglected.
- The gravitational field is uniform: g is constant over the region of motion, which also means treating the Earth's surface as flat and ignoring its rotation.
- The path is unobstructed, for instance by the ground, so the particle can actually travel to the points in question.

Checks

- At $x = 0$ the parabola gives $y = h = u^2/(2g)$, the greatest height of a vertical throw. On $y = 0$ it gives $x = 2h = u^2/g$, the familiar maximum range on level ground (at 45°, where $\tan\theta = 2h/x = 1$, as predicted).
- Numerically, with $g = 9.8$ and $h = 1.7$, we stepped the motion for 399 launch angles from almost straight down to almost straight up; the greatest height above the parabola anywhere on any path was about -5×10^{-13}, i.e. zero up to rounding, and maximising y over angle at fixed x reproduced $h - x^2/(4h)$ every time.

Common slips

- Working with θ in the range $\left[0, \frac{\pi}{2}\right]$ only. Points below O need downward projections ($T < 0$), and the argument has to cover them.
- Dividing by x (or x^2) without treating $x = 0$ separately.
- Showing only that one particular trajectory lies under the parabola; the claim is about *every* angle, which is why we fix the point and vary the angle.

Beyond STEP

The parabola here is the *envelope* of the one-parameter family of trajectories: the curve touching every member. For a family $F(x, y, T) = 0$ the envelope is found by eliminating T between $F = 0$ and $\partial F/\partial T = 0$, which for a quadratic in T is precisely the vanishing of the discriminant. Envelopes reappear in caustics in optics, in the theory of first-order differential equations (singular solutions) and in the "safe region" for all projectiles of a given speed. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile motion with and without air resistance.

WORKED SOLUTION

STEP I 1995, Question 10

Mechanics • Elastic Strings and Bouncing

STEP I 1995, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q10

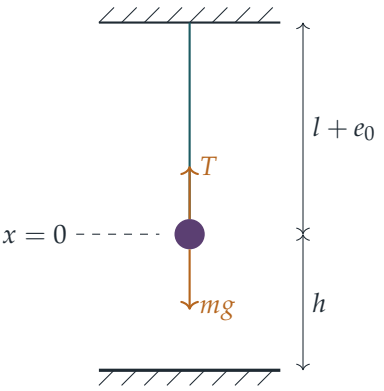
The key idea

Measured from the equilibrium position, the weight and the equilibrium part of the tension cancel exactly, so the only unbalanced force is a Hooke's-law restoring force proportional to the displacement x. Simple harmonic motion follows, and its energy equation is the speed formula. Everything after that is bookkeeping: the energy equation carries the ball between $x = 0$ and the surface, and Newton's experimental law reverses and scales the velocity at each bounce.

Solution

We model the ball as a particle and the string as light and elastic, obeying Hooke's law while it is stretched. We ignore air resistance, and take the bounce to be instantaneous. Throughout we write

$$\omega^2 = \frac{\lambda}{lm}.$$



The equilibrium length

Let the equilibrium extension be e_0, so the tension is $T_0 = \lambda e_0/l$. Resolving vertically for the ball in equilibrium, $T_0 = mg$, so

$$\frac{\lambda e_0}{l} = mg \quad \Longrightarrow \quad e_0 = \frac{mgl}{\lambda},$$

and the total length is $l + e_0 = l(1 + mg/\lambda)$, as required.

The speed formula

Let x be the distance of the ball below the equilibrium position, with downwards positive. For $x \geq 0$ the extension is $e_0 + x > 0$, so the string is taut with tension $\lambda(e_0 + x)/l$. Newton's second law downwards gives

$$m\ddot{x} = mg - \frac{\lambda(e_0 + x)}{l} = -\frac{\lambda x}{l},$$

because $\lambda e_0/l = mg$. So $\ddot{x} = -\omega^2 x$: simple harmonic motion about $x = 0$.

Writing $\ddot{x} = v\,\dfrac{\mathrm{d}v}{\mathrm{d}x}$ and integrating,

$$\int v\,\mathrm{d}v = -\omega^2 \int x\,\mathrm{d}x \quad \Longrightarrow \quad v^2 = -\omega^2 x^2 + C.$$

At $x = 0$ the speed is u_0, so $C = u_0^2$ and

$$v^2 + \frac{\lambda}{lm}x^2 = u_0^2,$$

as required. (Equivalently, this is conservation of energy: the gravitational and elastic energies combine into $\frac{1}{2} \cdot \frac{\lambda}{l}x^2$ plus a constant.)

The first bounce

The ball reaches the surface only if $v^2 > 0$ at $x = h$, that is, if $u_0^2 - \omega^2 h^2 > 0$. This is exactly the given condition $\lambda h^2 < lmu_0^2$. So the ball strikes the surface with speed w_0, where

$$w_0^2 = u_0^2 - \omega^2 h^2.$$

By Newton's experimental law, it leaves the surface moving upwards with speed ew_0.

On the way up from $x = h$ to $x = 0$ the string is still stretched, so the same equation of motion holds and the energy integral is $v^2 + \omega^2 x^2 = \text{constant}$. Only the constant is new. At $x = h$ we have $v^2 = e^2 w_0^2$, so the constant is $e^2 w_0^2 + \omega^2 h^2$, and at $x = 0$

$$\begin{aligned} u_1^2 &= e^2 w_0^2 + \omega^2 h^2 \\ &= e^2\left(u_0^2 - \omega^2 h^2\right) + \omega^2 h^2 \\ &= e^2 u_0^2 + \frac{\lambda}{lm}h^2\left(1 - e^2\right), \end{aligned}$$

as required. Here u_1 is directed upwards.

The second bounce

Ask yourself After passing $x = 0$ the ball rises above the equilibrium position, and the string may go slack. Does the ball still come back to $x = 0$ with speed u_1?

Yes. Above $x = 0$ the ball moves under its weight and the string tension, or under gravity alone if the string goes slack. Both forces are conservative: when the string is slack its elastic energy is simply zero. So the total mechanical energy is conserved for the whole of the flight above $x = 0$. When the ball comes back down to $x = 0$ it has the same potential energy as when it left, and therefore the same speed u_1, now directed downwards.

So the second descent is exactly like the first, with u_1 in place of u_0. We still need the ball to reach the surface, and it does, because

$$u_1^2 - \omega^2 h^2 = e^2\left(u_0^2 - \omega^2 h^2\right) > 0.$$

Repeating the first-bounce argument with u_0 replaced by u_1,

$$\begin{aligned} u_2^2 &= e^2 u_1^2 + \omega^2 h^2\left(1 - e^2\right) \\ &= e^2\left[e^2 u_0^2 + \omega^2 h^2(1 - e^2)\right] + \omega^2 h^2(1 - e^2) \\ &= e^4 u_0^2 + \omega^2 h^2(1 - e^2)(1 + e^2) \\ &= e^4 u_0^2 + \frac{\lambda}{lm} h^2\left(1 - e^4\right), \end{aligned}$$

which completes the question.

Checks

- If $e = 1$ there is no energy loss, and both formulae give $u_1 = u_2 = u_0$, as they should.
- If $e = 0$ the ball leaves the surface at rest, and then $u_1^2 = \omega^2 h^2$: the ball rises from rest at amplitude h and passes the centre at speed ωh, as simple harmonic motion predicts.
- A neat way to see the pattern: $u_n^2 - \omega^2 h^2 = e^{2n}\left(u_0^2 - \omega^2 h^2\right)$, because each bounce multiplies the squared impact speed by e^2.
- We stepped the motion numerically (Runge–Kutta, with the string allowed to go slack). For example, with $m = 1$, $l = 1$, $\lambda = 20$, $g = 9.8$, $u_0 = 6$, $h = 0.3$ and $e = 0.9$ the simulation gives $u_1 = 5.4316$ and $u_2 = 4.9233$, matching the formulae. In this example the string does go slack between bounces.

Common slips

- Measuring x from the natural length rather than from equilibrium. Then the weight doesn't cancel, and the formula comes out with extra linear terms.
- Applying the restitution factor e to the speed at $x = 0$ instead of to the impact speed at $x = h$.
- Assuming the string stays taut above equilibrium. It needn't, but energy conservation makes this irrelevant to the speed at $x = 0$.

Beyond STEP

The cancellation of gravity about the equilibrium point is a first example of a general principle: linearise about an equilibrium, and small oscillations are simple harmonic with ω^2 equal to the "stiffness" divided by the mass. The bouncing ball is an example of a piecewise-smooth or *impact* oscillator. These systems can behave remarkably richly (period doubling and even chaos) when the surface itself is moving. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats oscillators and energy methods at first-year level.

WORKED SOLUTION

STEP I 1995, Question 11

Mechanics • Friction Between Cylinders

STEP I 1995, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q11

The key idea

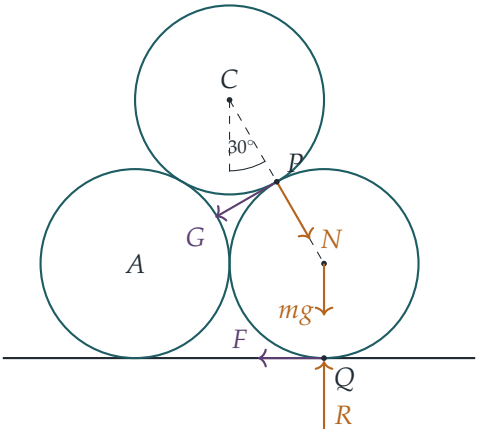
Look at one of the bottom cylinders. Its weight and the normal reaction from the top cylinder both pass through its centre, so taking moments about the centre involves only the two friction forces, and they must be equal in size. One horizontal resolution then fixes the ratio of friction to normal reaction at the contact between the cylinders, and the geometry of three touching circles (angles of 30° and 60°) does the rest.

Solution

The model

We treat the cylinders as uniform rigid cylinders of radius r and work in a cross-section perpendicular to the axes, so each cylinder is a circle with its weight acting at the centre. Every contact has the same coefficient of friction μ. Call the bottom cylinders A (left) and B (right), and the top one C. The centres of A and B are $2r$ apart at height r; the centre of C is $2r$ from each, so the triangle of centres is equilateral and the line from the centre of C to the centre of B makes 30° with the vertical.

For now we assume A and B exert no force on each other; we come back to this at the end. By symmetry it is enough to look at B and at the right-hand contact of C.



The forces on B are:

- its weight mg at its centre;
- at Q, the ground contact: a normal reaction R upwards and a friction force F horizontally towards A;
- at P, the contact with C: a normal reaction N along the line of centres (down and outwards, at 30° to the vertical), and a friction force G along the common tangent, which we take as acting down and towards A, that is at 30° below the horizontal.

If either friction force really acts the other way, its value will simply come out negative.

Moments about the centre of B

The weight passes through the centre of B, and so does N, since a normal reaction between two circles acts along the line of their centres. Both friction forces act along tangents, at perpendicular distance r from the centre. F, at the bottom and pointing towards A, turns B clockwise in the figure; G, at P and pointing down the tangent towards A, turns it anticlockwise. Hence

$$Fr = Gr, \qquad \text{so} \qquad F = G.$$

Resolving horizontally for B

N has an outward horizontal component $N\sin 30^\circ = \frac{1}{2}N$. Both F and the horizontal component $G\cos 30^\circ = \frac{\sqrt{3}}{2}G$ of G point inwards. So

$$\tfrac{1}{2}N = F + \tfrac{\sqrt{3}}{2}G = G\left(1 + \tfrac{\sqrt{3}}{2}\right),$$

and therefore

$$\frac{G}{N} = \frac{1}{2+\sqrt{3}} = \frac{2-\sqrt{3}}{(2+\sqrt{3})(2-\sqrt{3})} = 2-\sqrt{3}.$$

This is positive, so G does act in the direction we assumed, and so does $F = G$.

For the contact at P to hold without slipping we need $G \le \mu N$, that is

$$\mu \ge 2 - \sqrt{3}.$$

The contact with the ground

We must also check that the ground contact does not demand more friction. For C, the forces from B are N up and to the left at 30° to the vertical, and G up the tangent at 30° to the horizontal. Their vertical components are $\frac{\sqrt{3}}{2}N$ and $\frac{1}{2}G$, and by symmetry A contributes the same. Resolving vertically for C,

$$2\left(\tfrac{\sqrt{3}}{2}N + \tfrac{1}{2}G\right) = mg \implies N\left(\sqrt{3} + 2 - \sqrt{3}\right) = mg \implies N = \tfrac{1}{2}mg.$$

Resolving vertically for the whole system, which has weight $3mg$ shared equally by symmetry, $R = \frac{3}{2}mg$. Then

$$\frac{F}{R} = \frac{(2-\sqrt{3})\cdot\frac{1}{2}mg}{\frac{3}{2}mg} = \frac{2-\sqrt{3}}{3} < 2 - \sqrt{3},$$

so whenever $\mu \ge 2 - \sqrt{3}$ the ground contact holds as well. The contact between the cylinders is the critical one.

> **Ask yourself** A and B are touching. Why may we ignore any force between them?

They can only push each other apart, so suppose A pushes B outwards with a force $S \ge 0$ along the line of centres. This force passes through the centre of B, so the moment equation still gives $F = G$, but the horizontal resolution becomes $\frac{1}{2}N + S = G\left(1 + \frac{\sqrt{3}}{2}\right)$. Meanwhile the vertical equation for C still holds, and solving the two together shows that G/N increases as S increases (at $S = 0$ it is $2 - \sqrt{3}$). The problem is statically indeterminate, since S is not fixed by the equations, and the least friction is needed when $S = 0$. So the minimum possible coefficient of friction is $2 - \sqrt{3}$, as required.

Checks

- $2 - \sqrt{3} = \tan 15^\circ \approx 0.268$. So at the cylinder contact the total reaction makes exactly 15° with the normal, half the 30° angle in the geometry. That is a pleasing sign the algebra is right.
- The ground ratio $(2 - \sqrt{3})/3 \approx 0.089$ is a third of the cylinder ratio, which makes sense: the ground carries $\frac{3}{2}mg$, three times N, but the same size of friction.
- A numerical check, substituting into all the force and moment equations for B and C and minimising over the unknown push $S \ge 0$, gave a least μ of $0.267949\ldots$, agreeing with $2 - \sqrt{3}$.

Common slips

- Taking moments about the ground contact or about P instead of the centre. This works, but it brings in N and gives messier algebra; the centre removes both mg and N at once.
- Checking only the ground. The ground contact needs much less friction, so it gives the wrong critical value.
- Forgetting that the friction on C acts *up* the tangent, by Newton's third law, when resolving vertically for C.

Beyond STEP

The force S between the bottom cylinders cannot be found from the equilibrium equations: the structure is *statically indeterminate*, and the question sensibly asks for the minimum friction over all possible values of S. In engineering such systems are settled by modelling elastic deformation, and the same issues come up in the mechanics of granular materials such as stacked pipes or piles of sand. For a careful treatment of statics, friction and rigid-body equilibrium, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1995, Question 12

Probability and Statistics • Arrangements and Counting

STEP I 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q12

The key idea

Every arrangement of the n pupils is equally likely, so each probability is a count of favourable arrangements divided by $n!$. Better still, by symmetry we only need to care about *which positions* the hockey players occupy: each of the $\binom{n}{r}$ sets of r positions is equally likely. Each part then becomes a counting problem about choosing positions. For the last part, the hint about gaps is the standard "stars and bars" trick: line up the non-hockey-players first, and the hockey players must go into separate gaps between them.

Solution

Throughout, call the r hockey players H and the other $n-r$ pupils N. All $n!$ orderings of the (distinguishable) pupils are equally likely. Since every set of r positions for the Hs arises from the same number, $r!\,(n-r)!$, of orderings, the set of positions occupied by the Hs is equally likely to be any of the $\binom{n}{r}$ possible sets. We use whichever view is more convenient.

Part (i)

Fill the two end positions first. The pupil at the left end is a hockey player with probability $\frac{r}{n}$. Given that, the pupil at the right end is one of the remaining $n-1$ pupils, each equally likely, of whom $r-1$ play hockey. So

$$\mathrm{P}(\text{hockey player at each end}) = \frac{r}{n}\cdot\frac{r-1}{n-1} = \frac{r(r-1)}{n(n-1)}.$$

(We need $r \geq 2$ for this to be non-zero, which the question guarantees.)

Part (ii)

The hockey players stand together exactly when their positions form a block of r consecutive places. Such a block can start at position $1, 2, \ldots, n-r+1$, so there are $n-r+1$ favourable sets of positions out of $\binom{n}{r}$:

$$\mathrm{P}(\text{all together}) = \frac{n-r+1}{\binom{n}{r}} = \frac{r!\,(n-r+1)!}{n!}.$$

Equivalently, glue the r hockey players into one "super-pupil": there are $(n-r+1)!$ orderings of the $n-r+1$ units and $r!$ orderings inside the block, giving the same answer. As a check, when $r = n$ this is 1, as it should be.

Part (iii)

Stand the $n-r$ non-hockey-players in a row first. They create $n-r+1$ gaps: one before the first, one between each neighbouring pair, and one after the last.



No two hockey players are adjacent if and only if each gap receives at most one hockey player. (If two Hs were in the same gap they would be next to each other; if they are in different gaps, at least one N separates them.) So the favourable arrangements correspond to choosing r of the $n-r+1$ gaps, one H per chosen gap.

Count positions: a favourable set of H-positions is determined by which r gaps are used, so there are $\binom{n-r+1}{r}$ of them. Hence

$$\mathrm{P}(\text{no two together}) = \frac{\binom{n-r+1}{r}}{\binom{n}{r}} = \frac{(n-r+1)!\,(n-r)!}{(n-2r+1)!\,n!}.$$

(Counting orderings of the pupils directly gives the same thing: $(n-r)!$ ways to order the Ns, then $(n-r+1)(n-r)\cdots(n-2r+2) = \frac{(n-r+1)!}{(n-2r+1)!}$ ways to put the r distinct Hs into distinct gaps, all over $n!$.)

Ask yourself When is it impossible to keep all the hockey players apart?

We need r different gaps, and there are only $n-r+1$ of them. So:

- if $r \le n-r+1$, that is $n \ge 2r-1$, the probability is the non-zero value above;
- if $r > n-r+1$, that is $n \le 2r-2$ (too few non-hockey-players to separate them), the probability is 0.

The boundary case $n = 2r-1$ forces the pattern $HNHN\cdots NH$, and the formula gives $\frac{r!\,(r-1)!}{(2r-1)!} = 1/\binom{2r-1}{r}$, the chance that the Hs land in exactly those r alternate places.

Checks

- Take $n = 10$, $r = 3$. Then (i) gives $\frac{6}{90} = \frac{1}{15}$; (ii) gives $\frac{8}{120} = \frac{1}{15}$; (iii) gives $\binom{8}{3}/\binom{10}{3} = \frac{56}{120} = \frac{7}{15}$. All three agree with a computer enumeration of the 120 possible sets of positions, and the formulae were checked this way for every $2 \leq r \leq n \leq 14$ (and against all $n!$ orderings for $n \leq 7$).
- With $r = 2$, parts (ii) and (iii) are complementary events (the two players are either adjacent or not), and indeed $\frac{2}{n} + \frac{(n-1)(n-2)}{n(n-1)} = \frac{2}{n} + \frac{n-2}{n} = 1$.

Common slips

- Counting only $n - r - 1$ gaps (the ones strictly between non-hockey-players) and forgetting the two ends of the row.
- Mixing models: counting H-positions in the numerator but full orderings of pupils ($n!$) in the denominator. Choose one view and keep to it.
- Giving the condition for a non-zero answer as $n \geq 2r$; the case $n = 2r - 1$ (alternating, starting and ending with H) also works.

Beyond STEP

The gap argument is the "stars and bars" idea, a workhorse of enumerative combinatorics; the same count $\binom{n-r+1}{r}$ answers "how many r-subsets of $\{1, \ldots, n\}$ contain no two consecutive integers", and summing it over r gives the Fibonacci numbers. The step of replacing orderings of people by sets of positions is a first taste of exploiting symmetry to shrink a sample space, used constantly in university probability. Blitzstein and Hwang, *Introduction to Probability*, has excellent chapters on counting and the naive definition of probability.

WORKED SOLUTION

STEP I 1995, Question 13

Probability and Statistics • Poisson Thinning

STEP I 1995, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q13

The key idea

Part (i) is the *thinning* property of the Poisson distribution: condition on the true number of cancerous cells, and the marked cells are then binomial. Summing over the unknown true number collapses neatly into an exponential series. For part (ii), the event "the second mark is on slide k" is decided by how many marks there are on the first $k-1$ slides altogether (zero or one), and part (i) together with the additivity of independent Poisson variables tells us exactly how that total is distributed.

Solution

Throughout, write $q = 1 - p$ and $\lambda = \mu p$. We take the numbers of cancerous cells on different slides to be independent (the natural reading of "random"), and the scientist's detections to be independent of each other and of the cell counts.

Part (i)

On one slide let N be the number of cancerous cells and M the number she marks, so $N \sim \mathrm{Po}(\mu)$. Given $N = n$, each of the n cells is spotted independently with probability p, so $M \mid N = n \sim \mathrm{B}(n, p)$. For $m = 0, 1, 2, \ldots$, by the law of total probability (only $n \geq m$ contributes),

$$\begin{aligned} \mathrm{P}(M = m) &= \sum_{n=m}^{\infty} \mathrm{e}^{-\mu} \frac{\mu^n}{n!} \binom{n}{m} p^m q^{n-m} = \mathrm{e}^{-\mu} \frac{(\mu p)^m}{m!} \sum_{n=m}^{\infty} \frac{(\mu q)^{n-m}}{(n-m)!} \\ &= \mathrm{e}^{-\mu} \frac{(\mu p)^m}{m!} \mathrm{e}^{\mu q} = \mathrm{e}^{-\mu p} \frac{(\mu p)^m}{m!}, \end{aligned}$$

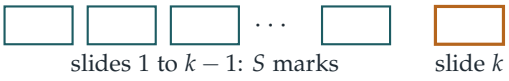
using $\binom{n}{m}/n! = 1/\big(m!\,(n-m)!\big)$, then the exponential series with $j = n - m$, and finally $-\mu + \mu q = -\mu p$. So $M \sim \mathrm{Po}(\mu p)$, as required.

Part (ii)

Let M_i be the number of marked cells on slide i. By part (i) and independence between slides, $M_1, M_2, \ldots$ are independent $\mathrm{Po}(\lambda)$ variables. Let $S = M_1 + \cdots + M_{k-1}$ be the number marked on the first $k-1$ slides; a sum of independent Poisson variables is Poisson with the means added, so $S \sim \mathrm{Po}\big((k-1)\lambda\big)$, and S is independent of M_k.

> **Ask yourself** Exactly what has to happen for the second mark to land on slide k?

Before slide k she must have made at most one mark, and by the end of slide k at least two. So there are two disjoint cases:



either $S = 0$ and $M_k \geq 2$, or $S = 1$ and $M_k \geq 1$

(For $k = 1$ there are no earlier slides, $S = 0$ automatically, and the formulas below still hold.) Using the independence of S and M_k:

$$\mathrm{P}(S = 0,\ M_k \geq 2) = \mathrm{e}^{-(k-1)\lambda}\big(1 - \mathrm{e}^{-\lambda} - \lambda\mathrm{e}^{-\lambda}\big),$$
$$\mathrm{P}(S = 1,\ M_k \geq 1) = (k-1)\lambda\,\mathrm{e}^{-(k-1)\lambda}\big(1 - \mathrm{e}^{-\lambda}\big).$$

Adding,

$$Q = \mathrm{e}^{-(k-1)\lambda}\Big\{\big(1 + (k-1)\lambda\big)\big(1 - \mathrm{e}^{-\lambda}\big) - \lambda\mathrm{e}^{-\lambda}\Big\}.$$

To match the required form, write $1 + (k-1)\lambda = (1 + k\lambda) - \lambda$:

$$\begin{aligned}\big(1 + (k-1)\lambda\big)\big(1 - \mathrm{e}^{-\lambda}\big) - \lambda\mathrm{e}^{-\lambda} &= (1 + k\lambda)\big(1 - \mathrm{e}^{-\lambda}\big) - \lambda + \lambda\mathrm{e}^{-\lambda} - \lambda\mathrm{e}^{-\lambda} \\ &= (1 + k\lambda)\big(1 - \mathrm{e}^{-\lambda}\big) - \lambda.\end{aligned}$$

With $\lambda = \mu p$ this gives

$$Q = \mathrm{e}^{-\mu p(k-1)}\Big\{(1 + k\mu p)\big(1 - \mathrm{e}^{-\mu p}\big) - \mu p\Big\},$$

which is the stated result.

Checks

- For $k = 1$ the formula gives $1 - \mathrm{e}^{-\lambda} - \lambda\mathrm{e}^{-\lambda} = \mathrm{P}(M_1 \geq 2)$, as it must.
- The probabilities should sum to 1 over $k = 1, 2, \ldots$ (she marks a second cell eventually with probability 1); summing numerically for several values of λ confirms this.
- A simulation with $\mu = 1.6$, $p = 0.45$ (so $\lambda = 0.72$) gave the second mark on slide 2 in about 26.0% of 200 000 runs; the formula gives $Q \approx 0.2591$.

Common slips

- Forgetting the case "one mark before slide k, at least one on slide k", or including $S = 1$, $M_k = 0$ by mistake.
- Working slide by slide with "first slide with a mark" arguments, which gets messy; the total S over the first $k - 1$ slides is the efficient object.
- Not saying where independence (between slides, and between S and M_k) is used.

Beyond STEP

Thinning is one face of the Poisson process: if events occur as a Poisson process and each is kept independently with probability p, the kept events again form a Poisson process, with rate scaled by p. Part (ii) is then a discrete version of the waiting time for the second event, which in continuous time has a gamma distribution. Grimmett and Welsh, *Probability: An Introduction*, develops Poisson processes and these waiting-time distributions.

WORKED SOLUTION

STEP I 1995, Question 14

Probability and Statistics • Estimating a Proportion

STEP I 1995, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q14

The key idea

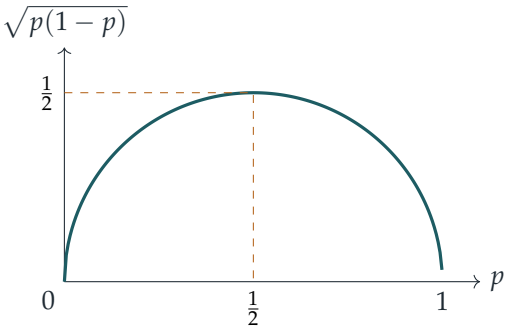
The sample proportion is a binomial count divided by n. For large n it is approximately normal with mean p and variance $p(1-p)/n$. The trouble is that the variance depends on the very p we don't know. Part (i) solves that: it gives the worst possible value of the standard deviation, so a sample size chosen for the worst case works whatever p is. Part (iii) then asks what changes when we already have a rough idea of p.

Solution

Part (i)

Let $f(p) = p(1-p) = \frac{1}{4} - \left(p - \frac{1}{2}\right)^2$, by completing the square. So $f(p) \leq \frac{1}{4}$ on $[0,1]$, with equality only at $p = \frac{1}{2}$. The square root is increasing, so

$$\max_{0 \leq p \leq 1} \sqrt{p(1-p)} = \sqrt{\tfrac{1}{4}} = \tfrac{1}{2}, \qquad \text{attained at } p = \tfrac{1}{2}.$$



Part (ii)

Let X be the number of females in the sample. Each person picked is female with probability p. We assume the population is large, so that the n choices are effectively independent (sampling without replacement from a huge population is almost the same as sampling with replacement). Then

$$X \sim \mathrm{B}(n,p), \qquad \hat{p} = \frac{X}{n}, \qquad \mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}.$$

For large n the central limit theorem (the normal approximation to the binomial) gives

$$\hat{p} \approx \mathrm{N}\left(p,\ \frac{p(1-p)}{n}\right), \qquad \text{so} \qquad Z = \frac{\hat{p}-p}{\sqrt{p(1-p)/n}} \approx \mathrm{N}(0,1).$$

We want $\mathrm{P}\big(|\hat{p}-p| > 0.01\big) < 0.01$, that is,

$$\mathrm{P}\left(|Z| > \frac{0.01\sqrt{n}}{\sqrt{p(1-p)}}\right) < 0.01.$$

By symmetry this is 1% split as 0.5% in each tail. Since $\Phi(2.576) = 0.995$, we need

$$\frac{0.01\sqrt{n}}{\sqrt{p(1-p)}} > 2.576 \quad \Longleftrightarrow \quad \sqrt{n} > 257.6\sqrt{p(1-p)}.$$

Ask yourself We don't know p, so how can we choose n from an inequality that involves it?

This is where part (i) comes in. Whatever p is, $\sqrt{p(1-p)} \le \frac{1}{2}$. So if $\sqrt{n} > 257.6 \times \frac{1}{2} = 128.8$, the inequality holds for every p. This condition is also necessary, because when $p = \frac{1}{2}$ (and the proportion of females is close to that) the bound is attained. Therefore we need

$$n > 128.8^2 = 16\,589.44,$$

and the required sample size is

$$\boxed{n = 16\,590} \qquad \text{(about } 16\,600\text{)}.$$

With a more precise percentage point, $z = 2.5758$, we get $n > 16\,587.2$. The answer is "about 16 600" either way, and the third significant figure depends on the tables used.

Part (iii)

In general the requirement is

$$n > \left(\frac{2.576}{0.01}\right)^2 p(1-p) \approx 66\,360\,p(1-p).$$

The value 16 590 came from the worst case $p = \frac{1}{2}$. If we know that p lies well away from $\frac{1}{2}$, we can use a smaller bound for $p(1-p)$.

(a) Left-handedness. Roughly 10% of people are left-handed. If we may safely assume $p \leq 0.15$, then $p(1-p) \leq 0.15 \times 0.85 = 0.1275$, since $p(1-p)$ is increasing on $[0, \frac{1}{2}]$. This gives $n > 8461$. If $p = 0.1$ exactly, $n > 66\,360 \times 0.09 \approx 5970$, so about 6000. Either way the sample is much smaller than the 16 590 needed when we know nothing. The normal approximation is still good, since np is in the hundreds.

(b) Millionaires. Now p is very small, perhaps 0.001 to 0.01. Taken literally, $p \leq 0.01$ gives $p(1-p) \leq 0.0099$ and $n > 657$. But this misses the point:

- An error of 0.01 is useless when p itself is about 0.001: the estimate $\hat{p} = 0$ already lies within 0.01 of the truth. What we really want is a *relative* accuracy, say an error of at most 1% of p. Then we need $n > 2.576^2(1-p)/(p \times 0.01^2)$, which for $p = 0.001$ is about 6.6×10^7. That is enormously larger.
- With p tiny and n only a few hundred, np is less than 1. The binomial is then very skewed, and the normal approximation is no longer valid; a Poisson approximation would be the better model.
- In practice millionaires are hard to reach, and people may not tell the truth about their wealth, so random sampling and honest responses are both doubtful.

So for left-handedness, knowing p roughly lets us use a smaller sample. For millionaires, any sensible accuracy requirement needs a far larger sample, or a different sampling method altogether.

Checks

- At $p = \frac{1}{2}$ and $n = 16\,590$ the normal approximation gives $\mathrm{P}(|\hat{p} - p| > 0.01) = 2\big(1 - \Phi(2.5762)\big) \approx 0.00999$, just under 1%.
- Summing the exact binomial probabilities for $n = 16\,590$, $p = \frac{1}{2}$ gives about 0.0102. The small excess comes from the discreteness of the binomial (the normal approximation without continuity correction). With $p = 0.3$ the exact value is about 0.005, comfortably inside, as part (i) predicts.
- For $p = 0.1$ and $n = 6000$ the exact binomial probability is about 0.0098, which confirms the figure in (iii)(a).

Common slips

- Using 1.96 or 2.326 instead of 2.576. The 1% is split between the two tails, so we need $\Phi(z) = 0.995$.
- Leaving n in terms of p. The question wants a single number that works for any p, which is why part (i) is there.
- Reading "$\sqrt{p(1-p)} \leq \frac{1}{2}$" as giving $n > 128.8$. It is $\sqrt{n}$, not n, that must exceed 128.8.

Beyond STEP

This is the standard calculation behind the "margin of error" quoted in opinion polls. A poll of about 1000 people gives roughly ± 3 percentage points at 95% confidence, by exactly the worst-case argument of part (i). At university it leads to confidence intervals, and to the question of how to estimate rare proportions, where Poisson and exact binomial methods take over from the normal approximation. Blitzstein and Hwang, *Introduction to Probability*, covers the central limit theorem and these approximations very readably.

WORKED SOLUTION

STEP II 1995, Question 1

Pure • Differentiating Geometric Series

STEP II 1995, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q1

The key idea

A finite geometric series has a closed form, and an identity between two functions of x can be differentiated to give a new identity. Differentiating $\sum x^k$ brings down a factor k, which is exactly what turns a geometric series into $\sum kx^{k-1}$; putting $x = -1$ then supplies the alternating signs. For the squares, the cleanest route is to let the answer's shape tell us what A and B must be, then prove it by induction.

Solution

Part (i)

Write $S = 1 + x + x^2 + \cdots + x^n$. Then

$$S(1-x) = (1 + x + \cdots + x^n) - (x + x^2 + \cdots + x^{n+1}) = 1 - x^{n+1},$$

since every power $x, x^2, \ldots, x^n$ appears once with each sign and cancels. If $x \neq 1$ then $1 - x \neq 0$, so we may divide by it:

$$1 + x + x^2 + \cdots + x^n = \frac{1 - x^{n+1}}{1 - x},$$

as required.

Part (ii)

Both sides of (i) are differentiable functions of x on $x \neq 1$ and agree there, so their derivatives agree. On the left,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(1 + x + \cdots + x^n\right) = 1 + 2x + 3x^2 + \cdots + nx^{n-1}.$$

On the right, by the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1 - x^{n+1}}{1 - x}\right) = \frac{-(n+1)x^n(1-x) + (1 - x^{n+1})}{(1-x)^2} = \frac{1 - (n+1)x^n + nx^{n+1}}{(1-x)^2}.$$

Now set $x = -1$, which is allowed because $-1 \neq 1$. The left-hand side becomes $1 - 2 + 3 - 4 + \cdots + (-1)^{n-1}n$, because the term kx^{k-1} becomes $(-1)^{k-1}k$. The right-hand side becomes

$$\frac{1 - (n+1)(-1)^n + n(-1)^{n+1}}{4} = \frac{1 - (-1)^n(2n+1)}{4},$$

using $(-1)^{n+1} = -(-1)^n$.

- If n is even, $(-1)^n = 1$ and the value is $\dfrac{1 - 2n - 1}{4} = -\dfrac{n}{2}$.
- If n is odd, $(-1)^n = -1$ and the value is $\dfrac{1 + 2n + 1}{4} = \dfrac{n+1}{2}$.

This is what we had to show.

Part (iii)

Let $T_n = 1^2 - 2^2 + 3^2 - \cdots + (-1)^{n-1}n^2$, and suppose $T_n = (-1)^{n-1}f(n)$ with $f(n) = An^2 + Bn$.

Ask yourself How can we find A and B before we know the formula is true?

Use the one fact we know for certain about T_n: consecutive sums differ by the last term, $T_n - T_{n-1} = (-1)^{n-1}n^2$. Since $T_{n-1} = (-1)^{n-2}f(n-1) = -(-1)^{n-1}f(n-1)$, this reads

$$(-1)^{n-1}\big(f(n) + f(n-1)\big) = (-1)^{n-1}n^2, \qquad \text{so} \qquad f(n) + f(n-1) = n^2.$$

With $f(n) = An^2 + Bn$,

$$\begin{aligned} f(n) + f(n-1) &= An^2 + Bn + A(n-1)^2 + B(n-1) \\ &= 2An^2 + (2B - 2A)n + (A - B). \end{aligned}$$

For this to equal n^2 for every n we need $2A = 1$ and $B = A$, that is

$$A = B = \tfrac{1}{2}, \qquad T_n = (-1)^{n-1}\,\frac{n(n+1)}{2}.$$

(These values are forced: $n = 1$ gives $A + B = 1$ and $n = 2$ gives $-(4A + 2B) = -3$, whose only solution is $A = B = \frac{1}{2}$.)

Proof by induction. For $n = 1$, $T_1 = 1 = (-1)^0 \cdot \frac{1\cdot 2}{2}$. Suppose $T_{n-1} = (-1)^{n-2}\frac{(n-1)n}{2}$ for some $n \geq 2$. Then

$$\begin{aligned} T_n = T_{n-1} + (-1)^{n-1}n^2 &= (-1)^{n-1}\left(n^2 - \frac{n(n-1)}{2}\right) \\ &= (-1)^{n-1}\,\frac{n(n+1)}{2}, \end{aligned}$$

which is the formula for n. So it holds for all $n \geq 1$, with $A = B = \frac{1}{2}$.

Alternatively, pair the terms: $(2m-1)^2-(2m)^2=-(4m-1)$. For $n=2m$ this gives $T_n=-\sum_{j=1}^{m}(4j-1)=-m(2m+1)=-\frac{n(n+1)}{2}$; for $n=2m+1$ add $(2m+1)^2$ to get $(2m+1)(m+1)=\frac{n(n+1)}{2}$. (One could also multiply the identity of (ii) by x and differentiate again, but the algebra is much heavier.)

Checks

For $n=4$: $1-2+3-4=-2=-\frac{4}{2}$, and $1-4+9-16=-10=-\frac{4\cdot5}{2}$. For $n=5$: $1-2+3-4+5=3=\frac{6}{2}$, and $-10+25=15=\frac{5\cdot6}{2}$. A computer check of all three identities for n up to 500 (for example $T_{10}=-55$ and $T_{11}=66$) agrees exactly.

Notice also that $|T_n|=1+2+\cdots+n$: the alternating sum of squares has the same size as the plain sum of the first n integers, a surprising coincidence that the pairing argument explains.

Common slips

- Forgetting that $x=-1$ is a legitimate value only because the identity in (i) holds for all $x\neq1$; at $x=1$ the right-hand side is undefined.
- Sign errors with $(-1)^{n+1}$: it equals $-(-1)^n$, not $(-1)^n$.
- Finding A and B from $n=1$ and $n=2$ and stopping there. That only shows these are the *only possible* values; the induction (or pairing) shows the formula actually holds for every n.

Beyond STEP

Differentiating $\sum x^k$ term by term is the finite version of a powerful idea: power series can be differentiated term by term inside their radius of convergence, giving $\sum kx^{k-1}=(1-x)^{-2}$ for $|x|<1$. At $x=-1$ the infinite series $1-2+3-4+\cdots$ diverges, yet the formula suggests the value $\frac{1}{4}$, and indeed the formula in (ii) can be written $\frac{1}{4}-(-1)^n\frac{2n+1}{4}$, so the partial sums swing symmetrically about $\frac{1}{4}$. Giving such divergent series a value is the idea behind Abel summation. Abbott's *Understanding Analysis* treats term-by-term differentiation of power series carefully.

WORKED SOLUTION

STEP II 1995, Question 2

Pure • Counting Colourings Recursively

STEP II 1995, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q2

This question has a full guided treatment in *Beyond STEP*, Section 9.3, *Colouring Fence Posts*: https://beyondstep.keithmcnulty.org/ch09/fence-posts.html

The key idea

Split the colourings that start with red according to the colour of the last post. Every colouring of a line is built by adding one post at a time, and each new post has exactly two allowed colours, so the total count is easy; the split between "ends red" and "ends not red" is what needs a recurrence. The circle is then just a line whose two ends must also differ.

Solution

Why $r_{n+1} = s_n$

Take a valid colouring of $n+1$ posts with the first and last posts red. Post n is adjacent to post $n+1$, which is red, so post n is not red. Deleting post $n+1$ therefore leaves a valid colouring of n posts with the first post red and the last post not red: one of the s_n colourings.

Conversely, given one of the s_n colourings of n posts, we may add a red post at the end, since post n is not red, and we get a colouring counted by r_{n+1}. These two operations undo each other, so they give a one-to-one correspondence and

$$r_{n+1} = s_n.$$

Finding $r_n + s_n$

Every valid colouring with the first post red ends either in red or in a different colour, so $r_n + s_n$ counts all valid colourings with the first post red. Paint the posts in order. The first is red (one choice), and each later post may be any colour except that of the

post just before it, which gives 2 choices every time, whatever happened earlier. Hence

$$r_n + s_n = 2^{n-1}.$$

Finding $r_{n+1} + r_n$

Substituting $s_n = r_{n+1}$,

$$r_{n+1} + r_n = 2^{n-1} \qquad (n \geq 1).$$

The induction

Let $P(n)$ be the statement $r_n = \frac{1}{3}\left(2^{n-1} + 2(-1)^{n-1}\right)$.

Base case. With one post painted red, that post is both first and last, so $r_1 = 1$. The formula gives $\frac{1}{3}(1+2) = 1$, so $P(1)$ holds.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$. By the previous result,

$$\begin{aligned} r_{n+1} &= 2^{n-1} - r_n \\ &= \frac{3 \cdot 2^{n-1} - 2^{n-1} - 2(-1)^{n-1}}{3} \\ &= \frac{2^n + 2(-1)^n}{3}, \end{aligned}$$

using $2 \cdot 2^{n-1} = 2^n$ and $-(-1)^{n-1} = (-1)^n$. This is exactly $P(n+1)$. By induction, $P(n)$ holds for all $n \geq 1$.

> **Ask yourself** Why not start the induction at $n = 2$ or use two base cases?

The recurrence $r_{n+1} = 2^{n-1} - r_n$ expresses each term using only the one before it, so a single base case is enough. A two-term recurrence would need two.

Posts in a circle

Number the posts $1, 2, \ldots, n$ round the circle, so that post n is now also adjacent to post 1. A valid circular colouring is precisely a valid colouring of the line $1, 2, \ldots, n$ in which, in addition, post n differs from post 1.

> **Ask yourself** How do we count colourings where the first post isn't fixed as red?

Use symmetry. Swapping the names of two colours maps valid colourings to valid colourings, so for each choice of colour for post 1 there are the same number of colourings. With post 1 red, the line colourings whose last post differs from the first are

exactly those counted by s_n. So the number of circular colourings is

$$\begin{aligned} 3s_n &= 3\left(2^{n-1} - r_n\right) \\ &= 3 \cdot 2^{n-1} - 2^{n-1} - 2(-1)^{n-1} \\ &= 2^n + 2(-1)^n. \end{aligned}$$

Here the posts are distinguishable (they stand in fixed places), so two colourings that differ by a rotation are counted separately.

Checks

- Small cases by hand: $r_2 = 0$ (two adjacent posts can't both be red) and $r_3 = 2$ (RWR, RBR), matching $\frac{1}{3}(2-2) = 0$ and $\frac{1}{3}(4+2) = 2$.
- A triangle of three posts must use all three colours, in $3! = 6$ ways, and $2^3 + 2(-1)^3 = 6$. Four posts in a square give $16 + 2 = 18$.
- A brute-force count of all colourings for $n \leq 12$ agrees with every formula above; for example $r_{10} = 170$, $s_{10} = 342$ and 1026 circular colourings of 10 posts.

Common slips

- Answering r_n or s_n for the circle, forgetting that post 1 need not be red; the factor 3 restores the other two colours.
- Dividing by n to remove rotations. The posts are in fixed positions, so rotated colourings are different paintings.

Beyond STEP

The circle answer $2^n + 2(-1)^n$ is the value at $k = 3$ of the *chromatic polynomial* of the cycle graph C_n, which in general is $(k-1)^n + (-1)^n(k-1)$ for k colours; the same "line, then close up the ends" argument proves it. Chromatic polynomials and recurrences like $r_{n+1} + r_n = 2^{n-1}$ are standard material in a first course on combinatorics or graph theory. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 1995, Question 3

Pure • Logical Deduction and Counting

STEP II 1995, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q3

The key idea

This is a logic puzzle in disguise, and the way in is *counting*. Every stage hands out $5+4+3+2+1=15$ points, so the five scores add up to a known total. Arouet's score is so close to the maximum that it pins down nearly all of his results. After that, Eluard's results squeeze Eluard's score from both sides, and Camus's "same position four times" leaves only a few cases to check.

Solution

We write A, B, C, D, E for the riders' total scores. We're told $A > B > C > D > E$ and $A = 24$.

The total, and Arouet's stages

Each stage gives out 15 points, so over five stages

$$A+B+C+D+E=75, \qquad \text{hence} \qquad B+C+D+E=51.$$

Arouet can get at most $5\times 5=25$ points. Getting 24 means losing exactly one point, so Arouet came *first in four stages and second in one*. Eluard won stage 5, so Arouet's second place must have come in stage 5, and Arouet won stages 1–4.

Why Eluard scored 11

Lower bound. Eluard got 3 points in stage 4 and 5 in stage 5, and at least 1 point in each of the other three stages. So $E \geq 3+5+3=11$.

Upper bound. The scores are distinct integers with $B > C > D > E$, so $D \geq E+1$, $C \geq E+2$ and $B \geq E+3$. Therefore

$$51 = B+C+D+E \geq 4E+6, \qquad \text{so} \qquad E \leq \tfrac{45}{4} < 12.$$

Combining the two bounds gives $E=11$. Equality in the lower bound also tells us that

Eluard came *last* in each of stages 1, 2 and 3.

What is left for Barthes, Camus and Diderot

We now know where Arouet and Eluard finished in every stage, so Barthes, Camus and Diderot shared out the remaining places:

Stage	Arouet, Eluard	Places left for B, C, D	
1, 2, 3	1st, 5th	2nd, 3rd, 4th	(4, 3, 2 points)
4	1st, 3rd	2nd, 4th, 5th	(4, 2, 1 points)
5	2nd, 1st	3rd, 4th, 5th	(3, 2, 1 points)

Also $B + C + D = 51 - 11 = 40$, and $B > C > D > 11$, so $D \geq 12$ and $C \geq 13$.

Ask yourself Camus finished in the same position four times. Which positions could that be, given the table?

The answer is 2nd, 3rd or 4th. Fifth place was available to Camus only in stages 4 and 5, which is two stages, not four. We take the three possibilities in turn.

Camus 2nd four times. Second place was available only in stages 1–4, so Camus was second in all of them. Then in stages 1–3 Barthes got at most 3 points each, in stage 4 at most 2, and in stage 5 at most 3. So $B \leq 9 + 2 + 3 = 14$. But $C \geq 4 \times 4 = 16 > B$, which is a contradiction.

Camus 4th four times. Fourth place is worth 2 points in every stage, so Camus's best total is $4 \times 2 + 4 = 12$ (a second place in the remaining stage). That contradicts $C \geq 13$.

Camus 3rd four times. Third place was available in stages 1, 2, 3 and 5, worth 3 points each time, so these are Camus's four third places. In stage 4 Camus scored $x \in \{4, 2, 1\}$, and $C = 12 + x$.

- $x = 4$: $C = 16$, so $B \geq 17$ and $D = 40 - B - C \leq 7$. This contradicts $D \geq 12$.
- $x = 2$: $C = 14$, so $B \geq 15$ and $D \leq 11$. Again a contradiction.
- $x = 1$: $C = 13$, so $B + D = 27$ with $B \geq 14$ and $12 \leq D \leq 12$. Hence $D = 12$ and $B = 15$.

So the scores are

$$\boxed{B = 15, \quad C = 13, \quad D = 12} \qquad (\text{with } A = 24,\ E = 11).$$

Barthes in the final stage

Ask yourself We know the totals, but not the stage-by-stage results. Is there enough information to fix stage 5?

There is, by a parity argument. With Camus third in stages 1, 2, 3 and 5 and fifth in stage 4, Barthes and Diderot shared the places worth 4 and 2 in each of stages 1–4. In

stage 5 they shared the places worth 2 and 1 (4th and 5th). Suppose Diderot took the 4-point place in k of stages 1–4 and scored $s \in \{1, 2\}$ in stage 5. Then

$$D = 4 \times 2 + 2k + s = 8 + 2k + s = 12.$$

So $s = 4 - 2k$ is even, which forces $s = 2$ (and $k = 1$). Diderot was fourth in the final stage, so **Barthes came last (fifth) in the final stage.**

Such a tour does exist. For example, take stage 1: A, D, C, B, E; stages 2 and 3: A, B, C, D, E; stage 4: A, B, E, D, C; stage 5: E, A, C, D, B. This gives $24, 15, 13, 12, 11$, as required.

Checks

- The totals add up: $24 + 15 + 13 + 12 + 11 = 75 = 5 \times 15$.
- In the example tour, Camus is third in four stages, and Eluard is third in stage 4 and first in stage 5.
- A computer search through all 120^5 possible sets of stage results, keeping those that meet every condition, finds only the scores $(24, 15, 13, 12, 11)$. Barthes is fifth in stage 5 in every case. (There are exactly four valid tours, depending on which of stages 1–4 Diderot finished second in.)

Common slips

- Showing only $E \geq 11$. The question asks why Eluard *must* have 11, so the upper bound from $B + C + D + E = 51$ and distinctness is essential.
- Assuming Camus was in the same place in four *consecutive* stages, or forgetting the case of fourth place.
- Finding the scores but not proving that the final-stage placing is forced. One example tour isn't enough; the parity argument shows that every possible tour has Barthes last in stage 5.

Beyond STEP

The argument has three moves: a conservation law (the fixed total), extremal reasoning (24 is nearly the maximum), and a parity argument. All three appear again and again in combinatorics. Bounding a quantity from both sides until it is forced is also the basic technique of analysis and of olympiad problem solving. For more puzzles in this spirit, try Siklos, *Advanced Problems in Mathematics*. Engel's *Problem-Solving Strategies* has whole chapters on the invariance and extremal principles.

WORKED SOLUTION

STEP II 1995, Question 4

Pure • Wallis Reduction Formula

STEP II 1995, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 4.4, *Wallis, Beta and the Factorial of a Half*: https://beyondstep.keithmcnulty.org/ch04/wallis-beta.html

The key idea

Integrating by parts turns $\cos^2 t$ into $1 - \sin^2 t$, which gives a reduction formula linking u_n to u_{n-2}. Multiplying that formula by u_{n-1} shows that nu_nu_{n-1} never changes as n increases, so it equals its value at $n = 1$. After that the question is a squeeze: since the u_n decrease, the product u_nu_{n-1} lies between u_n^2 and u_{n-1}^2.

Solution

The reduction formula

Let $n \geq 2$. We write $u_n = \int_0^{\pi/2} \sin t \, \sin^{n-1} t \, \mathrm{d}t$ and integrate by parts, differentiating $\sin^{n-1} t$ and integrating $\sin t$ to $-\cos t$:

$$\begin{aligned} u_n &= \Big[-\cos t \, \sin^{n-1} t\Big]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \cos^2 t \, \sin^{n-2} t \, \mathrm{d}t \\ &= 0 + (n-1)\int_0^{\pi/2} (1 - \sin^2 t)\sin^{n-2} t \, \mathrm{d}t \\ &= (n-1)(u_{n-2} - u_n). \end{aligned}$$

The boundary term vanishes because $\cos\frac{\pi}{2} = 0$ at the top and $\sin 0 = 0$ at the bottom (here we use $n - 1 \geq 1$). Collecting the u_n terms gives

$$nu_n = (n-1)u_{n-2} \qquad (n \geq 2).$$

Multiplying both sides by u_{n-1} gives

$$nu_nu_{n-1} = (n-1)u_{n-1}u_{n-2} \qquad (n \geq 2),$$

as required.

The constant product

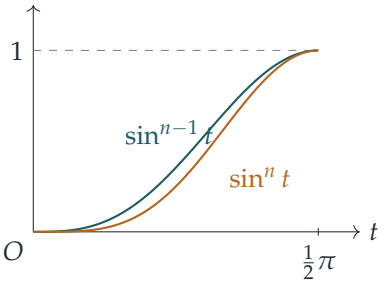
Write $p_n = nu_nu_{n-1}$ for $n \geq 1$. The last result says $p_n = p_{n-1}$ for every $n \geq 2$, so $p_n = p_1$ for every $n \geq 1$. Now

$$u_0 = \int_0^{\pi/2} 1 \,\mathrm{d}t = \tfrac{1}{2}\pi, \qquad u_1 = \int_0^{\pi/2} \sin t \,\mathrm{d}t = \Big[-\cos t\Big]_0^{\pi/2} = 1,$$

so $p_1 = 1 \cdot u_1u_0 = \frac{1}{2}\pi$. Hence

$$nu_nu_{n-1} = \tfrac{1}{2}\pi \qquad \text{for all } n \geq 1.$$

The sketch and $0 < u_n < u_{n-1}$



Both curves start at 0 when $t = 0$, increase, and reach 1 at $t = \frac{1}{2}\pi$ (the sketch shows $n = 4$; for $n = 1$ the upper curve is the line $y = 1$). For $0 < t < \frac{1}{2}\pi$ we have $0 < \sin t < 1$, so

$$0 < \sin^n t = \sin t \cdot \sin^{n-1} t < \sin^{n-1} t.$$

So the graph of $\sin^n t$ lies strictly below that of $\sin^{n-1} t$ except at the two ends. Both functions are continuous, so the areas under them satisfy

$$0 < u_n < u_{n-1} \qquad (n \geq 1).$$

Ask yourself Why are the inequalities strict, when the curves touch at the ends?

The difference $\sin^{n-1} t - \sin^n t$ is continuous, never negative, and positive at (say) $t = \frac{1}{4}\pi$. A continuous function that is positive at a point is positive on a small interval around it, so it encloses a positive area. Touching at two isolated points doesn't change that. The same argument shows $u_n > 0$.

The two-sided bound

Let $n \geq 1$. Since $u_n > 0$, multiplying $u_n < u_{n-1}$ by nu_n gives $nu_n^2 < nu_nu_{n-1}$. Since $u_{n-1} > 0$, multiplying by nu_{n-1} gives $nu_nu_{n-1} < nu_{n-1}^2$. The middle quantity is $\frac{1}{2}\pi$, so

$$nu_n^2 < \tfrac{1}{2}\pi < nu_{n-1}^2 \qquad (n \geq 1).$$

The squeeze and the limit

The right-hand inequality holds for every $n \geq 1$, so we may apply it with $n+1$ in place of n:

$$\tfrac{1}{2}\pi < (n+1)u_n^2, \qquad \text{so} \qquad nu_n^2 > \frac{n}{n+1} \cdot \tfrac{1}{2}\pi.$$

Combining this with the left-hand inequality gives

$$\left(\frac{n}{n+1}\right) \tfrac{1}{2}\pi < nu_n^2 < \tfrac{1}{2}\pi.$$

As $n \to \infty$, $\frac{n}{n+1} = 1 - \frac{1}{n+1} \to 1$, so the lower bound tends to $\frac{1}{2}\pi$. The upper bound is $\frac{1}{2}\pi$ itself. By the squeeze (sandwich) theorem,

$$nu_n^2 \to \tfrac{1}{2}\pi \quad \text{as } n \to \infty.$$

Checks

- The reduction formula gives $u_2 = \frac{1}{2}u_0 = \frac{1}{4}\pi$, which matches $\int_0^{\pi/2} \sin^2 t \, \mathrm{d}t = \frac{1}{4}\pi$ (the average of $\sin^2$ is $\frac{1}{2}$). Then $2u_2u_1 = \frac{1}{2}\pi$, as predicted.
- Numerically (Simpson's rule), $u_{10} \approx 0.38656$, which is $\frac{63}{512}\pi$. So $10u_{10}^2 \approx 1.4943$, which lies between $\frac{10}{11} \cdot \frac{1}{2}\pi \approx 1.4280$ and $\frac{1}{2}\pi \approx 1.5708$.
- Running the recursion to $n = 10^5$ gives $nu_n^2 \approx 1.570788$, already very close to $\frac{1}{2}\pi$.

Common slips

- Forgetting that the reduction formula needs $n \geq 2$, so the constant must be fixed using u_0 and u_1, not u_1 and u_2 alone.
- Claiming $u_n < u_{n-1}$ only because $\sin^n t \leq \sin^{n-1} t$. That gives $\leq$; strictness needs the inequality to be strict on an interval.
- Trying to get the lower bound from $\frac{1}{2}\pi < nu_{n-1}^2$ directly. That bounds u_{n-1}, not u_n, so shift the index first.

Beyond STEP

These are Wallis's integrals. Writing out u_{2m} and u_{2m+1} as products and using $nu_n^2 \to \frac{1}{2}\pi$ leads to Wallis's product $\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdots$. The same estimate fixes the constant $\sqrt{2\pi}$ in Stirling's formula $n! \sim \sqrt{2\pi n}\,(n/\mathrm{e})^n$. The squeeze argument used here is a standard tool of first-year analysis. Abbott's *Understanding Analysis* develops it carefully.

WORKED SOLUTION

STEP II 1995, Question 5

Pure • Optimisation and Concavity

STEP II 1995, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

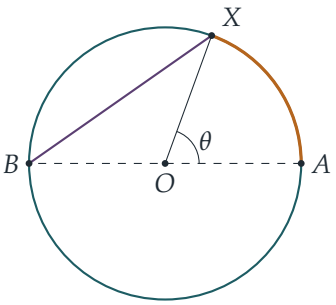
step.maths.org/questions/95-s2-q5

The key idea

Choices (i) and (iii) are the two ends of the family in (ii): $\theta \to 0$ means swimming straight across and $\theta \to \pi$ means running all the way. So we write the total time as a function of θ on $[0, \pi]$ and ask where its minimum is. The surprise is that the function is concave, so its only stationary point is a maximum, and the best route is always at one end.

Solution

Take the radius of the pool as 1 unit of length and her swimming speed as 1, so that she runs at speed k. Times are then in units of (radius)/(swimming speed), which doesn't affect any comparison.



The three times

(i) She swims the diameter AB, of length 2, taking time 2.

(iii) She runs a semicircle, of length π, taking time π/k.

(ii) She runs the arc AX, of length θ (radius 1), in time θ/k. Since $\angle AOB = \pi$, we have

$\angle XOB = \pi - \theta$. Triangle XOB is isosceles with $OX = OB = 1$, so the chord is

$$XB = 2\sin\tfrac{\pi-\theta}{2} = 2\cos\tfrac{\theta}{2}.$$

(Equivalently, $\angle AXB = \frac{\pi}{2}$ because AB is a diameter, and $\angle XAB = \frac{\pi-\theta}{2}$ from the isosceles triangle AOX, so $XB = 2\sin\angle XAB$.) Her total time is

$$T(\theta) = \frac{\theta}{k} + 2\cos\frac{\theta}{2}, \qquad 0 < \theta < \pi.$$

This formula also makes sense at the ends: $T(0) = 2$ is the time for (i) and $T(\pi) = \pi/k$ is the time for (iii). So all three choices are described by one continuous function on $[0, \pi]$.

Where is T smallest?

Differentiating,

$$T'(\theta) = \frac{1}{k} - \sin\frac{\theta}{2}, \qquad T''(\theta) = -\tfrac{1}{2}\cos\frac{\theta}{2}.$$

For $0 < \theta < \pi$ we have $0 < \frac{\theta}{2} < \frac{\pi}{2}$, so $\cos\frac{\theta}{2} > 0$ and $T''(\theta) < 0$: the function T is strictly concave on $[0, \pi]$.

Ask yourself If $k > 1$ there is a stationary point at $\sin\frac{\theta}{2} = \frac{1}{k}$. Isn't that the best place to dive in?

No: since $T'' < 0$ there, it is a *maximum*, the worst place to dive in. More generally, a strictly concave function on a closed interval lies strictly above the chord joining its end values, so for every θ with $0 < \theta < \pi$,

$$T(\theta) > \Big(1 - \frac{\theta}{\pi}\Big)T(0) + \frac{\theta}{\pi}\,T(\pi) \geq \min\big(T(0),\, T(\pi)\big).$$

So whatever the value of k, some interior choice of X is never even as quick as the better of (i) and (iii). (If $k \leq 1$ we can see this even more directly: then $T'(\theta) > 0$ on $(0, \pi)$, as $\sin\frac{\theta}{2} < 1 \leq \frac{1}{k}$, so T is increasing and swimming straight away is best.)

Comparing (i) and (iii)

It remains to compare $T(0) = 2$ with $T(\pi) = \pi/k$:

- if $k < \frac{\pi}{2}$, then $\pi/k > 2$, so she should **swim directly**, choice (i);
- if $k > \frac{\pi}{2}$, then $\pi/k < 2$, so she should **run all the way round**, choice (iii);
- if $k = \frac{\pi}{2}$, (i) and (iii) take the same time, and either is best.

The physical reason is simply that the semicircle is $\frac{\pi}{2}$ times as long as the diameter.

Finally, choice (ii) is the one she will never take: for every $k > 0$ and every $\theta \in (0, \pi)$ it is strictly slower than the better of (i) and (iii).

Checks

With $k = 3$, running takes $\pi/3 \approx 1.047$, beating swimming's 2; the stationary point is at $\theta = 2\arcsin\frac{1}{3} \approx 0.680$, where $T \approx 2.112$, which is slower than either end, as a maximum should be. A computer evaluating T on a fine grid of θ for thousands of random k between e^{-4} and e^{4} never found an interior value below $\min(2, \pi/k)$, and confirmed that the changeover is at $k = \pi/2 \approx 1.571$.

Common slips

- Setting $T'(\theta) = 0$ and declaring the result the optimum without checking its nature: here it is a maximum.
- Using the arc XB instead of the chord, or getting the chord as $2\sin\frac{\theta}{2}$ (that is AX, not XB).
- Forgetting the case $k \leq 1$, where there is no stationary point at all.

Beyond STEP

The principle at work is that a concave function on an interval attains its minimum at an endpoint; in higher dimensions this becomes the fact that a linear (or concave) objective over a polygon is minimised at a vertex, which is the foundation of linear programming. The question is also a cousin of Snell's law and Fermat's principle of least time, where the answer usually is an interior point because the time function is convex instead. For more problems of this flavour, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 1995, Question 6

Pure • Roots of Unity

STEP II 1995, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q6

The key idea

The seventh roots of unity are $1, \alpha, \alpha^2, \ldots, \alpha^6$, and they add up to zero. Split the six non-real ones into two groups of three, $\{\alpha, \alpha^2, \alpha^4\}$ and $\{\alpha^3, \alpha^5, \alpha^6\}$. The second group is just the complex conjugates of the first, so the two group sums are conjugates. Their sum and product are therefore real, and the first part turns them into the coefficients of the quadratic. Taking real and imaginary parts of the root then gives the trigonometric sums.

Solution

Roots and coefficients

If u and v are the roots of $z^2 + az + b = 0$, then the monic quadratic $z^2 + az + b$ has the factors $z - u$ and $z - v$, so

$$z^2 + az + b = (z - u)(z - v) = z^2 - (u + v)z + uv$$

for all z. Comparing coefficients of z and the constant terms gives $a = -u - v$ and $b = uv$, as required.

The roots of unity

A word of warning first. The paper says $z^6 - 1 = 0$, but this is a misprint for $z^7 - 1 = 0$. Indeed, $\alpha^6 = \cos(12\pi/7) + \mathrm{i}\sin(12\pi/7) \neq 1$, while the rest of the question is plainly about sevenths. We work with $z^7 - 1 = 0$.

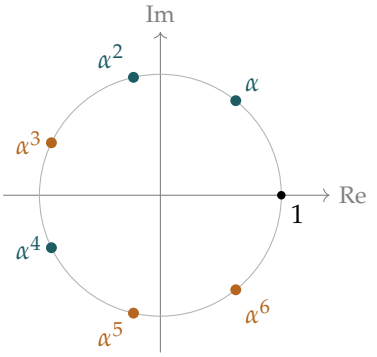
By de Moivre's theorem,

$$\alpha^7 = \cos 2\pi + \mathrm{i}\sin 2\pi = 1,$$

so α is a root of $z^7 - 1 = 0$. More generally, $\alpha^k = \cos(2k\pi/7) + \mathrm{i}\sin(2k\pi/7)$ and $(\alpha^k)^7 = (\alpha^7)^k = 1$. The seven numbers

$$1,\ \alpha,\ \alpha^2,\ \alpha^3,\ \alpha^4,\ \alpha^5,\ \alpha^6$$

are distinct, since their arguments $2k\pi/7$ for $k = 0, \ldots, 6$ are distinct in $[0, 2\pi)$. A polynomial of degree 7 has at most 7 roots, so these are all the roots of $z^7 - 1 = 0$. They sit at the vertices of a regular heptagon on the unit circle.



Two facts do all the work. First, the roots sum to zero: $z^7 - 1 = (z-1)(z^6 + z^5 + \cdots + z + 1)$, and $\alpha \neq 1$, so

$$1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0. \qquad (*)$$

(Equivalently, the coefficient of z^6 in $z^7 - 1$ is zero.) Second, since $|\alpha| = 1$ we have $\overline{\alpha^k} = \alpha^{-k} = \alpha^{7-k}$. So $\alpha^6 = \overline{\alpha}$, $\alpha^5 = \overline{\alpha^2}$ and $\alpha^3 = \overline{\alpha^4}$.

Finding A and B

Let

$$p = \alpha + \alpha^2 + \alpha^4, \qquad q = \alpha^3 + \alpha^5 + \alpha^6.$$

By the conjugate pairs above, $q = \overline{p}$. That is the natural guess for the other root: a quadratic with real coefficients and a non-real root p must also have $\overline{p}$ as a root.

> **Ask yourself** How do we know p isn't real? If it were, the "other root" needn't be $\overline{p}$.

The imaginary part of p is $\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7}$, which we show at the end is positive. So p is not real and the roots are p and $\overline{p} = q$. (In any case, once we have found real A and B with p and q as roots, the quadratic is determined.)

By $(*)$,

$$p + q = \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = -1.$$

For the product, expand and reduce the powers using $\alpha^7 = 1$:

$$\begin{aligned} pq &= (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6) \\ &= \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^8 + \alpha^7 + \alpha^9 + \alpha^{10} \\ &= \alpha^4 + \alpha^6 + 1 + \alpha^5 + 1 + \alpha + 1 + \alpha^2 + \alpha^3 \\ &= 3 + (\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6) = 3 - 1 = 2. \end{aligned}$$

Note that each of $\alpha, \ldots, \alpha^6$ appears exactly once in the nine terms, besides three 1s. By the first part, $A = -(p+q)$ and $B = pq$, so

$$A = 1, \qquad B = 2,$$

and p and q are the roots of $z^2 + z + 2 = 0$, namely

$$z = \frac{-1 \pm \mathrm{i}\sqrt{7}}{2}.$$

The trigonometric sums

Taking real and imaginary parts of $p = \alpha + \alpha^2 + \alpha^4$, with $\alpha^k = \cos(2k\pi/7) + \mathrm{i}\sin(2k\pi/7)$, gives

$$\operatorname{Re} p = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7},$$
$$\operatorname{Im} p = \sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7}.$$

Both roots of $z^2 + z + 2 = 0$ have real part $-\frac{1}{2}$, so

$$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7} = -\frac{1}{2},$$

as required. (Alternatively, $p + \overline{p} = -1$ gives $2\operatorname{Re} p = -1$ directly.)

The imaginary part is $\pm\frac{\sqrt{7}}{2}$, and we must decide which.

> **Ask yourself** Two of the three sines are positive and one is negative. How can we be sure which wins?

Compare the negative term with one of the positive ones. Since $\frac{8\pi}{7} = \pi + \frac{\pi}{7}$, we have $\sin\frac{8\pi}{7} = -\sin\frac{\pi}{7}$. Also, $\sin\frac{4\pi}{7} = \sin\left(\pi - \frac{4\pi}{7}\right) = \sin\frac{3\pi}{7}$. Sine is strictly increasing on $[0, \frac{\pi}{2}]$ and $0 < \frac{\pi}{7} < \frac{3\pi}{7} < \frac{\pi}{2}$, so $\sin\frac{3\pi}{7} > \sin\frac{\pi}{7}$. Hence

$$\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7} = \underbrace{\sin\frac{2\pi}{7}}_{>0} + \underbrace{\left(\sin\frac{3\pi}{7} - \sin\frac{\pi}{7}\right)}_{>0} > 0,$$

and therefore

$$\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7} = \frac{\sqrt{7}}{2}.$$

This also confirms that $p = \frac{-1+\mathrm{i}\sqrt{7}}{2}$ is not real, as we used above.

Checks

On a calculator (in radians), $\cos\frac{2\pi}{7} \approx 0.6235$, $\cos\frac{4\pi}{7} \approx -0.2225$ and $\cos\frac{8\pi}{7} \approx -0.9010$, which add to -0.5000. The sines are about 0.7818, 0.9749 and -0.4339, which add to 1.3229. That matches $\frac{\sqrt{7}}{2} \approx 1.3229$. You can also check that $p^2 + p + 2 = 0$ directly: with $p = \frac{-1+\mathrm{i}\sqrt{7}}{2}$, $p^2 = \frac{-6-2\mathrm{i}\sqrt{7}}{4}$, and $p^2 + p + 2 = 0$ follows.

Common slips

- Using $z^6 - 1 = 0$ as printed. Then α isn't a root, and the sum-to-zero fact $(*)$ fails.
- Forgetting to reduce powers with $\alpha^7 = 1$ when expanding pq, or miscounting the three terms equal to 1.
- Asserting that the sine sum is positive "because two terms are positive". You need a comparison such as $\sin\frac{3\pi}{7} > \sin\frac{\pi}{7}$.

Beyond STEP

The split $\{1,2,4\}$ versus $\{3,5,6\}$ is not arbitrary. The numbers $1,2,4$ are the non-zero squares modulo 7, and $2p+1 = \mathrm{i}\sqrt{7}$ is a *Gauss sum*. Gauss showed that for a prime ℓ of the form $4k+3$, the sum of α^r over the quadratic residues r is $\frac{-1+\mathrm{i}\sqrt{\ell}}{2}$. This is the first step of the theory of which regular polygons can be constructed with ruler and compass (Galois theory) and of quadratic reciprocity. Davenport's *The Higher Arithmetic* covers quadratic residues, and Needham's *Visual Complex Analysis* gives a geometric view of roots of unity.

WORKED SOLUTION

STEP II 1995, Question 7

Pure • Heron's Formula

STEP II 1995, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q7

The key idea

The incircle is the hero throughout. Its radii to the points of contact are perpendicular to the sides, so they are ready-made heights for the three small triangles that tile ABC, and that gives (i). Part (ii) is the classical derivation of Heron's formula: write $\sin^2\alpha = 1 - \cos^2\alpha$, use the cosine rule to remove the angle, and then factorise by the difference of two squares, twice. For the sphere, look at a vertical cross-section. The sphere's contact points with the edges of the hole turn out to be exactly the points where the incircle touches the sides.

Solution

Part (i)

Let the incircle touch BC, CA and AB at P, Q and R'. A radius to a point of tangency is perpendicular to the tangent, so $IP \perp BC$, $IQ \perp CA$ and $IR' \perp AB$, and each of these has length r. So triangle BIC has base a and perpendicular height r, and similarly for the other two:

$$[BIC] = \tfrac{1}{2}ar, \qquad [CIA] = \tfrac{1}{2}br, \qquad [AIB] = \tfrac{1}{2}cr.$$

Since I lies inside the triangle, these three triangles fit together exactly to make ABC, with no overlaps. Adding,

$$\Delta = \tfrac{1}{2}r(a+b+c) = rs,$$

as required.

Part (ii)

Squaring $\Delta = \tfrac{1}{2}bc\sin\alpha$ and using $\sin^2\alpha = 1 - \cos^2\alpha$,

$$\Delta^2 = \tfrac{1}{4}b^2c^2(1-\cos^2\alpha) = \tfrac{1}{16}\left[4b^2c^2 - (2bc\cos\alpha)^2\right],$$

which is the first result. The bracket is a difference of two squares:

$$16\Delta^2 = (2bc - 2bc\cos\alpha)(2bc + 2bc\cos\alpha).$$

By the cosine rule, $2bc\cos\alpha = b^2 + c^2 - a^2$. Substituting this into each factor,

$$2bc - 2bc\cos\alpha = a^2 - (b^2 - 2bc + c^2) = a^2 - (b-c)^2,$$
$$2bc + 2bc\cos\alpha = (b^2 + 2bc + c^2) - a^2 = (b+c)^2 - a^2,$$

so

$$\Delta^2 = \tfrac{1}{16}\left[a^2 - (b-c)^2\right]\left[(b+c)^2 - a^2\right],$$

the second result. Each bracket is again a difference of two squares:

$$16\Delta^2 = (a-b+c)(a+b-c)(b+c-a)(b+c+a).$$

Now we bring in s. Since $a+b+c = 2s$,

$$b+c-a = 2(s-a), \qquad a-b+c = 2(s-b), \qquad a+b-c = 2(s-c),$$

so the product is $2s\cdot 2(s-a)\cdot 2(s-b)\cdot 2(s-c) = 16\,s(s-a)(s-b)(s-c)$. Therefore $\Delta^2 = s(s-a)(s-b)(s-c)$. Since $\Delta > 0$, we take the positive square root:

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}.$$

Part (iii)

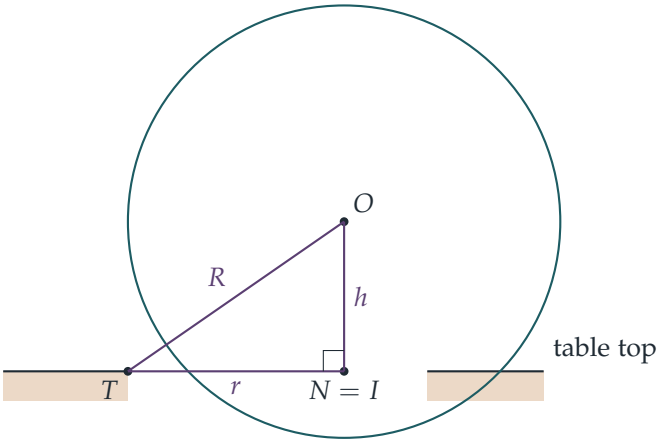
Let O be the centre of the sphere and h its height above the table top. Let N be the foot of the perpendicular from O to the plane of the table, so $ON = h$. The sphere rests on the three edges of the hole. Suppose it touches the edge BC at the point T.

Ask yourself Why must N be the incentre I?

At a point where a sphere touches a line, the radius is perpendicular to the line, so $OT \perp BC$ and $OT = R$. Also ON is perpendicular to the whole table plane, so $ON \perp BC$. So BC is perpendicular to two non-parallel lines in the plane ONT, which makes it perpendicular to that plane, and in particular to NT. So NT is the perpendicular distance from N to BC, and by Pythagoras in the right-angled triangle ONT,

$$NT^2 = R^2 - h^2.$$

The same argument for the edges CA and AB shows that N is at the same distance $\sqrt{R^2 - h^2}$ from all three sides. The point N lies inside the triangle (the sphere sits in the hole, above it), and the only point inside a triangle that is equidistant from all three sides is the incentre. So $N = I$, the common distance is r, and the points of contact are exactly where the incircle touches the sides. These lie on the edges themselves, not on their extensions, so the sphere really does rest on the three edges.



The diagram shows the vertical section through O and T; the edge BC passes through T, perpendicular to the page. Pythagoras in triangle ONT now gives $h^2 = R^2 - r^2$. From (i) and (ii),

$$r = \frac{\Delta}{s} = \frac{\sqrt{s(s-a)(s-b)(s-c)}}{s}, \qquad \text{so} \qquad r^2 = \frac{(s-a)(s-b)(s-c)}{s}.$$

Hence the centre of the sphere is at height

$$h = \sqrt{R^2 - \frac{(s-a)(s-b)(s-c)}{s}}$$

above the table top. This needs $R > r$: a sphere no bigger than the incircle falls straight through the hole.

Checks

Take the 3–4–5 triangle: $s = 6$, and Heron gives $\Delta = \sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6$, which matches $\frac{1}{2} \cdot 3 \cdot 4$. Then $r = \Delta/s = 1$, which you can confirm from the right-angle formula $r = \frac{1}{2}(3 + 4 - 5)$. With $R = 2$ the formula gives $h = \sqrt{4 - 1} = \sqrt{3} \approx 1.732$. As $R \to r^+$, $h \to 0$: the sphere sinks until its equator is level with the table and sits on the incircle. As a broader check, we tested 18,816 random triangles numerically. In every case, a centre placed at height h above the incentre was at distance exactly R from all three edge lines, with the nearest points inside the edges.

Common slips

- Taking $h = \sqrt{R^2 - d^2}$, where d is the distance from I to a *vertex*. The sphere touches the edges, not the corners.
- Assuming without reason that the centre is above the incentre. It needs the perpendicularity argument, or at least a clear symmetry reason.
- Mixing up the factor pairs in (ii): $b + c - a = 2(s - a)$, not $2(s - b)$ or $2(s - c)$.

Beyond STEP

Heron's formula belongs to a family of symmetric formulae for areas and volumes. Brahmagupta's formula does the same job for a cyclic quadrilateral, and the Cayley–Menger determinant gives the volume of a tetrahedron from its six edge lengths. The step "radius $\perp$ tangent line, so the foot of the perpendicular is the incentre" is the three-dimensional version of the theorem of the three perpendiculars, which is worth knowing for vector geometry. For many more questions of this kind, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 1995, Question 8

Pure • Logistic Differential Equation

STEP II 1995, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q8

This question has a full guided treatment in *Beyond STEP*, Section 8.3, *Harvesting a Population*: https://beyondstep.keithmcnulty.org/ch08/harvesting.html

The key idea

The growth law is a logistic equation, which is separable: partial fractions turn the integral into two logarithms. For the second part, completing the square in the new right-hand side shows that the shift $y = x - K + \alpha$ turns the new equation into *exactly the same* logistic equation, with α in place of K. So we can reuse the first answer without doing any more integration.

Solution

The population without removal

We take $K > 0$ (a population that starts at $x = K$ must have $K > 0$). The equation is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = (2K - x)x.$$

Starting from $x = K$, the rate is $K^2 > 0$. The solution can never reach $x = 0$ or $x = 2K$, because those constant functions are themselves solutions and solutions of this equation can't cross. So $0 < x < 2K$ throughout, and we may divide by $(2K - x)x$. In partial fractions,

$$\frac{1}{x(2K - x)} = \frac{1}{2K}\left(\frac{1}{x} + \frac{1}{2K - x}\right),$$

so separating the variables gives

$$\frac{1}{2K}\int\left(\frac{1}{x} + \frac{1}{2K - x}\right)\mathrm{d}x = \int \mathrm{d}t,$$

$$\frac{1}{2K}\Big(\ln x - \ln(2K - x)\Big) = t + C.$$

No modulus signs are needed, since x and $2K - x$ are both positive. At $t = 0$ we have $x = 2K - x = K$, so the left-hand side is 0 and $C = 0$. Hence

$$\ln \frac{x}{2K - x} = 2Kt \quad \Longrightarrow \quad x = (2K - x)\,\mathrm{e}^{2Kt} \quad \Longrightarrow \quad x = \frac{2K\mathrm{e}^{2Kt}}{1 + \mathrm{e}^{2Kt}} = \frac{2K}{1 + \mathrm{e}^{-2Kt}}.$$

To match the required form, write $2 = (1 + \mathrm{e}^{-2Kt}) + (1 - \mathrm{e}^{-2Kt})$:

$$x(t) = \frac{K(1 + \mathrm{e}^{-2Kt}) + K(1 - \mathrm{e}^{-2Kt})}{1 + \mathrm{e}^{-2Kt}} = K + K\,\frac{1 - \mathrm{e}^{-2Kt}}{1 + \mathrm{e}^{-2Kt}},$$

which is the result we needed. (Equivalently, $x = K + K\tanh Kt$.)

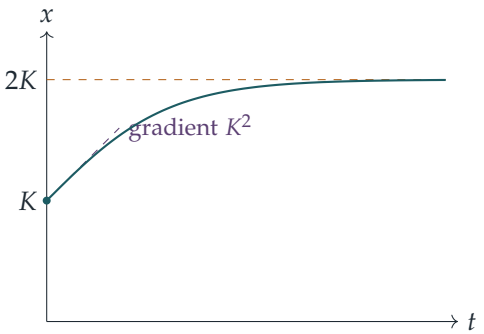
The sketch. For $t \geq 0$ we have $\mathrm{e}^{-2Kt} \in (0, 1]$, so $K \leq x < 2K$. Then $\dot{x} = (2K - x)x > 0$, so x is increasing. Differentiating the equation again,

$$\ddot{x} = (2K - 2x)\,\dot{x} \leq 0 \quad \text{since } x \geq K,$$

so the graph is concave, with gradient K^2 at $t = 0$ (where $\ddot{x} = 0$: the origin of time is the point of inflection of the full S-shaped logistic curve). As $t \to \infty$, $\mathrm{e}^{-2Kt} \to 0$, so

$$x(t) \to 2K.$$

The population levels off at $2K$, approaching it from below and never reaching it.



With removal at rate L

Removing L micrograms per hour subtracts L from the rate of change:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = (2K - x)x - L.$$

Complete the square in the right-hand side:

$$(2K - x)x - L = -(x^2 - 2Kx) - L = K^2 - L - (x - K)^2 = \alpha^2 - (x - K)^2,$$

where $\alpha = \sqrt{K^2 - L} > 0$ because $K^2 > L$.

Ask yourself Why is $y = x - K + \alpha$ the right substitution?

The right-hand side $\alpha^2 - (x-K)^2$ vanishes at $x = K \pm \alpha$, the two equilibrium populations. Shifting by $K - \alpha$ moves the lower equilibrium to 0, which is where the original logistic equation had its lower equilibrium too. With $x - K = y - \alpha$ and $\dot{y} = \dot{x}$,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \alpha^2 - (y-\alpha)^2 = 2\alpha y - y^2 = (2\alpha - y)\, y.$$

This is the first equation with α in place of K. The initial condition matches as well: $x(0) = K$ gives $y(0) = \alpha$, the analogue of $x(0) = K$. Since $\alpha > 0$, the first part applies word for word, and

$$y(t) = \frac{2\alpha}{1 + \mathrm{e}^{-2\alpha t}}, \qquad \text{so} \qquad x(t) = K - \alpha + \frac{2\alpha}{1 + \mathrm{e}^{-2\alpha t}}.$$

As $t \to \infty$, $\mathrm{e}^{-2\alpha t} \to 0$, so $y \to 2\alpha$ and

$$x(t) \to K - \alpha + 2\alpha = K + \alpha,$$

as required. Where $L > 0$ we have $\alpha < K$, so the harvested population settles at the lower level $K + \alpha < 2K$, as we'd expect.

Checks

- At $t = 0$ both formulae give $x = K$. Setting $L = 0$ gives $\alpha = K$, and the second formula reduces to $2K/(1 + \mathrm{e}^{-2Kt})$, the first.
- Differentiating $x = K + K\tanh Kt$ gives $\dot{x} = K^2 \operatorname{sech}^2 Kt = K^2(1 - \tanh^2 Kt) = (K - K\tanh Kt)(K + K\tanh Kt) = (2K - x)x$.
- Numerically, with $K = 1$, $x(1) = 2/(1 + \mathrm{e}^{-2}) \approx 1.7616$; with $K = 1$ and $L = 0.75$ (so $\alpha = \frac{1}{2}$), $x(1) \approx 1.2311$. Both agree with a direct numerical (Runge–Kutta) solution of the differential equations, and the second settles at $1.5 = K + \alpha$.

Common slips

- Losing the factor $\frac{1}{2K}$ in the partial fractions, which gives e^{-t} in place of e^{-2Kt}.
- In the second part, writing $+L$, or forgetting that the constant of integration must be found again. Here the substitution makes $y(0) = \alpha$, so no new constant is needed.

Beyond STEP

The second part is a first look at *stability of equilibria*. The equation $\dot{x} = \alpha^2 - (x-K)^2$ has equilibria at $K \pm \alpha$: the upper one attracts nearby solutions and the lower one repels them. You can see this from the sign of the right-hand side, without solving

anything. This phase-line approach, and what happens when $L > K^2$ (no equilibria, so the population is wiped out in finite time), are treated in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 1995, Question 9

Mechanics • Friction and Equilibrium

STEP II 1995, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

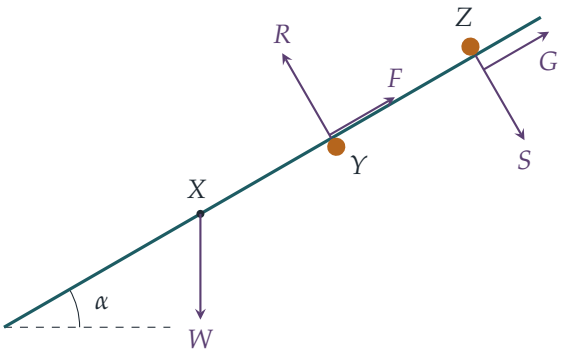
step.maths.org/questions/95-s2-q9

The key idea

The rod lies along a line of greatest slope, so it is natural to resolve along the rod and perpendicular to it. The bar the rod rests on pushes it up, away from the plane; the bar it passes under can only push it down. Taking moments about Y shows where the centre must be, and then the least μ comes from making both frictional forces act up the slope at their limiting values.

Solution

We model the rod as a uniform rigid rod of weight W and the bars as fixed thin bars, each giving a normal reaction perpendicular to the rod and a friction force along it. Since the rod is perpendicular to both horizontal bars, it lies along a line of greatest slope of their plane, and so it is inclined at α to the horizontal. Let R be the reaction at Y (pushing the rod away from the plane, since the rod lies on top of this bar) and S the reaction at Z (pushing the rod towards the plane, since the rod is underneath this bar), with $R, S \geq 0$. Let F and G be the friction forces at Y and Z, taken as positive up the slope.



Where the centre must be

Resolving perpendicular to the rod and taking moments about Y (distances measured along the rod):

$$R = S + W\cos\alpha,$$
$$S\,d = W\cos\alpha \cdot XY.$$

The moment of S about Y must be balanced by the moment of the weight component $W\cos\alpha$, which acts towards the plane just as S does. Two parallel forces in the same direction can only have cancelling moments about Y if they act on opposite sides of Y. Since Z is up the slope from Y, the centre X must be at or below Y, and then

$$XZ = XY + YZ, \qquad \text{so} \qquad l = XY + d \geq d.$$

(If X were between Y and Z, or beyond Z, the moment equation would force $S < 0$: the bar at Z would have to pull the rod, which it cannot do, and the rod would rotate off it.) Equality $l = d$ is the borderline case in which X is at Y and $S = 0$.

Ask yourself The paper says $l \leq d$. Why do we get the opposite inequality?

The inequality in the paper is a misprint: the question itself goes on to take $l = 2d$, which is only possible if $l \geq d$. The argument above shows that $l \geq d$ is what equilibrium requires, and that is what we use.

The least value of μ

With $XY = l - d$ the two equations give

$$S = \frac{W\cos\alpha\,(l-d)}{d}, \qquad R = \frac{W\cos\alpha\,l}{d}, \qquad \text{so} \qquad R + S = \frac{W\cos\alpha\,(2l-d)}{d}.$$

Resolving along the rod, friction must hold up the component of the weight down the slope:

$$F + G = W\sin\alpha.$$

The friction laws require $|F| \leq \mu R$ and $|G| \leq \mu S$, so $F + G \leq \mu(R+S)$. Hence equilibrium needs

$$W\sin\alpha \leq \mu\,\frac{W\cos\alpha\,(2l-d)}{d}, \qquad \text{that is,} \qquad \mu \geq \frac{d\tan\alpha}{2l-d}.$$

Conversely, if μ equals this value, then $F = \mu R$ and $G = \mu S$ (both up the slope, both limiting) satisfy every equation, so the rod can be in equilibrium. The least possible value is therefore

$$\mu_{\min} = \frac{d\tan\alpha}{2l-d}.$$

When $l = 2d$ this becomes $\mu_{\min} = \dfrac{d\tan\alpha}{3d} = \frac{1}{3}\tan\alpha$, so $\mu \geq \frac{1}{3}\tan\alpha$, as required.

Checks

- When $l = d$ the centre is over the lower bar, $S = 0$, and we get $\mu \geq \tan\alpha$: the familiar condition for a single particle resting on a rough slope.
- As $l \to \infty$ the reactions grow without limit, so very little friction is needed, and indeed $\mu_{\min} \to 0$.
- Numerically, with $\alpha = 0.7$ and $l = 2d$, the formula gives $0.28076\ldots$, which is $\frac{1}{3}\tan 0.7$. A computer also checked the force and moment equations about random points, and that equilibrium holds just above $\mu_{\min}$ and fails just below it.

Common slips

- Taking both reactions in the same direction. The bar the rod lies under can only push it towards the plane.
- Assuming friction is limiting at only one bar. To make μ as small as possible, both bars must provide their full friction, up the slope.

Beyond STEP

This is a small problem in statics with inequality constraints: the contact forces must satisfy linear equations together with the friction inequalities, and the least μ is the point at which the feasible set just becomes non-empty. The same structure, linear equations with linear inequalities, underlies linear programming and the analysis of frictional contact in engineering. For more on rigid-body statics and friction, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1995, Question 10

Mechanics • Collisions and Restitution

STEP II 1995, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q10

The key idea

Each collision is a two-equation problem: conservation of momentum and Newton's experimental law. Do the two collisions in turn, set the second sphere's final velocity equal to u, and solve for m_1; the inequality then comes from the simple fact that a mass must be positive. For the final part, notice that collisions only care about *relative* velocities, so watching the motion from a frame moving at speed v turns it into the first part.

Solution

We model the spheres as particles moving along one line on a smooth table, so the only horizontal forces are the impulses between spheres. Take the direction of the first sphere's motion as positive.



The first collision

Let the velocities of the first and second spheres just after the first collision be v_1 and w. Momentum and Newton's law (the speed of separation is e times the speed of approach) give

$$m_1 v_1 + m_2 w = m_1 u, \qquad w - v_1 = eu.$$

Substituting $v_1 = w - eu$ into the first equation, $(m_1 + m_2)w = m_1(1 + e)u$, so

$$w = \frac{m_1(1+e)u}{m_1 + m_2}, \qquad v_1 = \frac{(m_1 - em_2)u}{m_1 + m_2}.$$

The second collision

Now the second sphere, moving at w, hits the third, which is at rest. This is the same situation with m_1, m_2, u replaced by m_2, m_3, w, so the second sphere's velocity afterwards is

$$\frac{(m_2 - em_3)w}{m_2 + m_3}.$$

We are told this equals u. Substituting for w and cancelling $u \neq 0$,

$$\frac{(m_2 - em_3)\, m_1(1+e)}{(m_2+m_3)(m_1+m_2)} = 1, \qquad \text{so} \qquad m_1(1+e)(m_2 - em_3) = (m_1+m_2)(m_2+m_3).$$

Collecting the terms in m_1,

$$m_1\big[(1+e)m_2 - e(1+e)m_3 - m_2 - m_3\big] = m_2(m_2+m_3),$$

and the bracket simplifies to $em_2 - m_3(1+e+e^2)$. Hence

$$\boxed{m_1 = \frac{m_2(m_2+m_3)}{em_2 - m_3(1+e+e^2)}}.$$

Ask yourself Where does an inequality come from, when all we have is an equation?

From the physics: masses are positive. The numerator $m_2(m_2+m_3)$ is positive, and the bracket we divided by cannot be zero (otherwise the equation before it would read $0 = m_2(m_2+m_3)$). So $m_1 > 0$ forces the denominator to be positive:

$$em_2 - m_3(1+e+e^2) > 0, \qquad \text{that is,} \qquad m_2 e > m_3(1+e+e^2),$$

as required.

It is worth confirming that nothing else happens afterwards, since the final part counts collisions. After the second collision the second sphere moves at u, while the first moves at $v_1 = u - \frac{m_2(1+e)u}{m_1+m_2} < u$, so the first sphere never catches the second again. The third sphere moves away from the second with relative speed $ew > 0$. So there are exactly two collisions.

The final part

Now all three spheres share a common velocity v except the first, which moves at $u > v$. Look at the motion from a frame moving with constant velocity v along the line. In this frame:

- the first sphere moves at $u - v > 0$ towards the second, and the second and third are at rest;
- momentum is still conserved in each collision, because subtracting v from every velocity changes both sides of the momentum equation by the same amount, $(m_a + m_b)v$;

- Newton's law is unchanged, because it involves only differences of velocities.

So in the moving frame we have exactly the situation of the first part, with u replaced by $u - v$, and with the masses satisfying the same relation (which does not involve u at all). By the first part, after two collisions the second sphere moves at $u - v$ relative to the frame. Returning to the table's frame, we add v back on: the velocity of the second sphere after two collisions is

$$\boxed{u}.$$

The condition $u > v$ is what guarantees the first collision happens at all; the hypothesis $v > 0$ is not actually needed for the argument.

Checks

Take $e = \frac{1}{2}$, $m_3 = 1$ and $m_2 = 4$; then $em_2 = 2 > \frac{7}{4} = m_3(1 + e + e^2)$, and the formula gives $m_1 = \frac{4 \cdot 5}{1/4} = 80$. With $u = 7$, the first collision gives the second sphere $w = \frac{80 \cdot \frac{3}{2} \cdot 7}{84} = 10$, and the second collision then gives it $\frac{(4 - \frac{1}{2}) \cdot 10}{5} = 7 = u$, as it should. A computer also ran the collisions exactly (with fractions) for over 10^4 random admissible choices of masses and e, confirming in each case that there are exactly two collisions, that the second sphere ends at u, and that adding a common velocity $v < u$ to the second and third spheres gives the same final velocity u.

Common slips

- Writing Newton's law the wrong way round, e.g. $v_1 - w = eu$; the sphere in front must end up moving at least as fast as the one behind.
- In the final part, recomputing the collisions from scratch instead of using the change of frame, which the question explicitly invites.
- Forgetting to add v back at the end and giving $u - v$ as the answer.

Beyond STEP

The final part is an example of Galilean invariance: the laws of collisions look the same in every frame moving at constant velocity, so you can choose the frame that makes the problem easiest. In university mechanics the favourite choice is the centre-of-mass frame, in which the total momentum is zero and a collision simply reverses and scales the velocities by e. Kleppner and Kolenkow, *An Introduction to Mechanics*, uses this idea extensively in its treatment of collisions.

WORKED SOLUTION

STEP II 1995, Question 11

Mechanics • Linear Air Resistance

STEP II 1995, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

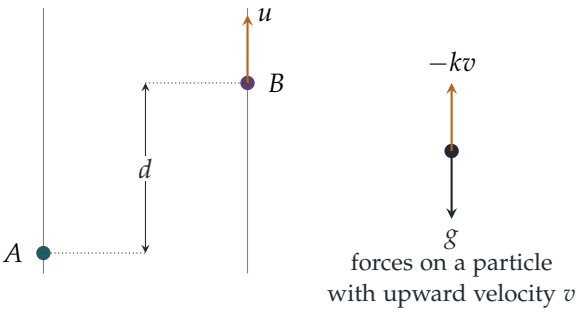
step.maths.org/questions/95-s2-q11

The key idea

Because the resistance is *linear* in velocity, the equation of motion is linear, and the same equation holds whether a particle is rising or falling. So subtract the two equations of motion: gravity cancels, and the *relative* velocity simply decays exponentially. Integrating that once gives the separation directly, without ever needing to know where either particle is.

Solution

Model each body as a particle of unit mass moving on a vertical line, acted on by its weight g and a resistance of magnitude $k|v|$ directed against its motion. Take the upward direction as positive, put the origin at A, and let particle 1 (from A) and particle 2 (from B) have heights x_1, x_2 and velocities v_1, v_2 at time t.



Equations of motion

For a particle with upward velocity v, the resistance is $-kv$: if $v > 0$ (rising) it points down, and if $v < 0$ (falling) it points up, which is exactly "against the motion" in both

cases. So Newton's second law for unit mass gives, for each particle,

$$\frac{\mathrm{d}v_1}{\mathrm{d}t} = -g - kv_1, \qquad \frac{\mathrm{d}v_2}{\mathrm{d}t} = -g - kv_2,$$

valid for all $t \geq 0$.

Ask yourself Particle 2 goes up, stops and comes down. Don't we need a separate equation for each stage?

With a resistance proportional to *speed squared* we would: the force would be $-kv|v|$, which changes form when v changes sign. Here the force is $-kv$, a single expression that already reverses direction when v does, so one equation covers the whole motion.

The relative motion

Let $w = v_2 - v_1$ be the velocity of particle 2 relative to particle 1. Subtracting the two equations, the weight terms cancel:

$$\frac{\mathrm{d}w}{\mathrm{d}t} = -kw, \qquad w(0) = u - 0 = u,$$

so $w = u\,\mathrm{e}^{-kt}$. Now let $s = x_2 - x_1$, with $s(0) = d$ since B is a height d above A. Then $\mathrm{d}s/\mathrm{d}t = w$, and

$$s = d + \int_0^t u\,\mathrm{e}^{-k\tau}\,\mathrm{d}\tau = d + \frac{u}{k}\left(1 - \mathrm{e}^{-kt}\right).$$

Since $u \geq 0$ and $1 - \mathrm{e}^{-kt} \geq 0$, we have $s \geq d > 0$: particle 2 is always above particle 1, and the particles never meet. So the distance between them at time t is

$$\boxed{d + \frac{u}{k}\left(1 - \mathrm{e}^{-kt}\right)}$$

As $t \to \infty$, $\mathrm{e}^{-kt} \to 0$ because $k > 0$, so the distance tends to $d + \dfrac{u}{k}$, as required.

The individual motions (if you prefer to go the long way)

Solving each equation separately (it is first-order linear, or separable) gives $v = -\frac{g}{k} + \left(v(0) + \frac{g}{k}\right)\mathrm{e}^{-kt}$, and integrating again,

$$x_1 = -\frac{gt}{k} + \frac{g}{k^2}\left(1 - \mathrm{e}^{-kt}\right),$$
$$x_2 = d - \frac{gt}{k} + \frac{1}{k}\left(u + \frac{g}{k}\right)\left(1 - \mathrm{e}^{-kt}\right).$$

Subtracting, the gt/k and g/k^2 terms cancel and we recover the same s. Both particles approach the terminal velocity $-g/k$, which is why their separation settles down.

Checks

- At $t = 0$ the distance is d; and for small t it is about $d + ut$, as it would be without resistance (both particles then fall equally under gravity).
- If $u = 0$ the particles are released together from rest, follow identical motions, and stay d apart, as the formula says.
- Units: k has units s^{-1} (force per unit mass per unit velocity), so u/k is a length.
- Numerically, stepping both equations of motion with a Runge–Kutta method for random k, u, d, t matched the formulas to about 10^{-12}. With $k = 0.5$, $u = 10$, $d = 3$, the distance is 10.87 at $t = 1$ and 22.999 at $t = 20$, approaching $d + u/k = 23$.

Common slips

- Splitting particle 2's motion into "up" and "down" phases with different signs for the resistance: unnecessary here, and a frequent source of sign errors.
- Writing the resistance as $+kv$ for the falling particle when v is already the (negative) upward velocity, which double-counts the sign.

Beyond STEP

The trick of subtracting the equations works because the equation of motion is linear: the difference of two solutions solves the homogeneous equation $\dot{w} = -kw$. This is the superposition principle that organises the whole theory of linear differential equations, where every solution is "particular solution plus complementary function". With quadratic drag the principle fails and the separation behaves quite differently. Boyce and DiPrima, *Elementary Differential Equations*, develops the linear theory; Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion with both kinds of resistance.

WORKED SOLUTION

STEP II 1995, Question 12

Probability and Statistics • Sequential Duels

STEP II 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.13, *The Truel*: https://beyondstep.keithmcnulty.org/ch12/truel.html

The key idea

In a duel that goes on until someone is hit, the game "starts again" every time both players miss. So the probability we want satisfies a simple equation in terms of itself, and there's no need to sum an infinite series (though you can). For the three-way duel, each option for Algernon's first shot quickly turns into a two-person duel, and the first result handles those. The surprise is that Algernon's best move is to hit nobody.

Solution

The two-person duel

Let P be the probability that Percy wins when he throws first. Look at the first pair of throws:

- Percy hits with his first throw (probability p), and he wins at once;
- Percy misses and then Rodney hits (probability $(1-p)r$), and Percy loses;
- both miss (probability $(1-p)(1-r)$), and we are back where we started with Percy to throw. Throws are independent, so from here Percy wins with probability P again.

Hence

$$P = p + (1-p)(1-r)P \quad\Longrightarrow\quad P\big(1-(1-p)(1-r)\big) = p \quad\Longrightarrow\quad P = \frac{p}{p+r-pr},$$

since $1-(1-p)(1-r) = p+r-pr$, which is positive because $p > 0$. This is the required result.

Ask yourself Does the game definitely end, so that Percy losing really is the same as Rodney winning?

Yes. The chance that the first n rounds all end in double misses is $\big((1-p)(1-r)\big)^n$, which tends to 0 because $p > 0$. Equivalently, summing the geometric series $\sum_{k\geq 0} p\big((1-p)(1-r)\big)^k$ gives the same answer.

In the three-way duel, A, B and C hit with probabilities $\frac{2}{5}$, $\frac{3}{5}$ and 1. A two-person duel between A and B with A throwing first will come up twice, so we work it out now using the result above with $p = \frac{2}{5}$, $r = \frac{3}{5}$:

$$\frac{\frac{2}{5}}{\frac{2}{5}+\frac{3}{5}-\frac{6}{25}} = \frac{\frac{2}{5}}{\frac{19}{25}} = \frac{10}{19}.$$

Part (i)

If Algernon hits Cuthbert, then only Algernon and Bertie are left, and Bertie throws next, aiming at Algernon. Algernon survives only if Bertie misses (probability $\frac{2}{5}$). After that it is a duel between A and B with A throwing first, which Algernon wins with probability $\frac{10}{19}$. So

$$\mathrm{P}(\text{A wins}) = \frac{2}{5}\cdot\frac{10}{19} = \frac{4}{19}.$$

Part (ii)

If Algernon hits Bertie, Cuthbert throws next with only Algernon left to aim at. Cuthbert never misses, so Algernon's chance of winning is 0.

Part (iii)

If Algernon misses, all three are still in. Bertie throws next, and aims at Cuthbert.

- With probability $\frac{3}{5}$ Bertie hits Cuthbert. Algernon and Bertie are left with Algernon to throw, and Algernon wins with probability $\frac{10}{19}$.
- With probability $\frac{2}{5}$ Bertie misses. Cuthbert then aims at Bertie and is certain to hit him. Algernon and Cuthbert are left with Algernon to throw. Algernon must hit with this throw, because otherwise Cuthbert hits him. So he wins with probability $\frac{2}{5}$.

Therefore

$$\mathrm{P}(\text{A wins}) = \frac{3}{5}\cdot\frac{10}{19} + \frac{2}{5}\cdot\frac{2}{5} = \frac{6}{19} + \frac{4}{25} = \frac{150+76}{475} = \frac{226}{475}.$$

This is the same however the miss comes about, whether deliberate or a failed aim, because Bertie and Cuthbert play the same way afterwards.

Algernon's best plan

Ask yourself Algernon can't choose to *hit* someone, only to *aim* at them. What does each choice really give him?

If he aims at someone, he hits with probability $\frac{2}{5}$ and misses with probability $\frac{3}{5}$, and a miss leads to part (iii). So:

$$\begin{aligned}
\text{aim at Cuthbert:} &\quad \frac{2}{5}\cdot\frac{4}{19}+\frac{3}{5}\cdot\frac{226}{475}=\frac{200+678}{2375}=\frac{878}{2375}\approx 0.370,\\
\text{aim at Bertie:} &\quad \frac{2}{5}\cdot 0+\frac{3}{5}\cdot\frac{226}{475}=\frac{678}{2375}\approx 0.285,\\
\text{deliberately miss:} &\quad \frac{226}{475}=\frac{1130}{2375}\approx 0.476.
\end{aligned}$$

Even a certain hit on Cuthbert would only give $\frac{4}{19}=\frac{100}{475}$, which is less than $\frac{226}{475}$. Hitting either opponent can only lower his chances. Algernon should therefore **deliberately miss with his first shot**. Then Bertie and Cuthbert shoot at each other, and Algernon gets the first throw against whoever survives. With this plan his probability of winning is $\frac{226}{475}$, as required.

Checks

- If $r=0$ the duel formula gives 1, and if $p=1$ it gives 1, both as they should.
- The three options rank as $0.476 > 0.370 > 0.285$. A Monte Carlo simulation of the whole three-way duel (3×10^5 games per strategy) gave 0.476, 0.369 and 0.284, matching the exact values.
- Part (iii) as a sanity check: $\frac{6}{19}\approx 0.316$ and $\frac{4}{25}=0.16$ add to 0.476.

Common slips

- Getting the order of throws wrong after the first shot. In part (i) Bertie throws *next*, so the duel formula only applies after Bertie has missed.
- Treating "aim at Cuthbert" as if it meant "hit Cuthbert". Parts (i)–(iii) are conditional on the outcome, and the plan has to weight them by $\frac{2}{5}$ and $\frac{3}{5}$.

Beyond STEP

Writing a probability in terms of itself after one step is called *first-step analysis*. It is the basic tool for Markov chains, where it gives hitting probabilities and expected hitting times as solutions of linear equations. The "truel" is a well-known puzzle in game theory, and the weakest player often does best by staying out of the fight. For more, see

Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP II 1995, Question 13

Probability and Statistics • Poisson Distribution Optimisation

STEP II 1995, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q13

The key idea

Despite the database tag, this is a probability question dressed up as business: the no-shows are a binomial count with a large number of trials and a tiny success probability, which is exactly when the Poisson distribution takes over. After that, the trick for optimising over a whole number k is not calculus but the *first difference* $u_{k+1} - u_k$: as long as selling one more ticket increases the expected profit, keep going. Once that difference is shown to be decreasing, the best k is where it first stops being positive.

Solution

The Poisson model

Suppose each of the $N + k$ ticket holders independently fails to turn up with probability ϵ. Then the number R who fail to turn up has the binomial distribution $\mathrm{B}(N + k, \epsilon)$, with mean $(N + k)\epsilon$. Since k is very small compared with N, this mean is approximately $N\epsilon = \lambda$. A binomial distribution with a large number of trials and a very small success probability is well approximated by a Poisson distribution with the same mean: for fixed r, as $n \to \infty$ with $n\epsilon = \lambda$ fixed,

$$\binom{n}{r}\epsilon^r(1-\epsilon)^{n-r} = \frac{n(n-1)\cdots(n-r+1)}{n^r}\cdot\frac{\lambda^r}{r!}\left(1-\frac{\lambda}{n}\right)^{n-r} \;\to\; \frac{\lambda^r}{r!}\,\mathrm{e}^{-\lambda},$$

because the first factor tends to 1 and $(1 - \lambda/n)^n \to \mathrm{e}^{-\lambda}$. So, approximately, $R \sim \mathrm{Po}(\lambda)$. From now on we take this as exact and write

$$p_r = \mathrm{P}(R = r) = \frac{\lambda^r}{r!}\,\mathrm{e}^{-\lambda}.$$

The profit

The airline sells $N + k$ tickets at £A profit each, which gives $A(N + k)$ whatever happens. If r passengers fail to turn up, then $N + k - r$ arrive for N seats.

- If $r \geq k$, at most N people arrive and everyone flies: nobody is compensated, and $\max(0, k-r) = 0$.
- If $r < k$, then $k-r$ people cannot be flown, each costing £B, and $\max(0, k-r) = k-r$.

In both cases the profit in pounds is $A(N+k) - B\max(0, k-r)$.

The expected profits $u_0, \ldots, u_3$

Taking expectations, only $r = 0, 1, \ldots, k-1$ contribute to the penalty, so

$$u_k = A(N+k) - B\sum_{r=0}^{k-1}(k-r)\,p_r.$$

With $p_0 = \mathrm{e}^{-\lambda}$, $p_1 = \lambda\mathrm{e}^{-\lambda}$ and $p_2 = \frac{1}{2}\lambda^2\mathrm{e}^{-\lambda}$:

$$\begin{aligned}
u_0 &= AN,\\
u_1 &= A(N+1) - B\mathrm{e}^{-\lambda},\\
u_2 &= A(N+2) - B\mathrm{e}^{-\lambda}(2+\lambda),\\
u_3 &= A(N+3) - B\mathrm{e}^{-\lambda}\left(3 + 2\lambda + \tfrac{1}{2}\lambda^2\right).
\end{aligned}$$

The differences v_k

Subtracting the general formulae,

$$\begin{aligned}
v_k = u_{k+1} - u_k &= A - B\Big[\sum_{r=0}^{k}(k+1-r)\,p_r - \sum_{r=0}^{k-1}(k-r)\,p_r\Big]\\
&= A - B\Big[p_k + \sum_{r=0}^{k-1}\big((k+1-r) - (k-r)\big)p_r\Big]\\
&= A - B\sum_{r=0}^{k} p_r,
\end{aligned}$$

where the $r = k$ term of the first sum, $(k+1-k)p_k = p_k$, has no partner in the second sum. In words, $v_k = A - B\,\mathrm{P}(R \leq k)$. This makes sense: the $(k+1)$th extra ticket earns £A for sure, and it costs an extra £B exactly when that passenger ends up bumped, which happens when $R \leq k$.

Ask yourself Why does the bracket collapse to a plain sum of probabilities?

For each $r \leq k-1$, the coefficient of p_r goes up by exactly one when k increases by one (one more bumped passenger in that outcome), and the outcome $r = k$ newly starts to cost one payment. So every r from 0 to k contributes p_r once.

Then

$$v_k - v_{k+1} = B\,p_{k+1} = B\,\frac{\lambda^{k+1}}{(k+1)!}\,\mathrm{e}^{-\lambda} > 0,$$

since $B > 0$ and $\lambda > 0$. So $v_k > v_{k+1}$ for every k: the differences strictly decrease.

Finally, $\sum_{r=0}^{k} p_r$ is a partial sum of $e^{-\lambda}\sum_{r\geq 0}\lambda^r/r! = e^{-\lambda}e^{\lambda} = 1$, so it tends to 1 as $k \to \infty$, and

$$v_k \to A - B \qquad \text{as } k \to \infty,$$

as required. Since $B > A$, this limit is negative.

Advice to Fly By Night

Write $F(k) = P(R \leq k) = \sum_{r=0}^{k} p_r$, so $v_k = A - BF(k)$, and

$$u_{k+1} > u_k \iff v_k > 0 \iff F(k) < \frac{A}{B}.$$

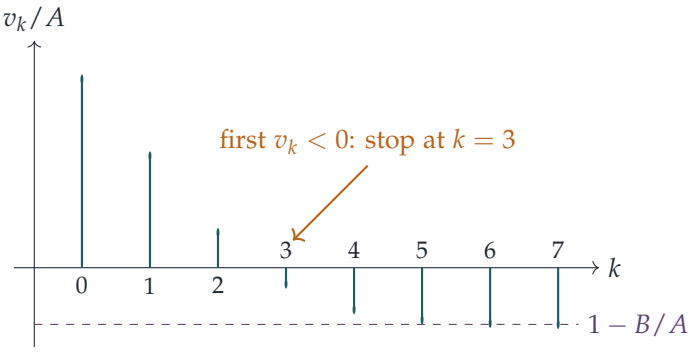
Because v_k is strictly decreasing with a negative limit, the sequence $v_0, v_1, v_2, \ldots$ is positive up to some point and negative (or zero once) from then on. So u_k rises and then falls, and its maximum is at the first k for which the next ticket no longer pays:

$$\boxed{\text{choose the smallest } k \text{ with } F(k) = \sum_{r=0}^{k} \frac{\lambda^r}{r!}e^{-\lambda} \geq \frac{A}{B}.}$$

In particular, if $e^{-\lambda} \geq A/B$ (very reliable passengers, or a very heavy penalty), then $v_0 \leq 0$ and the airline should not overbook at all: $k = 0$. If $F(k) = A/B$ exactly, then $v_k = 0$ and k and $k+1$ give the same expected profit. Since $A/B < 1$ and $F(k) \to 1$, such a k always exists.

So the advice is: keep selling extra tickets while the probability that at most k people fail to turn up is less than the ratio A/B of ticket profit to compensation; stop at the first k where it reaches A/B.

For example, with $\lambda = 2$ and $B = 1.25A$ (so $A/B = 0.8$), we get $F(0) \approx 0.135$, $F(1) \approx 0.406$, $F(2) \approx 0.677$ and $F(3) \approx 0.857$, so the best choice is $k = 3$. The plot shows v_k/A for this case.



Checks

- From the general formula, $u_1 - u_0 = A - B\mathrm{e}^{-\lambda} = A - Bp_0$ and $u_2 - u_1 = A - B\mathrm{e}^{-\lambda}(1+\lambda) = A - B(p_0 + p_1)$, matching v_0 and v_1.
- Limiting cases: if $B \to A$ then $A/B \to 1$ and k grows without bound (bumping is almost free); if $B \to \infty$ then $A/B \to 0$ and $k = 0$.
- Numerically, for hundreds of random choices of N, A, B, λ we computed u_k directly from the profit formula and confirmed the formulae for $u_0, \ldots, u_3$ and v_k, that $v_k - v_{k+1} = Bp_{k+1}$, and that the k maximising u_k is the smallest k with $F(k) \geq A/B$. For $N = 400$, $k = 3$, $\epsilon = 0.01$, the binomial and Poisson probabilities of $r = 3$ no-shows are 0.194 and 0.195.

Common slips

- Including the $r \geq k$ terms in the penalty sum: they contribute nothing, since then everyone who turns up gets a seat.
- Differentiating u_k with respect to k. The variable is an integer, and the difference v_k is the right tool.
- Stating the answer as "$v_k = 0$". In general no k makes v_k exactly zero; the answer is the first k where $v_k \leq 0$.

Beyond STEP

The argument "the differences decrease, so stop where they change sign" is the discrete version of a concave function having its maximum where the derivative vanishes. The rule "stop when $\mathrm{P}(R \leq k)$ reaches A/B" is the classic *newsvendor* (critical fractile) rule from operations research, and airline overbooking is a standard application of it. The Poisson limit of the binomial, made rigorous above, is one of the first limit theorems in a university probability course. Grimmett and Welsh, *Probability: An Introduction*, covers both the Poisson approximation and expectations of functions like $\max(0, k - R)$.

WORKED SOLUTION

STEP II 1995, Question 14

Probability and Statistics • Hypothesis Testing

STEP II 1995, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q14

The key idea

Write the density as $x^2\phi(x)$, where ϕ is the standard normal density. Then every integral we need becomes a moment of a standard normal Z, which is where the hint about $\mathrm{E}(Z^4)$ comes in. To decide whether data "shakes our faith", we ask how probable a result at least this extreme would be if our hypothesis ($\mu = 0$) were true. If that probability is tiny, and a shift in μ would make the data unremarkable, the data count against us.

Solution

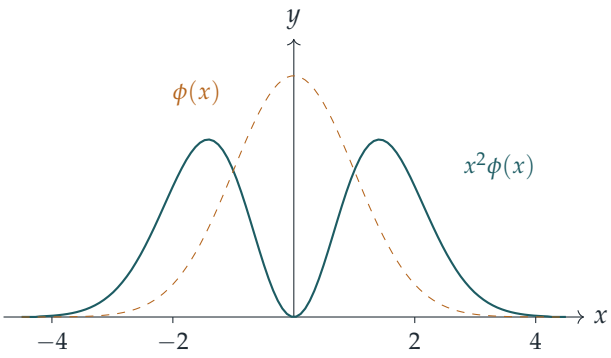
Finding A

Integrating by parts with $u = x$ and $\mathrm{d}v = x\mathrm{e}^{-x^2/2}\,\mathrm{d}x$, so $v = -\mathrm{e}^{-x^2/2}$:

$$\int_{-\infty}^{\infty} x^2\mathrm{e}^{-x^2/2}\,\mathrm{d}x = \left[-x\mathrm{e}^{-x^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\,\mathrm{d}x = 0 + \sqrt{2\pi},$$

using the standard normal integral. The total probability must be 1, so

$$A = \frac{1}{\sqrt{2\pi}}, \qquad \text{that is,} \qquad \mathrm{f}(x) = x^2\phi(x), \quad \phi(x) = \tfrac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2}.$$



The density is symmetric with two humps at $x = \pm\sqrt{2}$ and is zero at $x = 0$. By symmetry $\mathrm{E}(X) = 0$, and

$$\mathrm{E}(X^2) = \int_{-\infty}^{\infty} x^4\phi(x)\,\mathrm{d}x = \mathrm{E}(Z^4) = 3, \qquad \text{so} \qquad \mathrm{Var}(X) = 3.$$

Part (i)

Under our hypothesis, how likely is a result as far out as $a = 87.3$? Integrating by parts as before (now $v = -\phi(x)$, since $\phi'(x) = -x\phi(x)$),

$$\mathrm{P}(X \geq a) = \int_a^{\infty} x \cdot x\phi(x)\,\mathrm{d}x = a\phi(a) + \mathrm{P}(Z \geq a).$$

For $a > 0$ we also have $\mathrm{P}(Z \geq a) = \int_a^{\infty} \phi(x)\,\mathrm{d}x \leq \int_a^{\infty} \frac{x}{a}\phi(x)\,\mathrm{d}x = \frac{\phi(a)}{a}$, since $x/a \geq 1$ on the range. Hence, by symmetry,

$$\mathrm{P}(|X| \geq 87.3) \leq 2\left(a + \frac{1}{a}\right)\phi(a) \approx 70\,\mathrm{e}^{-3810.6} \approx 10^{-1653}.$$

This is absurdly small: if our hypothesis were true, we should essentially never see such a value. On the other hand, if μ were around 85 or 89 (so that 87.3 sits on one of the humps of $\mathrm{f}(x - \mu)$), a result of 87.3 would be entirely ordinary. So a single result of 87.3 should shake our faith completely. The only escape is to doubt the result itself (a recording error, say), and a sensible scientist would check that before abandoning the hypothesis.

Ask yourself Isn't it unfair to judge from one observation, when every particular value has probability zero?

The point is not the probability of the exact value 87.3, but of a value *at least this far* from where our hypothesis puts the mass. Equally, the density at 87.3 under our hypothesis is roughly 10^{-1650} times the density there under a suitable alternative. Either way, one observation is ample when it is this extreme.

Part (ii)

Let $X_1, \ldots, X_{1000}$ be the independent results and $\bar{X}$ their mean. Under our hypothesis each X_i has mean 0 and variance 3, so

$$\mathrm{E}(\bar{X}) = 0, \qquad \mathrm{Var}(\bar{X}) = \frac{3}{1000}, \qquad \mathrm{sd}(\bar{X}) = \sqrt{0.003} \approx 0.0548.$$

The variance is finite, and 1000 is large, so by the central limit theorem $\bar{X}$ is approximately $\mathrm{N}(0, 0.003)$, even though f itself looks nothing like a normal density. The observed mean is

$$\frac{0.23}{0.0548} \approx 4.2$$

standard deviations from 0, and

$$\mathrm{P}(|\bar{X}| \geq 0.23) \approx \mathrm{P}(|Z| \geq 4.2) \approx 3 \times 10^{-5}.$$

So if our hypothesis were true, a sample mean this far from 0 would turn up about three times in a hundred thousand repetitions of the whole experiment. Under the alternative with $\mu \approx 0.23$, the same data would be completely typical. This should also shake our faith seriously, though less overwhelmingly than in (i). We would reject $\mu = 0$ at any conventional significance level, e.g. 1% or even 0.1%.

Ask yourself Could we avoid the central limit theorem by using Chebyshev's inequality?

Chebyshev's inequality gives $\mathrm{P}(|\bar{X}| \geq 0.23) \leq 0.003/0.23^2 \approx 0.057$ without any approximation. That is suggestive but not decisive, since results this extreme would then be ruled out only to about 6%. The central limit theorem gives the much sharper figure, and with 1000 observations of a distribution whose tails decay like $\mathrm{e}^{-x^2/2}$, the approximation can be trusted.

Checks

- Simpson's rule gives $\int x^2 \mathrm{e}^{-x^2/2}\,\mathrm{d}x = 2.506628\ldots = \sqrt{2\pi}$, and $\int x^2 \mathrm{f}(x)\,\mathrm{d}x = 3.000$.
- The tail formula $\mathrm{P}(|X| \geq a) = 2(a\phi(a) + \mathrm{P}(Z \geq a))$ agrees with numerical integration at $a = 3$ (both 0.029291).
- $|X|$ has the same density as the length of a vector of three independent $\mathrm{N}(0,1)$ variables. Simulating X this way and averaging 1000 values 4000 times gave about 4.9% of means beyond 2 sd, close to the normal prediction of 4.6%, and none beyond 0.23.

Common slips

- Taking $\mathrm{Var}(X) = 1$ because the exponential looks like a standard normal. The extra x^2 pushes mass outwards, giving variance 3.

- In (ii), applying a normal test to a single X rather than to $\bar{X}$, or forgetting to divide the variance by 1000.
- In (i), arguing that "one observation proves nothing". With an outcome this extreme, it does.

Beyond STEP

This is hypothesis testing seen from first principles. A first-year statistics course formalises it with p-values, likelihood ratios (the Neyman–Pearson lemma) and the central limit theorem. Incidentally, $|X|$ has the chi distribution with 3 degrees of freedom, which is the Maxwell–Boltzmann distribution of molecular speeds in a gas. Grimmett and Welsh, *Probability: An Introduction*, covers the limit theorems used here.

WORKED SOLUTION

STEP III 1995, Question 1

Pure • Simultaneous Linear Equations

STEP III 1995, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q1

The key idea

The second and third equations have the same y and z terms, so subtracting one from the other removes two unknowns at once and leaves a single equation for x. Most of the question is about the values of a and b for which that step, or a later division, breaks down. The determinant of the coefficients tells us in advance where those values are, so we compute it first and then treat each exceptional case on its own.

Solution

Label the equations

$$a^2x + ay + z = a^2 \tag{1}$$
$$ax + y + bz = 1 \tag{2}$$
$$a^2bx + y + bz = b. \tag{3}$$

Where can things go wrong?

Expanding the determinant of the coefficient matrix along the first row,

$$\begin{vmatrix} a^2 & a & 1 \\ a & 1 & b \\ a^2b & 1 & b \end{vmatrix} = a^2(b-b) - a(ab - a^2b^2) + (a - a^2b)$$
$$= a^3b^2 - 2a^2b + a = a(ab-1)^2.$$

So there is exactly one solution unless $a = 0$ or $ab = 1$. In those cases there are either no solutions or infinitely many, and we have to look at the equations themselves to decide which.

The general case: $a \neq 0$ and $ab \neq 1$

Subtracting (2) from (3) gets rid of y and z:

$$(a^2b - a)x = b - 1, \qquad \text{that is,} \qquad a(ab-1)\,x = b - 1.$$

Since $a \neq 0$ and $ab \neq 1$ we may divide, giving

$$x = \frac{1-b}{a(1-ab)}.$$

Next, (1) gives $z = a^2 - a^2x - ay$. Substituting this into (2):

$$ax + y + a^2b - a^2bx - aby = 1 \quad \Longrightarrow \quad (1-ab)\,y = (1-a^2b) - ax(1-ab).$$

Dividing by $1 - ab \neq 0$ and using $ax = \dfrac{1-b}{1-ab}$,

$$y = \frac{1-a^2b}{1-ab} - \frac{1-b}{1-ab} = \frac{b-a^2b}{1-ab} = \frac{b(1-a^2)}{1-ab}.$$

Finally,

$$\begin{aligned} z = a^2 - a(ax) - ay &= a^2 - \frac{a(1-b)}{1-ab} - \frac{ab(1-a^2)}{1-ab} \\ &= \frac{a^2 - a^3b - a + ab - ab + a^3b}{1-ab} = \frac{a(a-1)}{1-ab}. \end{aligned}$$

So when $a \neq 0$ and $ab \neq 1$ the unique solution is

$$\boxed{x = \frac{1-b}{a(1-ab)}, \qquad y = \frac{b(1-a^2)}{1-ab}, \qquad z = \frac{a(a-1)}{1-ab}.}$$

The case $a = 0$

The equations become

$$z = 0, \qquad y + bz = 1, \qquad y + bz = b,$$

so $z = 0$, $y = 1$, and the last equation then says $1 = b$. Hence:

- if $a = 0$ and $b \neq 1$, there are **no solutions**;
- if $a = 0$ and $b = 1$, the solutions are $x = \lambda$, $y = 1$, $z = 0$ for any real λ: a line of solutions, since x no longer appears in any equation.

The case $ab = 1$ (so $a \neq 0$)

> **Ask yourself** When $ab = 1$, the equation $a(ab-1)x = b - 1$ has left-hand side 0. What does that tell us?

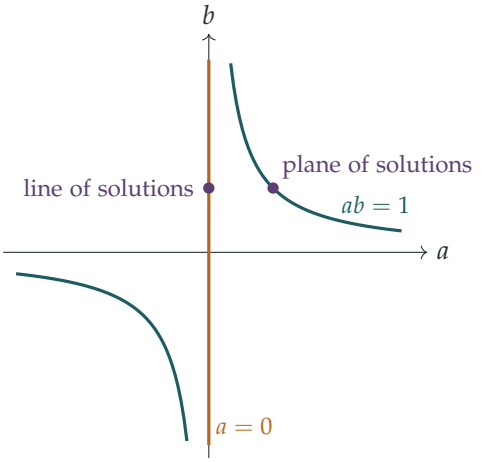
Subtracting (2) from (3) is still valid; it now reads $0 = b - 1$. So if $b \neq 1$ the equations are inconsistent. With $ab = 1$, the condition $b = 1$ forces $a = 1$. Hence:

- if $ab = 1$ and $a \neq 1$, there are **no solutions**;
- if $a = b = 1$, all three equations become $x + y + z = 1$, so the solutions are $x = \lambda$, $y = \mu$, $z = 1 - \lambda - \mu$ for any real λ, μ: a whole plane of solutions.

This fits with the squared factor $(ab-1)^2$ in the determinant. At $a = b = 1$ the matrix drops all the way to rank 1, so there are two free parameters, while at $a = 0$, $b = 1$ it has rank 2 and there is only one.

Summary

The picture below shows the (a, b)-plane. Away from the line $a = 0$ and the hyperbola $ab = 1$ there is a unique solution. On them there are no solutions, except at the two marked points.



- $a \neq 0$, $ab \neq 1$: the unique solution boxed above.
- $a = 0$, $b = 1$: $(x, y, z) = (\lambda, 1, 0)$, $\lambda \in \mathbb{R}$.
- $a = b = 1$: $(x, y, z) = (\lambda, \mu, 1 - \lambda - \mu)$, $\lambda, \mu \in \mathbb{R}$.
- $a = 0$ with $b \neq 1$, or $ab = 1$ with $a \neq 1$: no solutions.

Checks

Take $a = 2$, $b = 3$. The formulae give $x = \frac{1}{5}$, $y = \frac{9}{5}$, $z = -\frac{2}{5}$, and indeed $4 \cdot \frac{1}{5} + 2 \cdot \frac{9}{5} - \frac{2}{5} = 4 = a^2$, $\frac{2}{5} + \frac{9}{5} - \frac{6}{5} = 1$ and $\frac{12}{5} + \frac{9}{5} - \frac{6}{5} = 3 = b$. Another quick test is $b = 1$, $a \neq 0, 1$: the formulae give $x = 0$, $y = 1 + a$, $z = -a$, and these satisfy all three equations. We also checked the determinant and the general formulae with exact rational arithmetic for 2000 random pairs (a, b).

Common slips

- Dividing by $a(ab-1)$ without first setting aside $a=0$ and $ab=1$. "For all possible values" means the special cases are the point of the question.
- Saying "determinant zero, so no solutions". A zero determinant only rules out a *unique* solution; here two of the special parameter values give infinitely many.
- Missing that $ab=1$ and $b=1$ together force $a=1$, so there is just one special point on the hyperbola.

Beyond STEP

This is a small example of how the rank of a matrix controls a linear system. The system is consistent exactly when the coefficient matrix and the augmented matrix have the same rank, and then the solutions form a set of dimension $3-\text{rank}$. That is why we got a single point, a line or a plane. Gaussian elimination makes all of this systematic, and Strang's *Introduction to Linear Algebra* develops it with many worked examples.

WORKED SOLUTION

STEP III 1995, Question 2

Pure • Reduction Formulae

STEP III 1995, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q2

The key idea

The factors $\sqrt{x}$ and $\sqrt{a-x}$ both suggest the substitution $x = a\sin^2\theta$, which turns them into $\sqrt{a}\sin\theta$ and $\sqrt{a}\cos\theta$ at the same time. (There is also a neat geometric shortcut for I_0.) For the reduction formula, integrate by parts so that the power of x drops by one. The power of $a-x$ then goes up by one, and we split it off using $a-x$ again. This recovers I_n itself, so we can collect terms.

Throughout, $a > 0$.

Solution

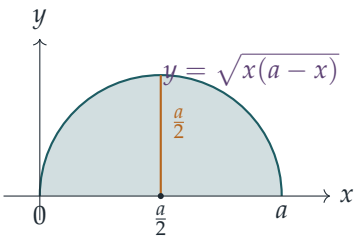
Finding I_0

We need $\mathrm{I}_0 = \int_0^a x^{1/2}(a-x)^{1/2}\,\mathrm{d}x$. Put $x = a\sin^2\theta$, so $\mathrm{d}x = 2a\sin\theta\cos\theta\,\mathrm{d}\theta$. As x runs from 0 to a, θ runs from 0 to $\frac{\pi}{2}$, and on this range $\sin\theta, \cos\theta \geq 0$. So $\sqrt{x} = \sqrt{a}\sin\theta$ and $\sqrt{a-x} = \sqrt{a}\cos\theta$, with no sign trouble. Then

$$\begin{aligned}\mathrm{I}_0 &= \int_0^{\pi/2} (\sqrt{a}\sin\theta)(\sqrt{a}\cos\theta)\,2a\sin\theta\cos\theta\,\mathrm{d}\theta = 2a^2\int_0^{\pi/2}\sin^2\theta\cos^2\theta\,\mathrm{d}\theta \\ &= \frac{a^2}{2}\int_0^{\pi/2}\sin^2 2\theta\,\mathrm{d}\theta = \frac{a^2}{4}\int_0^{\pi/2}(1-\cos 4\theta)\,\mathrm{d}\theta = \frac{a^2}{4}\cdot\frac{\pi}{2} = \frac{\pi a^2}{8},\end{aligned}$$

as required. Here we used $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$ and $\sin^2 2\theta = \frac{1}{2}(1-\cos 4\theta)$, and $\cos 4\theta$ integrates to zero over $[0, \frac{\pi}{2}]$.

A picture. If $y = \sqrt{x(a-x)}$, then $y^2 = ax - x^2$, which rearranges to $\left(x - \frac{a}{2}\right)^2 + y^2 = \left(\frac{a}{2}\right)^2$ with $y \geq 0$. So I_0 is the area of a semicircle of radius $\frac{a}{2}$, which is $\frac{1}{2}\pi\left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{8}$.



The reduction formula

Take $n \geq 1$. Integrate by parts with $u = x^{n+\frac{1}{2}}$ and $\mathrm{d}v = (a-x)^{1/2}\,\mathrm{d}x$. Then $\mathrm{d}u = \left(n+\frac{1}{2}\right)x^{n-\frac{1}{2}}\,\mathrm{d}x$ and $v = -\frac{2}{3}(a-x)^{3/2}$. This gives

$$\mathrm{I}_n = \left[-\tfrac{2}{3}x^{n+\frac{1}{2}}(a-x)^{3/2}\right]_0^a + \frac{2}{3}\left(n+\frac{1}{2}\right)\int_0^a x^{n-\frac{1}{2}}(a-x)^{3/2}\,\mathrm{d}x.$$

The boundary term vanishes at both ends: at $x = a$ because of the factor $(a-x)^{3/2}$, and at $x = 0$ because of $x^{n+\frac{1}{2}}$. Note that $\frac{2}{3}\left(n+\frac{1}{2}\right) = \frac{2n+1}{3}$.

> **Ask yourself** The new integral has $(a-x)^{3/2}$ in it, which is not of the form I_{n-1}. What do we do with it?

Write $(a-x)^{3/2} = (a-x)\,(a-x)^{1/2}$ and expand the first factor:

$$\int_0^a x^{n-\frac{1}{2}}(a-x)^{3/2}\,\mathrm{d}x = a\int_0^a x^{n-\frac{1}{2}}(a-x)^{1/2}\,\mathrm{d}x - \int_0^a x^{n+\frac{1}{2}}(a-x)^{1/2}\,\mathrm{d}x = a\,\mathrm{I}_{n-1} - \mathrm{I}_n.$$

So

$$\mathrm{I}_n = \frac{2n+1}{3}\left(a\,\mathrm{I}_{n-1} - \mathrm{I}_n\right) \quad\Longrightarrow\quad (2n+4)\,\mathrm{I}_n = (2n+1)\,a\,\mathrm{I}_{n-1},$$

after multiplying by 3 and collecting the I_n terms ($3 + 2n + 1 = 2n + 4$). This is the required result.

Evaluating I_n

Applying $\mathrm{I}_k = \dfrac{(2k+1)a}{2k+4}\,\mathrm{I}_{k-1}$ for $k = n, n-1, \ldots, 1$ and ending at I_0 gives

$$\mathrm{I}_n = a^n \,\frac{\prod_{k=1}^{n}(2k+1)}{\prod_{k=1}^{n}(2k+4)} \cdot \frac{\pi a^2}{8}.$$

Now we tidy the two products. The top is the product of the odd numbers $3 \cdot 5 \cdots (2n+1)$. We put in the missing even numbers $2 \cdot 4 \cdots (2n) = 2^n n!$ to get

$$\prod_{k=1}^{n}(2k+1) = \frac{(2n+1)!}{2^n\, n!}.$$

The bottom is $\prod_{k=1}^{n} 2(k+2) = 2^n \cdot 3 \cdot 4 \cdots (n+2) = 2^n \dfrac{(n+2)!}{2}$. Therefore

$$\mathrm{I}_n = \frac{\pi a^{n+2}}{8} \cdot \frac{(2n+1)!}{2^n\, n!} \cdot \frac{2}{2^n\,(n+2)!} = \boxed{\frac{\pi\, a^{n+2}\,(2n+1)!}{4^{n+1}\, n!\,(n+2)!}}$$

This also gives the correct value $\frac{\pi a^2}{8}$ when $n = 0$.

Checks

- $n = 1$: the recurrence gives $6\mathrm{I}_1 = 3a\mathrm{I}_0$, so $\mathrm{I}_1 = \frac{\pi a^3}{16}$. The formula gives $\pi a^3 \cdot 3!/(16 \cdot 1! \cdot 3!) = \frac{\pi a^3}{16}$, which agrees.
- Symmetry: replacing x by $a - x$ shows $\mathrm{I}_1 = \int_0^a x^{1/2}(a-x)^{3/2}\,\mathrm{d}x$ too. Adding the two expressions gives $2\mathrm{I}_1 = a\mathrm{I}_0$, which confirms $\mathrm{I}_1 = \frac{\pi a^3}{16}$ independently.
- Numerically with $a = 1$, Simpson or midpoint quadrature gives $\mathrm{I}_0 \approx 0.392699 = \pi/8$ and $\mathrm{I}_3 \approx 0.085903$. The formula gives $\pi \cdot 7!/(4^4 \cdot 3! \cdot 5!) = 35\pi/1280 \approx 0.085903$.

Common slips

- Getting the sign of v wrong: $\int (a-x)^{1/2}\,\mathrm{d}x = -\frac{2}{3}(a-x)^{3/2}$, with a minus sign from the chain rule.
- Trying to reach I_{n-1} directly instead of noticing that I_n reappears on the right and must be collected.
- Dropping the factor of 2 from $3 \cdot 4 \cdots (n+2) = (n+2)!/2$.

Beyond STEP

These integrals are special cases of the Beta function $B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,\mathrm{d}t$. Substituting $x = at$ shows $\mathrm{I}_n = a^{n+2}B\left(n+\frac{3}{2}, \frac{3}{2}\right)$. The recurrence we proved is one case of the identity $B(p+1,q) = \frac{p}{p+q}B(p,q)$. In a first-year course you meet the formula $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$, which explains where the π comes from: $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Siklos, *Advanced Problems in Mathematics*, has more reduction-formula questions in this style.

WORKED SOLUTION

STEP III 1995, Question 3

Pure • Damped Harmonic Motion

STEP III 1995, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q3

The key idea

Try $x = \mathrm{e}^{\lambda t}$: the differential equation becomes a quadratic in λ, and the three cases are just the three possibilities for its discriminant (two real roots, a repeated root, complex roots). For the damped case, the stationary points come round at equal time intervals of half a period, and over each such interval the oscillating factor simply flips sign while the exponential factor shrinks by the same amount. That is why the ratio of successive extremes is constant.

Solution

Substituting $x = \mathrm{e}^{\lambda t}$ gives the auxiliary equation

$$\lambda^2 + 2k\lambda + 1 = 0, \qquad \lambda = -k \pm \sqrt{k^2 - 1}.$$

Throughout, A and B are arbitrary constants.

Part (i): $k > 1$

The roots are real and distinct, so

$$x = A\,\mathrm{e}^{(-k+\sqrt{k^2-1})\,t} + B\,\mathrm{e}^{(-k-\sqrt{k^2-1})\,t}.$$

(Both exponents are negative, since $\sqrt{k^2-1} < k$: this is *overdamping*, with no oscillation.)

Part (ii): $k = 1$

The auxiliary equation is $(\lambda+1)^2 = 0$, a repeated root $\lambda = -1$. The second independent solution is $t\,\mathrm{e}^{-t}$, so

$$x = (A + Bt)\,\mathrm{e}^{-t}.$$

(Check: if $x = te^{-t}$ then $x' = (1-t)e^{-t}$ and $x'' = (t-2)e^{-t}$, and $(t-2) + 2(1-t) + t = 0$.)

Part (iii): $0 < k < 1$

Now $k^2 - 1 < 0$, so the roots are $\lambda = -k \pm \mathrm{i}\omega$ where

$$\omega = \sqrt{1 - k^2} > 0.$$

Taking real combinations of $\mathrm{e}^{(-k \pm \mathrm{i}\omega)t}$,

$$x = \mathrm{e}^{-kt}\big(A\cos\omega t + B\sin\omega t\big).$$

The successive extremes

With $x(0) = 0$ we need $A = 0$, so $x = B\,\mathrm{e}^{-kt}\sin\omega t$, and $B \neq 0$ (otherwise $x \equiv 0$ and there are no maxima or minima). Differentiating,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = B\,\mathrm{e}^{-kt}\big(\omega\cos\omega t - k\sin\omega t\big).$$

Since $k^2 + \omega^2 = 1$, we can write $k = \sin\varphi$, $\omega = \cos\varphi$ with $0 < \varphi < \frac{\pi}{2}$, and then

$$\omega\cos\omega t - k\sin\omega t = \cos(\omega t + \varphi).$$

So $\dfrac{\mathrm{d}x}{\mathrm{d}t} = B\,\mathrm{e}^{-kt}\cos(\omega t + \varphi)$, which is zero exactly when $\omega t + \varphi$ is an odd multiple of $\frac{\pi}{2}$, and changes sign at each such time. The stationary points (for $t > 0$) are therefore at

$$t_n = t_1 + \frac{(n-1)\pi}{\omega}, \qquad n = 1, 2, 3, \ldots,$$

equally spaced π/ω apart, and since $\mathrm{d}x/\mathrm{d}t$ changes sign at each one they really are alternately maxima and minima.

Ask yourself How do we compare $x(t_{n+1})$ with $x(t_n)$ without finding t_1 explicitly?

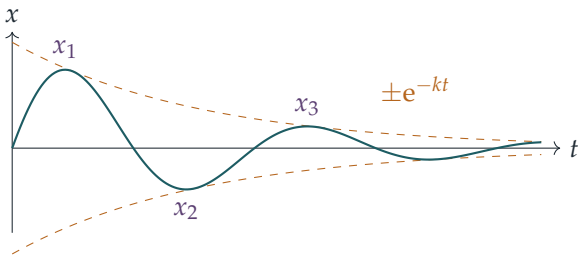
Moving on by π/ω in time adds π to ωt, and $\sin(\theta + \pi) = -\sin\theta$. So

$$x_{n+1} = B\,\mathrm{e}^{-k(t_n + \pi/\omega)}\sin(\omega t_n + \pi) = -\,\mathrm{e}^{-k\pi/\omega}\,B\,\mathrm{e}^{-kt_n}\sin\omega t_n = -\,\mathrm{e}^{-k\pi/\omega}\,x_n.$$

Also $x_n \neq 0$, because $\sin\omega t_n = 0$ would force $\cos\omega t_n = \pm 1$ and then $\mathrm{d}x/\mathrm{d}t = \pm B\omega\,\mathrm{e}^{-kt_n} \neq 0$. Hence

$$\left|\frac{x_{n+1}}{x_n}\right| = \alpha, \qquad \text{where } \alpha = \mathrm{e}^{-k\pi/\omega} = \exp\left(-\frac{k\pi}{\sqrt{1-k^2}}\right),$$

which does not depend on n, as required. (The minus sign confirms the alternation between maxima and minima.)



The formula for k^2

Taking logarithms, $\ln\alpha = -\dfrac{k\pi}{\sqrt{1-k^2}}$, so

$$(\ln\alpha)^2 = \frac{k^2\pi^2}{1-k^2} \implies (\ln\alpha)^2 - k^2(\ln\alpha)^2 = k^2\pi^2 \implies k^2 = \frac{(\ln\alpha)^2}{\pi^2 + (\ln\alpha)^2},$$

which is the required result.

Checks

- As $k \to 0$ (no damping), $\alpha \to 1$ and $\ln\alpha \to 0$, so the formula gives $k^2 \to 0$, as it should: the extremes all have the same size.
- As $k \to 1^-$, $\alpha \to 0$ and $(\ln\alpha)^2 \to \infty$, so $k^2 \to 1$: at critical damping the motion no longer swings back.
- Numerically, with $k = 0.5$ and $x(0) = 0$, stepping the equation forward gives successive extreme ratios 0.163034, matching $\mathrm{e}^{-0.5\pi/\sqrt{0.75}} \approx 0.163034$; and then $(\ln\alpha)^2/(\pi^2 + (\ln\alpha)^2) = 0.25 = k^2$.

Common slips

- Taking the time between a maximum and the next minimum to be a full period $2\pi/\omega$ rather than half a period π/ω; this gives α^2 instead of α.
- Thinking the extremes occur where $\sin\omega t = \pm 1$. They don't: the decaying factor shifts them earlier, to where $\tan\omega t = \omega/k$. The argument above never needs their exact position, only their spacing.

Beyond STEP

The quantity $-\ln\alpha$ is the *logarithmic decrement*: engineers measure it from a recorded oscillation, and this question's formula turns it into the damping k. Boyce and DiPrima, *Elementary Differential Equations*, treats over-, critical and under-damping in detail.

WORKED SOLUTION

STEP III 1995, Question 4

Pure • Trigonometric Series Summation

STEP III 1995, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q4

The key idea

A sum of cosines is the real part of a sum of complex exponentials, and $\sum e^{\mathrm{i}k\theta}$ is a geometric series. Summing it and then pulling out a "half-angle" factor $e^{\mathrm{i}\theta/2}$ from top and bottom turns the awkward $e^{\mathrm{i}m\theta}-1$ terms into sines, which is where the $\sin(\frac{1}{2}\theta)$ in the denominator comes from. For the inequality, the trick is to rewrite $\mathrm{C}_n(\theta)$ so that $\frac{1}{2}$ visibly splits off.

Solution

The sums C_n and S_n

Put $z = e^{\mathrm{i}\theta}$. Since $0 < \theta < 2\pi$, we have $z \neq 1$, so the geometric series formula applies:

$$\mathrm{C}_n(\theta) + \mathrm{i}\,\mathrm{S}_n(\theta) = \sum_{k=0}^{n} e^{\mathrm{i}k\theta} = \frac{e^{\mathrm{i}(n+1)\theta} - 1}{e^{\mathrm{i}\theta} - 1}.$$

Now use $e^{\mathrm{i}\phi} - 1 = e^{\mathrm{i}\phi/2}\left(e^{\mathrm{i}\phi/2} - e^{-\mathrm{i}\phi/2}\right) = 2\mathrm{i}\,e^{\mathrm{i}\phi/2}\sin(\frac{1}{2}\phi)$ on the top (with $\phi = (n+1)\theta$) and on the bottom (with $\phi = \theta$):

$$\mathrm{C}_n(\theta) + \mathrm{i}\,\mathrm{S}_n(\theta) = \frac{2\mathrm{i}\,e^{\mathrm{i}(n+1)\theta/2}\sin\left(\frac{1}{2}(n+1)\theta\right)}{2\mathrm{i}\,e^{\mathrm{i}\theta/2}\sin(\frac{1}{2}\theta)} = e^{\mathrm{i}n\theta/2}\,\frac{\sin\left(\frac{1}{2}(n+1)\theta\right)}{\sin(\frac{1}{2}\theta)}.$$

The denominator is non-zero because $0 < \frac{1}{2}\theta < \pi$. The fraction is real, so taking real and imaginary parts (with $e^{\mathrm{i}n\theta/2} = \cos\frac{1}{2}n\theta + \mathrm{i}\sin\frac{1}{2}n\theta$) gives

$$\mathrm{C}_n(\theta) = \frac{\cos(\frac{1}{2}n\theta)\sin\left(\frac{1}{2}(n+1)\theta\right)}{\sin(\frac{1}{2}\theta)}, \qquad \mathrm{S}_n(\theta) = \frac{\sin(\frac{1}{2}n\theta)\sin\left(\frac{1}{2}(n+1)\theta\right)}{\sin(\frac{1}{2}\theta)}.$$

The first is the required result, and the second is the corresponding expression for S_n.

Ask yourself Is there a way to see this without complex numbers?

Yes: multiply each term by $2\sin(\frac{1}{2}\theta)$ and use the product-to-sum formula

$$2\sin(\tfrac{1}{2}\theta)\cos k\theta = \sin\big((k+\tfrac{1}{2})\theta\big) - \sin\big((k-\tfrac{1}{2})\theta\big).$$

Summing from $k=0$ to n, everything telescopes except the last and first terms:

$$2\sin(\tfrac{1}{2}\theta)\,\mathrm{C}_n(\theta) = \sin\big((n+\tfrac{1}{2})\theta\big) - \sin(-\tfrac{1}{2}\theta) = \sin\big((n+\tfrac{1}{2})\theta\big) + \sin(\tfrac{1}{2}\theta). \qquad (*)$$

By the sum-to-product formula $\sin A + \sin B = 2\sin\frac{1}{2}(A+B)\cos\frac{1}{2}(A-B)$, with $A = (n+\frac{1}{2})\theta$, $B = \frac{1}{2}\theta$, the right-hand side is $2\sin\big(\frac{1}{2}(n+1)\theta\big)\cos(\frac{1}{2}n\theta)$, which agrees with what we found above.

The inequality

Equation $(*)$ is exactly the form we want. Dividing by $2\sin(\frac{1}{2}\theta) \neq 0$,

$$\mathrm{C}_n(\theta) = \frac{1}{2} + \frac{\sin\big((n+\frac{1}{2})\theta\big)}{2\sin(\frac{1}{2}\theta)}.$$

(Starting from the given formula instead, use $2\cos(\frac{1}{2}n\theta)\sin\big(\frac{1}{2}(n+1)\theta\big) = \sin\big((n+\frac{1}{2})\theta\big) + \sin(\frac{1}{2}\theta)$ to get the same line.) Hence

$$\Big|\mathrm{C}_n(\theta) - \frac{1}{2}\Big| = \frac{\big|\sin\big((n+\frac{1}{2})\theta\big)\big|}{2\sin(\frac{1}{2}\theta)} \leq \frac{1}{2\sin(\frac{1}{2}\theta)},$$

where we have used that $\sin(\frac{1}{2}\theta) > 0$ for $0 < \theta < 2\pi$ (so no modulus is needed in the denominator) and that $|\sin x| \leq 1$. This is the required inequality, with equality exactly when $\sin\big((n+\frac{1}{2})\theta\big) = \pm 1$.

Ask yourself Why is the positivity of $\sin(\frac{1}{2}\theta)$ important?

If θ were allowed to range outside $(0, 2\pi)$, $\sin(\frac{1}{2}\theta)$ could be negative and the right-hand side of the inequality would be negative, which is impossible for a modulus. In the given range, $\frac{1}{2}\theta \in (0, \pi)$, so the bound is a genuine positive number.

Checks

- $n=1$: $\mathrm{C}_1 = 1+\cos\theta = 2\cos^2(\frac{1}{2}\theta)$, and the formula gives $\cos(\frac{1}{2}\theta)\sin\theta/\sin(\frac{1}{2}\theta) = 2\cos^2(\frac{1}{2}\theta)$. Similarly $\mathrm{S}_1 = \sin\theta$ matches $\sin(\frac{1}{2}\theta)\sin\theta/\sin(\frac{1}{2}\theta)$.
- As $\theta \to 0$, the formula for C_n tends to $(n+1)$, which is the sum of $n+1$ ones; the bound $1/(2\sin\frac{1}{2}\theta)$ blows up, as it must.
- Numerically, with $n=3$, $\theta=1$: the direct sum is $\mathrm{C}_3(1) \approx 0.1342$, matching the formula, and $|\mathrm{C}_3 - \frac{1}{2}| \approx 0.366 \leq 1/(2\sin\frac{1}{2}) \approx 1.043$. Equality occurs, for example, at $n=2$, $\theta = \pi/5$, where both sides equal $1.618\ldots$.

Common slips

- Forgetting that the sums start at $k = 0$, so there are $n + 1$ terms: the geometric series has $e^{\mathrm{i}(n+1)\theta}$ on top, not $e^{\mathrm{i}n\theta}$.
- Taking real parts of $\left(e^{\mathrm{i}(n+1)\theta} - 1\right) / \left(e^{\mathrm{i}\theta} - 1\right)$ without first making the denominator real; the half-angle factorisation does this cleanly.
- Writing $|\sin(\frac{1}{2}\theta)|$ and not saying why it equals $\sin(\frac{1}{2}\theta)$.

Beyond STEP

The function $\mathrm{C}_n(\theta) - \frac{1}{2} = \sin\left((n + \frac{1}{2})\theta\right) / \left(2\sin\frac{1}{2}\theta\right)$ is, up to a factor, the *Dirichlet kernel*, the central object in the theory of Fourier series: the nth partial Fourier sum of a function is its average against this kernel. The bound proved here, together with the oscillation of the numerator, is the starting point for proving when Fourier series converge (and for Dirichlet's test for series such as $\sum \cos(k\theta)/k$). Abbott's *Understanding Analysis* has a readable chapter on Fourier series that uses exactly these sums.

WORKED SOLUTION

STEP III 1995, Question 5

Pure • Maclaurin Series

STEP III 1995, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q5

The key idea

Write $\theta = \sin^{-1} x$, so that $\mathrm{d}\theta/\mathrm{d}x = 1/\sqrt{1-x^2}$. Multiplying $y^{(1)}$ by $\sqrt{1-x^2}$ cancels that awkward factor, and differentiating once more gives the second-order equation with no square roots in it. Leibniz's rule then turns the equation into a recurrence, and at $x = 0$ the factor $1 - x^2$ becomes 1 and the term in x disappears. The recurrence then hands us every derivative at 0.

Solution

The differential equation

Let $\theta = \sin^{-1} x$, so $\sin\theta = x$ and, for $|x| < 1$, $\cos\theta = \sqrt{1-x^2} > 0$, with $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{1}{\sqrt{1-x^2}}$.

Then $y = \sin^2 m\theta$, and by the chain rule

$$y^{(1)} = 2m \sin m\theta \cos m\theta \cdot \frac{1}{\sqrt{1-x^2}} = \frac{m \sin 2m\theta}{\sqrt{1-x^2}}, \qquad \text{so} \qquad \sqrt{1-x^2}\, y^{(1)} = m \sin 2m\theta.$$

Differentiate this with respect to x, using the product rule on the left:

$$\sqrt{1-x^2}\, y^{(2)} - \frac{x}{\sqrt{1-x^2}}\, y^{(1)} = 2m^2 \cos 2m\theta \cdot \frac{1}{\sqrt{1-x^2}}.$$

Multiply through by $\sqrt{1-x^2}$, and use $\cos 2m\theta = 1 - 2\sin^2 m\theta = 1 - 2y$:

$$(1-x^2)\, y^{(2)} = x\, y^{(1)} + 2m^2(1-2y),$$

which is the required equation.

Ask yourself Why not square $\sqrt{1-x^2}\, y^{(1)} = m \sin 2m\theta$ first?

Squaring gives $(1-x^2)\left(y^{(1)}\right)^2 = 4m^2 y(1-y)$, and differentiating that leaves a factor $2y^{(1)}$ on every term. We would then have to divide by $y^{(1)}$, which is zero at $x = 0$,

the very point we care about later. Differentiating before squaring avoids the division altogether.

The recurrence

Differentiate the equation n times, with $n \geq 1$, using Leibniz's rule. Since $1 - x^2$ has only two non-zero derivatives ($-2x$ and -2), and x has only one,

$$\begin{aligned}
\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[(1-x^2)y^{(2)}\right] &= (1-x^2)y^{(n+2)} - 2nx\,y^{(n+1)} - n(n-1)\,y^{(n)},\\
\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[x\,y^{(1)}\right] &= x\,y^{(n+1)} + n\,y^{(n)},\\
\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[2m^2(1-2y)\right] &= -4m^2\,y^{(n)}.
\end{aligned}$$

The last line needs $n \geq 1$, because only then does the constant $2m^2$ differentiate to zero. Equating the two sides,

$$(1-x^2)y^{(n+2)} - 2nx\,y^{(n+1)} - n(n-1)y^{(n)} = x\,y^{(n+1)} + n\,y^{(n)} - 4m^2y^{(n)},$$

and collecting terms ($2n+1$ lots of $xy^{(n+1)}$, and $n(n-1)+n = n^2$) gives

$$(1-x^2)\,y^{(n+2)} = (2n+1)\,x\,y^{(n+1)} + (n^2-4m^2)\,y^{(n)} \qquad (n \geq 1),$$

as required.

The Maclaurin series

At $x = 0$ we have $\theta = 0$, so

$$y(0) = \sin^2 0 = 0, \qquad y^{(1)}(0) = m\sin 0 = 0, \qquad y^{(2)}(0) = 0 + 2m^2(1-0) = 2m^2,$$

the last from the differential equation. Putting $x = 0$ in the recurrence gives

$$y^{(n+2)}(0) = (n^2-4m^2)\,y^{(n)}(0) \qquad (n \geq 1).$$

Odd derivatives. Since $y^{(1)}(0) = 0$, the recurrence gives $y^{(3)}(0) = 0$, then $y^{(5)}(0) = 0$, and so on: every odd derivative vanishes at 0. That makes sense, because y is an even function of x.

Even derivatives. Take $n = 2j$:

$$y^{(2j+2)}(0) = (4j^2-4m^2)\,y^{(2j)}(0) = -4(m^2-j^2)\,y^{(2j)}(0) \qquad (j \geq 1).$$

Starting from $y^{(2)}(0) = 2m^2$ and applying this for $j = 1, 2, \ldots, k-1$,

$$y^{(2k)}(0) = 2m^2\prod_{j=1}^{k-1}\bigl(-4(m^2-j^2)\bigr) = (-1)^{k-1}\,2^{2k-1}\prod_{j=0}^{k-1}(m^2-j^2),$$

where we absorbed the leading m^2 as the $j = 0$ factor. (For $k = 1$ the formula gives $2m^2$, as it should.)

Hence the Maclaurin series is

$$y = \sin^2\!\left(m \sin^{-1} x\right) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}\, 2^{2k-1}}{(2k)!}\, m^2(m^2-1^2)(m^2-2^2)\cdots\left(m^2-(k-1)^2\right) x^{2k},$$

and its first few terms are

$$y = m^2x^2 - \frac{m^2(m^2-1)}{3}\, x^4 + \frac{2m^2(m^2-1)(m^2-4)}{45}\, x^6 - \cdots .$$

If m is an integer, every term with $k > |m|$ contains the factor $m^2 - m^2 = 0$, so the series stops and y is a polynomial of degree $2|m|$.

Checks

- $m = 1$: $y = \sin^2(\sin^{-1} x) = x^2$, and the series gives x^2 and then zeros. $m = 2$: the series gives $4x^2 - 4x^4$, and indeed $\sin^2 2\theta = 4\sin^2\theta(1 - \sin^2\theta) = 4x^2 - 4x^4$.
- $m = \frac{1}{2}$, $x = \frac{1}{2}$: $y = \sin^2(\pi/12) = \frac{1}{2}\left(1 - \frac{\sqrt{3}}{2}\right) \approx 0.066987$. The series is $\frac{1}{4}x^2 + \frac{1}{16}x^4 + \frac{1}{32}x^6 + \cdots$, whose first three terms give 0.066895, and adding more terms closes the gap.
- We checked numerically that the differential equation holds at random x and m, that the series coefficients satisfy it exactly, and that the series (to 300 terms) agrees with y to about 10^{-15} for $|x| \le 0.8$.

Common slips

- Using the recurrence with $n = 0$. It holds only for $n \ge 1$, because the constant $2m^2$ survives when $n = 0$. The case $n = 0$ is the original equation, which is how we got $y^{(2)}(0) = 2m^2$.
- Getting the Leibniz coefficients wrong: the term $-n(n-1)y^{(n)}$ comes from $\binom{n}{2} \cdot (-2)$, not from $\binom{n}{2} \cdot (-2x)$.
- Writing the general term without the alternating sign. Each step multiplies by $4(j^2 - m^2)$, which is negative when $m^2 > j^2$.

Beyond STEP

A Maclaurin series built from derivatives might not equal the function it came from, but here it does for $|x| < 1$. The series has radius of convergence 1 (by the ratio test, when m is not an integer), and $\sin^{-1} x$ stops being well behaved at $x = \pm 1$. The polynomials you get for integer m are relatives of the Chebyshev polynomials, since $\cos 2m\theta$ is a polynomial in $\sin\theta$. To see why such series really do converge to their functions, read about Taylor's theorem with remainder in Abbott, *Understanding Analysis*. Needham's *Visual Complex Analysis* explains why the radius of convergence is set by the trouble at ± 1.

WORKED SOLUTION

STEP III 1995, Question 6

Pure • Loci and Inversion

STEP III 1995, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q6

The key idea

The circle $|z - \mathrm{i}| = 1$ passes through the origin, so in polar form its equation is especially simple: the modulus is a single trigonometric function of the argument. Once we have z in modulus–argument form, $1/z$ is immediate (take the reciprocal of the modulus and negate the argument), and writing it in Cartesian form reveals that the circle has become a straight line. The final part is the same phenomenon run in reverse, and it is quickest to do it directly with $w = x + \mathrm{i}y$.

Solution

The modulus of z

Write $z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r > 0$. Then $z - \mathrm{i} = r\cos\theta + \mathrm{i}(r\sin\theta - 1)$, so

$$\begin{aligned} |z - \mathrm{i}|^2 = 1 &\iff r^2\cos^2\theta + r^2\sin^2\theta - 2r\sin\theta + 1 = 1 \\ &\iff r^2 - 2r\sin\theta = 0 \\ &\iff r = 2\sin\theta, \end{aligned}$$

where in the last step we divided by r, which is allowed because $z \neq 0$. So

$$|z| = 2\sin\theta.$$

Since $r > 0$, this forces $\sin\theta > 0$, so $0 < \theta < \pi$: the circle (centre i, radius 1) lies in the upper half-plane and touches the real axis only at the origin, which is excluded. Conversely, every $\theta \in (0, \pi)$ gives a point $z = 2\sin\theta\,(\cos\theta + \mathrm{i}\sin\theta)$ of the circle, so as θ runs through $(0, \pi)$, z traces the whole circle except the origin.

? **Ask yourself** Why is $|z| = 2\sin\theta$ "obvious" from a picture?

The segment from 0 to 2i is a diameter of the circle. By the angle in a semicircle, the triangle with vertices 0, z, 2i has a right angle at z, and the angle at 0 is $\frac{\pi}{2} - \theta$. So $|z| = 2\cos(\frac{\pi}{2} - \theta) = 2\sin\theta$.

The modulus and argument of $1/z$

Since $z = 2\sin\theta\,(\cos\theta + \mathrm{i}\sin\theta)$,

$$\frac{1}{z} = \frac{1}{2\sin\theta}\big(\cos(-\theta) + \mathrm{i}\sin(-\theta)\big),$$

so

$$\left|\frac{1}{z}\right| = \frac{1}{2\sin\theta}, \qquad \arg\frac{1}{z} = -\theta.$$

Because $0 < \theta < \pi$, the angle $-\theta$ lies in $(-\pi, 0)$, so it is the principal argument.

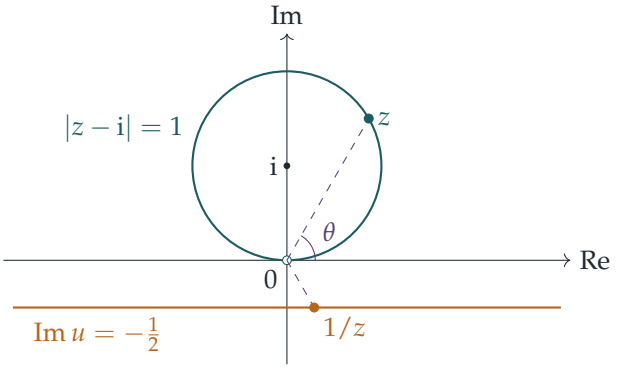
The loci

To see what $1/z$ traces, convert to Cartesian form:

$$\frac{1}{z} = \frac{\cos\theta - \mathrm{i}\sin\theta}{2\sin\theta} = \tfrac{1}{2}\cot\theta - \tfrac{1}{2}\mathrm{i}.$$

The imaginary part is always $-\frac{1}{2}$, and as θ runs through $(0, \pi)$ the real part $\frac{1}{2}\cot\theta$ takes every real value exactly once (it decreases from $+\infty$ to $-\infty$). So $1/z$ traces the *whole* straight line $\operatorname{Im} u = -\frac{1}{2}$.

So z lies on the circle with centre i and radius 1 (minus the origin), and $1/z$ lies on the horizontal line through $-\frac{1}{2}\mathrm{i}$. The diagram shows both, with the pair of points for $\theta = \frac{\pi}{3}$: here $z = \frac{\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}$ and $1/z = \frac{1}{2\sqrt{3}} - \frac{1}{2}\mathrm{i}$. The two points lie on rays at angles θ and $-\theta$, and their distances from 0 multiply to 1. As z approaches the origin, $1/z$ runs off to infinity along the line; the top of the circle, $z = 2\mathrm{i}$, goes to the point $-\frac{1}{2}\mathrm{i}$, the nearest point of the line to the origin.



The locus C

Let $w = x + \mathrm{i}y \neq 0$ (we need $w \neq 0$ for $1/w$ to exist). Then

$$\frac{1}{w} = \frac{\bar{w}}{|w|^2} = \frac{x - \mathrm{i}y}{x^2 + y^2}, \qquad \text{so} \qquad \operatorname{Re}\frac{1}{w} = \frac{x}{x^2 + y^2}.$$

Hence, for $w \neq 0$,

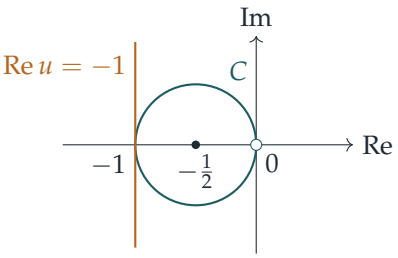
$$\begin{aligned} \operatorname{Re}\frac{1}{w} = -1 &\iff x = -(x^2+y^2) \\ &\iff x^2 + x + y^2 = 0 \\ &\iff \left(x+\tfrac{1}{2}\right)^2 + y^2 = \tfrac{1}{4}. \end{aligned}$$

Every step is reversible (we multiplied by x^2+y^2, which is non-zero), so this is a genuine "if and only if". Therefore C is the circle

$$\left|w + \tfrac{1}{2}\right| = \tfrac{1}{2},$$

with centre $-\frac{1}{2}$ and radius $\frac{1}{2}$, *with the origin removed*. The origin does lie on this circle, but $1/w$ is undefined there, so it is not in C.

This mirrors the first part: there a circle through 0 was sent by $z \mapsto 1/z$ to a line not through 0; here the line $\operatorname{Re} u = -1$ is sent by the same map (which is its own inverse) to a circle through 0.



Checks

- Take $\theta = \frac{\pi}{6}$: then $|z| = 1$ and $z = \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$, and indeed $|z - \mathrm{i}| = \left|\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}\right| = 1$. Its reciprocal is its conjugate, $\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i} \approx 0.866 - 0.5\mathrm{i}$, which has imaginary part $-\frac{1}{2}$ as predicted.
- $\theta = \frac{\pi}{2}$ gives $z = 2\mathrm{i}$, the top of the circle, and $1/z = -\frac{1}{2}\mathrm{i}$.
- For C: $w = -1$ is on the circle and $1/w = -1$; $w = -\frac{1}{2} + \frac{1}{2}\mathrm{i}$ is on the circle and $1/w = -1 - \mathrm{i}$, with real part -1.

A numerical test of 10 000 random points on each locus confirmed all three results.

Common slips

- Forgetting the range $0 < \theta < \pi$: the formula $|z| = 2\sin\theta$ only makes sense where $\sin\theta > 0$.
- Describing the locus of $1/z$ as a half-line or segment. Since $\cot\theta$ takes all real values on $(0, \pi)$, it is the whole line.

- Including the origin in C. It satisfies the circle's equation but $1/w$ does not exist there.

Beyond STEP

The map $z \mapsto 1/z$ is an example of a Möbius transformation, $z \mapsto (az + b)/(cz + d)$. A central fact about these maps is that they send "generalised circles" (circles and straight lines) to generalised circles, and the circles through the origin are exactly the ones that $1/z$ sends to lines. This is the beginning of the geometry of the Riemann sphere, on which a line is just a circle through the point at infinity. Needham's *Visual Complex Analysis* develops all of this beautifully, with pictures.

WORKED SOLUTION

STEP III 1995, Question 7

Pure • Groups and Isomorphisms

> **STEP III 1995, Question 7**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/95-s3-q7

The key idea

Associativity is given, so for each set there are only three things to check: closure, an identity, and an inverse for every element. A set fails as soon as one of them breaks, and a single explicit counterexample is enough. For the isomorphisms, look for structural features that any isomorphism must preserve: the number of elements, and the orders of the elements.

Solution

Throughout, "multiplication" means ordinary multiplication of complex numbers, multiplication modulo n, or matrix multiplication, as appropriate.

Part (i)

Let $S = \{z \in \mathbb{C} : |z| = 1\}$.

- *Closure:* if $|z| = |w| = 1$ then $|zw| = |z||w| = 1$.
- *Identity:* $1 \in S$ and $1z = z1 = z$.
- *Inverses:* if $|z| = 1$ then $z \neq 0$, and $z^{-1} = \bar{z}$ (since $z\bar{z} = |z|^2 = 1$), which also has modulus 1.

So S is a group (an infinite one).

Part (ii)

Under multiplication modulo 4 the set $\{0, 1, 2, 3\}$ is closed and has identity 1, but 0 has no inverse, since $0 \times b \equiv 0 \not\equiv 1$ for every b. So it is *not* a group. (Removing 0 doesn't rescue it: 2 still has no inverse, as $2b$ is always even, and in fact $2 \times 2 \equiv 0$, so $\{1, 2, 3\}$ isn't even closed.)

Part (iii)

Using the compound-angle formulae,

$$\mathrm{M}(\alpha)\mathrm{M}(\beta) = \begin{pmatrix} c_\alpha c_\beta - s_\alpha s_\beta & -(s_\alpha c_\beta + c_\alpha s_\beta) \\ s_\alpha c_\beta + c_\alpha s_\beta & c_\alpha c_\beta - s_\alpha s_\beta \end{pmatrix} = \mathrm{M}(\alpha+\beta),$$

writing $c_\alpha = \cos\alpha$, $s_\alpha = \sin\alpha$ and so on. Since cos and sin have period 2π, $\mathrm{M}(\alpha+\beta) = \mathrm{M}(\gamma)$ where γ is $\alpha+\beta$ reduced to lie in $[0, 2\pi)$. So:

- *Closure:* the product of two matrices in the set is again in the set.
- *Identity:* $\mathrm{M}(0) = I$.
- *Inverses:* $\mathrm{M}(\theta)\mathrm{M}(2\pi-\theta) = \mathrm{M}(2\pi) = I$ for $0 < \theta < 2\pi$, and $\mathrm{M}(0)$ is its own inverse.

So this set is a group.

Part (iv)

Modulo 8,

$$3^2 = 9 \equiv 1, \qquad 5^2 = 25 \equiv 1, \qquad 7^2 = 49 \equiv 1,$$
$$3\times 5 = 15 \equiv 7, \qquad 3\times 7 = 21 \equiv 5, \qquad 5\times 7 = 35 \equiv 3.$$

With the trivial products involving 1, this gives the whole multiplication table (multiplication is commutative), and every product lies in $\{1,3,5,7\}$, so the set is closed. The identity is 1, and each element is its own inverse. So this is a group, of order 4, in which every element other than 1 has order 2.

Part (v)

The identity matrix I has integer entries, and the set is closed (products and sums of integers are integers), but inverses fail. The zero matrix has no inverse at all. Even an invertible matrix can fail: $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ has inverse $\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$ in the real matrices, and inverses are unique, so A has no inverse with integer entries. So this set is *not* a group.

> **Ask yourself** Is pointing at the zero matrix really enough, or do we need to say something about all the matrices?

One element without an inverse is enough to show that the set is not a group. The second example shows that the problem goes deeper than singular matrices. An integer matrix has an integer inverse exactly when its determinant is ± 1, because $\det A \det A^{-1} = 1$ with both determinants integers. Restricting to those matrices does give a group.

Part (vi)

Modulo 5, the powers of 2 are

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8 \equiv 3, \quad 2^4 = 16 \equiv 1.$$

So $\{1,2,3,4\} = \{2^0, 2^1, 2^2, 2^3\}$, and $2^a 2^b = 2^{a+b}$ with exponents read modulo 4. Hence the set is closed, 1 is the identity, and 2^a has inverse 2^{4-a}. It is a group: the cyclic group of order 4, generated by 2 (or by 3).

Isomorphisms

The groups are (i), (iii), (iv) and (vi), which gives six pairs.

(i) and (iii) are isomorphic. Every unit complex number can be written uniquely as $e^{i\theta}$ with $0 \leq \theta < 2\pi$. Define $\phi(e^{i\theta}) = M(\theta)$. This is a bijection, since distinct θ in $[0, 2\pi)$ give distinct matrices (the first column $(\cos\theta, \sin\theta)$ determines θ). It respects multiplication: $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$ and $M(\alpha)M(\beta) = M(\alpha+\beta)$, where in both groups the angle is reduced modulo 2π. Geometrically, both groups are the rotations of the plane about the origin.

(i) or (iii) with (iv) or (vi): not isomorphic. An isomorphism is a bijection, so the two groups must have the same number of elements. Groups (i) and (iii) are infinite, while (iv) and (vi) have four elements each. That disposes of four pairs.

> **Ask yourself** (iv) and (vi) both have four elements and both are commutative. What could tell them apart?

(iv) and (vi): not isomorphic. An isomorphism ϕ preserves the order of every element: $\phi(g)^n = \phi(g^n)$, which equals the identity exactly when g^n is the identity, because ϕ is injective and sends identity to identity. In (vi) the element 2 has order 4. In (iv) every element g satisfies $g^2 = 1$, so no element has order 4. So no isomorphism exists.

In summary, the only isomorphic pair is (i) and (iii).

Checks

A computer confirmed the closure, identity and inverse properties for (iv) and (vi) by brute force over the tables. It found that 0 and 2 have no inverse modulo 4, and that the element orders are $1,2,2,2$ in (iv) and $1,4,4,2$ (for $1,2,3,4$) in (vi). It also checked $M(\alpha)M(\beta) = M(\alpha+\beta)$ and the correspondence with $e^{i\theta}$ at 10^4 random angles. You can check (iv) quickly by hand: squaring any odd number gives 1 modulo 8, since $(2k+1)^2 = 4k(k+1)+1$ and $k(k+1)$ is even.

Common slips

- Saying (iv) and (vi) are isomorphic "because both have four elements". Equal size is necessary but not sufficient.

- In (v), checking only that $\det A \neq 0$. The inverse must also have integer entries.
- Reading (ii) as addition modulo 4. Under addition it would be a cyclic group of order 4, isomorphic to (vi), but the question specifies multiplication.

Beyond STEP

Up to isomorphism there are exactly two groups of order 4: the cyclic group C_4, as in (vi), and the Klein four-group $C_2 \times C_2$, as in (iv). Classifying small groups by element orders is one of the first things a university group theory course does. The circle group of (i) and (iii) is the simplest example of a Lie group, a group that is also a smooth curve or surface. The integer matrices of determinant ± 1 from (v) form the group $\mathrm{GL}_2(\mathbb{Z})$, which is important in number theory. Allenby's *Rings, Fields and Groups* is a good place to read more.

WORKED SOLUTION

STEP III 1995, Question 8

Pure • Planes and Spheres

STEP III 1995, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q8

The key idea

When a plane cuts a sphere, the circle it makes has its centre at the foot of the perpendicular from the sphere's centre to the plane. So the sphere's centre must lie on the normal line through the centre of *each* circle, and Pythagoras ties its distance along each normal to the sphere's radius. Everything else in the question is a matter of dotting that one vector equation with well-chosen vectors.

Solution

The normal and the distance

The normal to π through X is the line through $\mathbf{x}$ in the direction $\mathbf{n}$:

$$\mathbf{r} = \mathbf{x} + t\,\mathbf{n}, \qquad t \in \mathbb{R}.$$

It meets π where $(\mathbf{x} + t\mathbf{n}) \cdot \mathbf{n} = p$, that is (since $\mathbf{n} \cdot \mathbf{n} = 1$) where $t = p - \mathbf{x} \cdot \mathbf{n}$. The foot of the perpendicular is therefore $\mathbf{x} + (p - \mathbf{x} \cdot \mathbf{n})\mathbf{n}$, and because $\mathbf{n}$ is a unit vector the perpendicular distance of X from π is

$$|\mathbf{x} \cdot \mathbf{n} - p|.$$

A sphere through both circles

Suppose a sphere with centre $\mathbf{c}$ and radius R contains both circles. Fix i. Every point $\mathbf{y}$ of C_i satisfies $|\mathbf{y} - \mathbf{r}_i| = 1$ and $(\mathbf{y} - \mathbf{r}_i) \cdot \mathbf{n}_i = 0$; as $\mathbf{y}$ runs round C_i, the vector $\mathbf{u} = \mathbf{y} - \mathbf{r}_i$ runs through *every* unit vector perpendicular to $\mathbf{n}_i$.

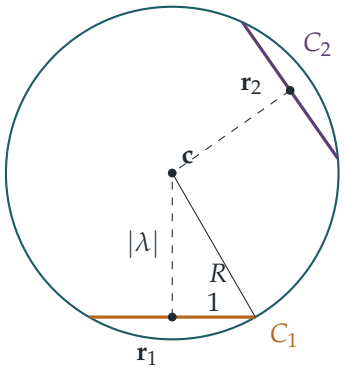
Split $\mathbf{c} - \mathbf{r}_i = \mathbf{a} + \mu_i \mathbf{n}_i$ with $\mathbf{a} \perp \mathbf{n}_i$. Then

$$\begin{aligned} R^2 = |\mathbf{y} - \mathbf{c}|^2 &= |\mathbf{u} - \mathbf{a} - \mu_i \mathbf{n}_i|^2 \\ &= 1 - 2\,\mathbf{u} \cdot \mathbf{a} + |\mathbf{a}|^2 + \mu_i^2, \end{aligned}$$

using $\mathbf{u}\cdot\mathbf{n}_i = \mathbf{a}\cdot\mathbf{n}_i = 0$. The left side is the same for every point of C_i, so $\mathbf{u}\cdot\mathbf{a}$ takes the same value for every allowed $\mathbf{u}$. But $-\mathbf{u}$ is allowed whenever $\mathbf{u}$ is, so $\mathbf{u}\cdot\mathbf{a} = -\mathbf{u}\cdot\mathbf{a} = 0$ for every unit $\mathbf{u}$ in the plane, and choosing $\mathbf{u}$ parallel to $\mathbf{a}$ (if $\mathbf{a} \neq \mathbf{0}$) gives a contradiction. Hence $\mathbf{a} = \mathbf{0}$ and

$$\mathbf{c} = \mathbf{r}_i + \mu_i \mathbf{n}_i, \qquad R^2 = 1 + \mu_i^2.$$

In words: the centre of the sphere lies on the normal to π_i through the centre of C_i, which is exactly the line found in the first part.



Applying this with $i = 1$ and $i = 2$ gives $1 + \mu_1^2 = R^2 = 1 + \mu_2^2$, so $\mu_2 = \pm\mu_1$. Writing $\lambda = \mu_1$,

$$\mathbf{r}_1 + \lambda\mathbf{n}_1 = \mathbf{c} = \mathbf{r}_2 \pm \lambda\mathbf{n}_2,$$

so such a λ must exist whenever a suitable sphere exists, as required.

The two necessary conditions

Rearranging,

$$\mathbf{r}_1 - \mathbf{r}_2 = -\lambda\mathbf{n}_1 \pm \lambda\mathbf{n}_2.$$

Dot with $\mathbf{n}_1 \times \mathbf{n}_2$, which is perpendicular to both $\mathbf{n}_1$ and $\mathbf{n}_2$:

$$(\mathbf{r}_1 - \mathbf{r}_2)\cdot(\mathbf{n}_1 \times \mathbf{n}_2) = 0.$$

? **Ask yourself** Where does p_1 come from? It doesn't appear in the vector equation.

The centre $\mathbf{r}_1$ lies in π_1, so $\mathbf{n}_1\cdot\mathbf{r}_1 = p_1$; likewise $\mathbf{n}_2\cdot\mathbf{r}_2 = p_2$. Write $k = \mathbf{n}_1\cdot\mathbf{n}_2$. Dotting $\mathbf{r}_1 + \lambda\mathbf{n}_1 = \mathbf{r}_2 \pm \lambda\mathbf{n}_2$ with $\mathbf{n}_1$ and then with $\mathbf{n}_2$ gives

$$p_1 + \lambda = \mathbf{n}_1\cdot\mathbf{r}_2 \pm \lambda k,$$
$$\mathbf{n}_2\cdot\mathbf{r}_1 + \lambda k = p_2 \pm \lambda.$$

So

$$p_1 - \mathbf{n}_1\cdot\mathbf{r}_2 = \lambda(\pm k - 1), \qquad p_2 - \mathbf{n}_2\cdot\mathbf{r}_1 = \lambda(k \mp 1).$$

With the upper sign both right-hand sides equal $\lambda(k-1)$; with the lower sign they are $-\lambda(k+1)$ and $\lambda(k+1)$. Either way the two left-hand sides are equal or opposite, so

$$(p_1 - \mathbf{n}_1 \cdot \mathbf{r}_2)^2 = (p_2 - \mathbf{n}_2 \cdot \mathbf{r}_1)^2.$$

Geometric meaning

First condition. It says that $\mathbf{r}_1 - \mathbf{r}_2$, $\mathbf{n}_1$ and $\mathbf{n}_2$ are coplanar (the scalar triple product vanishes). Equivalently, the normal to π_1 through the centre of C_1 and the normal to π_2 through the centre of C_2 lie in a common plane: they are not skew, so they meet (at the sphere's centre) or are parallel.

Second condition. By the first part, $|p_1 - \mathbf{n}_1 \cdot \mathbf{r}_2|$ is the distance of the centre of C_2 from π_1, and $|p_2 - \mathbf{n}_2 \cdot \mathbf{r}_1|$ is the distance of the centre of C_1 from π_2. So the condition says these two distances are equal: each circle's centre is as far from the other circle's plane as the other centre is from its plane. This reflects the symmetry of the picture: both circles are at the same distance $|\lambda|$ from $\mathbf{c}$, so a reflection in a suitable plane through $\mathbf{c}$ swaps them, and swaps the two distances.

Checks

- Take $\pi_1 : z = 0$ and $\pi_2 : z = 2$ with unit circles centred on the z-axis. They lie on the sphere centred at $(0, 0, 1)$ of radius $\sqrt{2}$; here $\lambda = 1$ with the lower sign, $\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}$, and both distances in the second condition equal 2.
- Numerically, we built 5000 random spheres of radius $R > 1$ with two random planes each cutting a unit circle, confirmed every circle point lies on the sphere, and found both conditions held to within 10^{-9} in every case, with both signs occurring. Random unrelated pairs of circles failed the first condition every time.

Common slips

- Assuming the sphere's centre lies on the normal through $\mathbf{r}_i$ without justification; that is the heart of the proof.
- Forgetting the $\pm$: the sphere's centre may be on the same side of both planes as the normals point, or on opposite sides.
- Forgetting that $\mathbf{n}_1 \cdot \mathbf{r}_1 = p_1$ because the centre of C_1 lies in π_1.

Beyond STEP

The conditions here are only necessary: for a full "if and only if" you would also need the two normals actually to meet at a point equidistant from both circles. Questions like this are first steps into analytic geometry of quadrics, where planes, spheres and their intersections are handled with linear algebra; the scalar triple product reappears as a 3×3 determinant measuring volume. Strang's *Introduction to Linear Algebra* develops

these ideas carefully.

WORKED SOLUTION

STEP III 1995, Question 9

Mechanics • Compound Pendulum

STEP III 1995, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

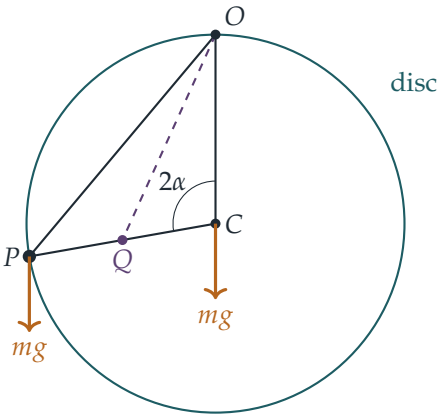
step.maths.org/questions/95-s3-q9

The key idea

The disc and the particle have the *same* mass, so the centre of mass of the whole system is the midpoint Q of CP. Almost everything follows from that: the stable equilibrium has Q directly below O, and the energy equation in part (iii) only needs the height of Q. The rest is the geometry of the isosceles triangle OCP and a standard moment of inertia integral.

Solution

We model the disc as a rigid lamina and P as a particle fixed to its rim; the axis is smooth, so mechanical energy is conserved. Throughout, $OC = CP = r$ and $\angle OCP = 2\alpha$.



Part (i)

Since the disc (centre of mass C) and the particle (at P) both have mass m, the centre of mass of the system is the midpoint Q of CP. The only forces are gravity and the reaction at the axis, so the system is in equilibrium exactly when Q is vertically in line

with O, and the equilibrium is stable when Q is *below* O (any displacement then raises Q, increasing the potential energy). So in stable equilibrium OQ is vertical, and β is the angle $\angle COQ$.

Use axes in the plane of the disc with origin C and O at $(r, 0)$; then $P = r(\cos 2\alpha, \sin 2\alpha)$ and $Q = \frac{r}{2}(\cos 2\alpha, \sin 2\alpha)$. Hence

$$\overrightarrow{OC} = (-r,\, 0), \qquad \overrightarrow{OQ} = \tfrac{r}{2}\big(-(2 - \cos 2\alpha),\ \sin 2\alpha\big).$$

Since $2 - \cos 2\alpha > 0$, the vector $\overrightarrow{OQ}$ has a positive component $\frac{r}{2}(2 - \cos 2\alpha)$ along $\overrightarrow{OC}$ and a component $\frac{r}{2}\sin 2\alpha$ perpendicular to it, so β is acute and

$$\tan\beta = \frac{\frac{r}{2}\sin 2\alpha}{\frac{r}{2}(2 - \cos 2\alpha)} = \frac{\sin 2\alpha}{2 - \cos 2\alpha},$$

as required.

Part (ii)

We take $\rho x / r$ to be mass per unit area. The ring between radii x and $x + \mathrm{d}x$ has area $2\pi x\,\mathrm{d}x$, so

$$m = \int_0^r \frac{\rho x}{r}\, 2\pi x\,\mathrm{d}x = \frac{2\pi\rho r^2}{3}.$$

Every point of that ring is distance x from the axis through C perpendicular to the disc, so the moment of inertia about that axis is

$$I_C = \int_0^r \frac{\rho x}{r}\, 2\pi x \cdot x^2\,\mathrm{d}x = \frac{2\pi\rho r^4}{5} = \frac{3}{5}mr^2,$$

using $2\pi\rho r^2 = 3m$. The axis through O is parallel to this one (both are perpendicular to the vertical plane of the disc) at distance $OC = r$, so by the parallel axis theorem

$$I_O = \frac{3}{5}mr^2 + mr^2 = \frac{8}{5}mr^2.$$

Part (iii)

Moment of inertia of the system. Triangle OCP is isosceles with apex angle 2α, so the chord is $OP = 2r\sin\alpha$. The particle adds $m(2r\sin\alpha)^2$, giving

$$I = \frac{8}{5}mr^2 + 4mr^2\sin^2\alpha = \frac{4}{5}mr^2\,(2 + 5\sin^2\alpha).$$

Drop of the centre of mass. At release OQ is horizontal, so Q is level with O. In the final position P is vertically below O, so OP is vertical, and the depth of any point below O is its distance along OP from O.

- P is at depth $OP = 2r\sin\alpha$.
- The base angles of triangle OCP are $\frac{1}{2}(\pi - 2\alpha) = \frac{\pi}{2} - \alpha$, so $\angle COP = \frac{\pi}{2} - \alpha$ and C is at depth $r\cos(\frac{\pi}{2} - \alpha) = r\sin\alpha$.

Q is the midpoint of CP, so its depth is the average, $\frac{3}{2}r\sin\alpha$. The system (total mass $2m$) has therefore lost potential energy $2mg\cdot\frac{3}{2}r\sin\alpha = 3mgr\sin\alpha$.

Ask yourself Does the system actually reach the position with P below O?

Released from rest with Q level with O, the system swings like a compound pendulum between the two positions with OQ horizontal, passing through every position in which Q is below the level of O. In the final position Q is at depth $\frac{3}{2}r\sin\alpha > 0$, so it is reached (and the condition that C starts lower than P just fixes which way the swing goes).

Energy. Starting from rest, with angular speed ω,

$$\tfrac{1}{2}I\omega^2 = 3mgr\sin\alpha \quad\Longrightarrow\quad \omega^2 = \frac{6gr\sin\alpha}{\frac{4}{5}r^2(2+5\sin^2\alpha)} = \frac{15g\sin\alpha}{2r(2+5\sin^2\alpha)}.$$

C moves on a circle of radius $OC = r$ about O, so $v = r\omega$ and

$$v^2 = \frac{15gr\sin\alpha}{2(2+5\sin^2\alpha)},$$

as required.

Maximum. Put $s = \sin\alpha$; since $0 < \alpha \le \pi/2$, $s \in (0,1]$. Let $f(s) = \dfrac{s}{2+5s^2}$. Then

$$f'(s) = \frac{(2+5s^2) - 10s^2}{(2+5s^2)^2} = \frac{2-5s^2}{(2+5s^2)^2},$$

which is positive for $s < \sqrt{2/5}$ and negative for $s > \sqrt{2/5}$. As $\sqrt{2/5} < 1$ lies in the allowed range, f has its maximum there, $f = \dfrac{\sqrt{2/5}}{4} = \dfrac{\sqrt{10}}{20}$. Hence

$$v^2_{\max} = \frac{15gr}{2}\cdot\frac{\sqrt{10}}{20} = \frac{3\sqrt{10}}{8}\,gr \approx 1.19\,gr, \qquad \text{when } \sin\alpha = \sqrt{2/5}.$$

Checks

- If $\alpha \to 0$, P sits at O and the disc alone hangs with OC vertical: indeed $\tan\beta \to 0$. If $\alpha = \pi/2$, P is diametrically opposite O, Q lies on OC and again $\beta = 0$.
- With $\alpha = \pi/2$: $I = \frac{28}{5}mr^2$, drop $\frac{3}{2}r$ for mass $2m$, so $v^2 = r^2\cdot 3mgr\cdot\frac{10}{28mr^2} = \frac{15}{14}gr$, matching the formula.
- We integrated the equation of motion numerically (RK4) from the release position for several α; for example at $\alpha = 0.7$, $r = 1.3$, $g = 9.81$ the simulation gives $v^2 = 15.1206$, agreeing with the formula to six figures. A direct search also confirms the maximum $3\sqrt{10}/8 \approx 1.1859$ (in units of gr).

Common slips

- Using the uniform-disc value $\frac{1}{2}mr^2$ for I_C; the density here increases outwards, which pushes I_C up to $\frac{3}{5}mr^2$.

- Forgetting the particle's contribution $m \cdot OP^2$ to the moment of inertia in part (iii).
- Giving ω^2 instead of v^2: the question asks for the speed of C, which is $r\omega$.

Beyond STEP

A rigid body pivoted about a horizontal axis is a *compound pendulum*: it behaves like a simple pendulum whose bob is at the centre of mass, but with inertia I instead of Md^2. Small oscillations about the equilibrium in part (i) have period $2\pi\sqrt{I/(Mgd)}$, where $d = OQ$, and the "equivalent simple pendulum" length $I/(Md)$ leads to Kater's reversible pendulum, once used to measure g precisely. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and the parallel axis theorem in this spirit.

WORKED SOLUTION

STEP III 1995, Question 10

Mechanics • Projectiles on Inclined Planes

STEP III 1995, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q10

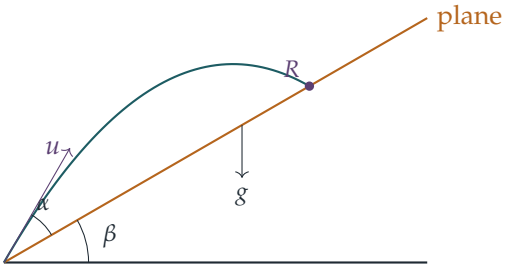
The key idea

On an inclined plane the natural axes are not horizontal and vertical but *along* and *perpendicular to* the plane. In those axes gravity has two constant components, the time of flight comes from the perpendicular motion alone, and the range is a product of trigonometric functions. Turning that product into a difference of sines with the factor formula makes the maximum obvious, with no calculus needed.

Solution

Setting up

Model the cannon ball as a particle moving freely under gravity, with no air resistance, fired from the foot of the plane up its line of greatest slope. Let it be fired at angle α *above the plane*, so that its angle of elevation above the horizontal is $\theta = \alpha + \beta$; we need $0 < \alpha < \frac{\pi}{2} - \beta$ for the ball to go up the plane at all.



Take the x-axis up the line of greatest slope and the y-axis perpendicular to the plane, pointing away from it. Resolving the acceleration g (vertically downwards) into these directions,

$$\ddot{x} = -g\sin\beta, \qquad \ddot{y} = -g\cos\beta,$$

and the initial velocity has components $u\cos\alpha$ along the plane and $u\sin\alpha$ perpendicular

to it. Hence at time t

$$x = ut\cos\alpha - \tfrac{1}{2}gt^2\sin\beta, \qquad y = ut\sin\alpha - \tfrac{1}{2}gt^2\cos\beta.$$

Time of flight and range

The ball lands when $y = 0$ again with $t > 0$:

$$T = \frac{2u\sin\alpha}{g\cos\beta}.$$

The distance travelled up the plane is then

$$\begin{aligned} R = x(T) &= \frac{2u^2\sin\alpha\cos\alpha}{g\cos\beta} - \tfrac{1}{2}g\sin\beta\cdot\frac{4u^2\sin^2\alpha}{g^2\cos^2\beta} \\ &= \frac{2u^2\sin\alpha}{g\cos^2\beta}\big(\cos\alpha\cos\beta - \sin\alpha\sin\beta\big) \\ &= \frac{2u^2\sin\alpha\cos(\alpha+\beta)}{g\cos^2\beta}. \end{aligned}$$

Ask yourself How can we maximise $\sin\alpha\cos(\alpha+\beta)$ without differentiating?

The factor formula $2\sin A\cos B = \sin(A+B) + \sin(A-B)$, with $A = \alpha$ and $B = \alpha+\beta$, gives

$$2\sin\alpha\cos(\alpha+\beta) = \sin(2\alpha+\beta) - \sin\beta,$$

so

$$R = \frac{u^2\big(\sin(2\alpha+\beta) - \sin\beta\big)}{g\cos^2\beta}.$$

Only the term $\sin(2\alpha+\beta)$ depends on α, and it is at most 1. As α runs over $(0, \frac{\pi}{2}-\beta)$, the angle $2\alpha+\beta$ runs over $(\beta, \pi-\beta)$, which contains $\frac{\pi}{2}$; so the maximum value 1 is attained, exactly when $2\alpha+\beta = \frac{\pi}{2}$, that is,

$$\alpha = \frac{\pi}{4} - \frac{\beta}{2}, \qquad \text{equivalently} \qquad \theta = \alpha+\beta = \frac{\pi}{4} + \frac{\beta}{2}.$$

So the cannon should be fired at $\frac{\pi}{4} + \frac{\beta}{2}$ above the horizontal: the firing direction bisects the angle between the plane and the vertical.

The maximum range

With $\sin(2\alpha+\beta) = 1$, and using $\cos^2\beta = 1 - \sin^2\beta = (1-\sin\beta)(1+\sin\beta)$,

$$R_{\max} = \frac{u^2(1-\sin\beta)}{g(1-\sin\beta)(1+\sin\beta)} = \frac{u^2}{g(1+\sin\beta)},$$

where cancelling $1-\sin\beta$ is valid because $\beta < \frac{\pi}{2}$. This is the required distance.

Checks

- Flat ground, $\beta = 0$: the best angle is 45° and the range u^2/g, the familiar results.
- As $\beta \to \frac{\pi}{2}$ (a vertical "plane"), the best direction tends to straight up and $R_{\max} \to u^2/(2g)$, the greatest height reached by a ball fired vertically.
- Numerically, with $u = 20\,\mathrm{m\,s^{-1}}$, $g = 9.81\,\mathrm{m\,s^{-2}}$ and $\beta = 30^\circ$, the formula gives $R_{\max} \approx 27.18\,\mathrm{m}$ at elevation 60°. Stepping the motion in horizontal and vertical coordinates until the ball crosses the plane reproduces $R(\alpha)$ for random u, β, α, and a fine search over α puts the maximum at $\frac{\pi}{4} - \frac{\beta}{2}$.

Common slips

- Mixing up the angle above the horizontal with the angle above the plane; the answer $\frac{\pi}{4} + \frac{\beta}{2}$ is measured from the horizontal.
- Forgetting the component $-g\sin\beta$ along the plane, which gives the flat-ground range formula scaled wrongly.

Beyond STEP

There is an elegant geometric view: all trajectories with speed u from a point lie inside a "parabola of safety" (their envelope) with focus at the launch point, and the maximum range on any plane is where the plane meets that parabola. The focus–directrix property then gives $R_{\max} = u^2/(g(1+\sin\beta))$ in one line, and explains why the best direction bisects the angle between the plane and the vertical. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile problems treated with vector methods.

WORKED SOLUTION

STEP III 1995, Question 11

Mechanics • Relative Velocity

STEP III 1995, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q11

The key idea

Work in the ship's frame. Relative to the ship, the plane has to fly straight out along the north-west line and straight back again. The air carries the plane, so in this frame the plane's velocity is its airspeed vector plus a "current": the wind's velocity minus the ship's velocity. Once we know that current, each leg is a river-crossing problem. We choose the heading so that the resultant points along the line, and then the out-and-back time fixes the distance.

Solution

Throughout, take x east and y north, and measure speeds in knots (nautical miles per hour), so that distances come out in nautical miles. The ship's velocity is $(-V, 0)$. If the air moves with velocity $\mathbf{w}$ and the plane's velocity relative to the air is $\mathbf{a}$ with $|\mathbf{a}| = kV$, then the plane's velocity relative to the ship is

$$\mathbf{a} + \mathbf{c}, \qquad \mathbf{c} = \mathbf{w} - (-V, 0) = \mathbf{w} + (V, 0).$$

The plane stays north-west of the ship, so $\mathbf{a} + \mathbf{c}$ must point along $\mathbf{u} = \frac{1}{\sqrt{2}}(-1, 1)$ on the way out and along $-\mathbf{u}$ on the way back.

With the wind: the outward flight

The wind blows *from* north *to* south, so $\mathbf{w} = (0, -V)$ and

$$\mathbf{c} = (V, -V) = -\sqrt{2}\, V\, \mathbf{u}.$$

This current has speed $\sqrt{2}\, V$ and points due south-east, exactly along the patrol line. So the pilot simply points the plane along the line. Heading out (north-west) she flies straight into the current; heading back she flies with it:

$$\text{outward speed} = (k - \sqrt{2})V, \qquad \text{return speed} = (k + \sqrt{2})V.$$

Here we need $k > \sqrt{2}$: otherwise the plane could never get away from the ship along the north-west line.

If the plane reaches a distance d_w from the ship and each out-and-back trip takes $\frac{1}{2}$ hour, then

$$\frac{d_w}{(k-\sqrt{2})V} + \frac{d_w}{(k+\sqrt{2})V} = \frac{1}{2}$$
$$\frac{d_w}{V} \cdot \frac{2k}{k^2-2} = \frac{1}{2},$$

so

$$d_w = \frac{(k^2-2)V}{4k}.$$

The outward flight therefore lasts

$$\frac{d_w}{(k-\sqrt{2})V} = \frac{k^2-2}{4k(k-\sqrt{2})} = \frac{k+\sqrt{2}}{4k} \text{ hours} = 15\left(1+\frac{\sqrt{2}}{k}\right) \text{ minutes.}$$

This is always more than 15 minutes, as it should be: the outward leg runs against the current.

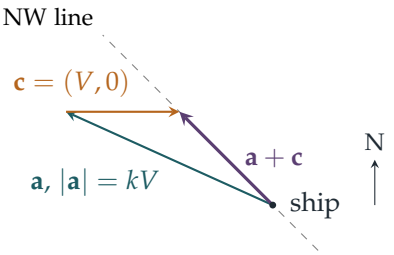
Ask yourself Why can't the plane just point north-west in calm air as well?

In calm air the current is no longer along the patrol line. If the pilot pointed the plane north-west, the eastward part of the current would carry it off the line. She has to aim slightly into the current, and that uses up some of her airspeed.

In calm air

Now $\mathbf{w} = \mathbf{0}$, so $\mathbf{c} = (V, 0)$: relative to the ship, the air streams east at V. Split $\mathbf{c}$ along the line, using $\mathbf{u}$, and across it, using $\mathbf{v} = \frac{1}{\sqrt{2}}(1,1)$:

$$\mathbf{c} = -\frac{V}{\sqrt{2}}\,\mathbf{u} + \frac{V}{\sqrt{2}}\,\mathbf{v}.$$



The resultant must have no $\mathbf{v}$-component, so $\mathbf{a}$ must have $\mathbf{v}$-component $-V/\sqrt{2}$. Since $|\mathbf{a}| = kV$, its component along the direction of travel is then

$$\sqrt{k^2V^2 - \tfrac{1}{2}V^2} = V\sqrt{k^2 - \tfrac{1}{2}}.$$

Along the line, the current contributes $-V/\sqrt{2}$ on the way out and $+V/\sqrt{2}$ on the way back (because $\mathbf{c}\cdot(-\mathbf{u}) = V/\sqrt{2}$). Write $A = V\sqrt{k^2 - \tfrac{1}{2}}$ and $B = V/\sqrt{2}$. Then

$$\text{outward speed} = A - B, \qquad \text{return speed} = A + B.$$

The outward speed is positive because $A^2 - B^2 = V^2(k^2 - 1) > 0$.

With the same half-hour round trip,

$$\frac{d_c}{A-B} + \frac{d_c}{A+B} = \frac{2A\,d_c}{A^2 - B^2} = \frac{1}{2} \quad\Longrightarrow\quad d_c = \frac{A^2 - B^2}{4A} = \frac{V(k^2-1)}{4\sqrt{k^2 - \tfrac{1}{2}}}.$$

The ratio

Dividing,

$$\frac{d_w}{d_c} = \frac{(k^2-2)V}{4k}\cdot\frac{4\sqrt{k^2-\tfrac{1}{2}}}{V(k^2-1)} = \frac{(k^2-2)\sqrt{k^2-\tfrac{1}{2}}}{k(k^2-1)}.$$

Since $\sqrt{k^2 - \tfrac{1}{2}} = \tfrac{1}{2}\sqrt{4k^2-2}$,

$$\frac{d_w}{d_c} = \frac{k^2-2}{2k(k^2-1)}\sqrt{4k^2-2},$$

which is the required result.

Checks

- As $k \to \infty$ the wind and the ship hardly matter. The outward flight tends to 15 minutes, and $d_w/d_c \to 1$ (top and bottom of the ratio both grow like $2k^3$).
- As $k \to \sqrt{2}^{+}$ the windy outward speed falls to zero, so $d_w \to 0$ and the ratio $\to 0$, as the formula says.
- For $k = 2$ the outward flight takes $15(1 + 1/\sqrt{2}) \approx 25.6$ minutes, and $d_w/d_c = \tfrac{2}{12}\sqrt{14} \approx 0.624$. We confirmed this numerically by solving for the heading directly and by stepping the motion.

Common slips

- Getting the wind's direction backwards: a wind "from north to south" blows *towards* the south.
- Adding the ship's velocity instead of subtracting it. Relative to a westbound ship, still air drifts *east*.
- In calm air, pointing the plane north-west and using $kV \pm V/\sqrt{2}$ as the speeds. The cross-current has to be cancelled first.

Beyond STEP

This is a Galilean change of reference frame: velocities measured in frames moving uniformly relative to each other differ by a constant vector. The same bookkeeping underlies the Michelson–Morley "boat in a river" argument. There, the across-and-back trip and the along-and-back trip take different times in exactly the way they do here, and that is the classical calculation that special relativity overturned. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats relative motion and then relativity in this spirit.

WORKED SOLUTION

STEP III 1995, Question 12

Probability and Statistics • Radially Symmetric Densities

STEP III 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.22, *A Distribution With No Mean*: https://beyondstep.keithmcnulty.org/ch12/cauchy.html

The key idea

A radially symmetrical density depends on (u, v) only through the distance $r = \sqrt{u^2 + v^2}$ from the origin, so it looks the same after any rotation about the origin. In particular, the probability of landing in a sector with vertex at the origin depends only on the angle of the sector. The event $U/V < k$ is a pair of opposite sectors cut out by the line $u = kv$ and the u-axis, so its probability is just an angle divided by 2π.

Solution

The joint density and radial symmetry

Each of X and Y has density $\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$. Since X and Y are independent, their joint density is the product of the marginal densities:

$$f(x, y) = \phi(x)\,\phi(y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)},$$

for all real x and y, as required.

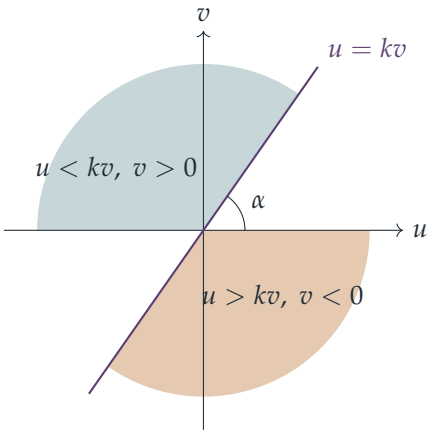
Saying that this density is *radially symmetrical* means that its value at the point (x, y) depends only on the distance $r = \sqrt{x^2 + y^2}$ of the point from the origin, and not on its direction. Here $f(x, y) = \frac{1}{2\pi} e^{-r^2/2}$. Equivalently, the density is unchanged by any rotation of the plane about the origin, and its contours are circles centred on the origin.

The identity for $\mathrm{P}(U/V < k)$

Let U and V have a radially symmetrical joint density $h(u,v) = \rho\big(\sqrt{u^2+v^2}\big)$. The line $v = 0$ has probability zero (it has zero area), so we may ignore the case $V = 0$. Multiplying the inequality $U/V < k$ by V, and reversing it when $V < 0$,

$$\left\{\frac{U}{V} < k\right\} = \{U < kV,\ V > 0\} \cup \{U > kV,\ V < 0\},$$

and the two events on the right are disjoint.



As the diagram shows (drawn for $k > 0$), the line $u = kv$ through the origin and the u-axis split the plane into four sectors, and the two shaded ones make up the event $U/V < k$. The map $(u,v) \mapsto (-u,-v)$ is a rotation through π about the origin, and it takes the region $\{u < kv,\ v > 0\}$ exactly onto $\{u > kv,\ v < 0\}$ (multiply both inequalities by -1). A radially symmetrical density satisfies $h(-u,-v) = h(u,v)$, so these two regions have the same probability. Hence

$$\mathrm{P}\left(\frac{U}{V} < k\right) = 2\,\mathrm{P}(U < kV,\ V > 0),$$

which is what we had to show.

The density of U/V

Use polar coordinates $u = r\cos\theta$, $v = r\sin\theta$. The region $\{u < kv,\ v > 0\}$ is the sector $\alpha < \theta < \pi$, where $\alpha \in (0,\pi)$ is the angle the ray $u = kv$, $v > 0$ makes with the positive u-axis, so $\cot\alpha = k$. (Indeed, for $0 < \theta < \pi$ we have $\sin\theta > 0$, and $u < kv$ becomes $\cot\theta < k$; since cot is decreasing on $(0,\pi)$, this is $\theta > \alpha$.)

Ask yourself What is the probability that (U,V) lies in a sector of angle β with vertex at the origin, when we don't know the function ρ?

In polar coordinates the area element is $r\,\mathrm{d}r\,\mathrm{d}\theta$, so a sector $\theta_1 < \theta < \theta_1 + \beta$ has probability

$$\int_{\theta_1}^{\theta_1+\beta}\int_0^\infty \rho(r)\,r\,\mathrm{d}r\,\mathrm{d}\theta = \beta\int_0^\infty \rho(r)\,r\,\mathrm{d}r = \frac{\beta}{2\pi},$$

because taking $\beta = 2\pi$ gives the whole plane, so $2\pi\int_0^\infty \rho(r)\,r\,\mathrm{d}r = 1$. In other words, the direction of (U,V) is uniformly distributed, whatever ρ is.

Our sector has angle $\pi - \alpha$, so

$$\mathrm{G}(k) = \mathrm{P}\Big(\frac{U}{V} < k\Big) = 2\cdot\frac{\pi-\alpha}{2\pi} = 1 - \frac{\alpha}{\pi}.$$

Since $\cot\alpha = k$ with $\alpha \in (0,\pi)$, we have $\alpha = \frac{\pi}{2} - \arctan k$ (both sides lie in $(0,\pi)$ and have cotangent k, because $\cot(\frac{\pi}{2} - \psi) = \tan\psi$). Therefore

$$\mathrm{G}(k) = \frac{1}{2} + \frac{1}{\pi}\arctan k, \qquad -\infty < k < \infty.$$

This rises from 0 to 1, as a distribution function must. Differentiating gives the density of U/V:

$$\mathrm{g}(k) = \mathrm{G}'(k) = \frac{1}{\pi(1+k^2)}, \qquad -\infty < k < \infty,$$

as required. In particular this applies to X/Y for the independent standard normals of the first part.

Checks

- $\mathrm{G}(0) = \frac{1}{2}$: by symmetry U/V is as likely to be positive as negative.
- $\int_{-\infty}^{\infty}\frac{\mathrm{d}k}{\pi(1+k^2)} = \frac{1}{\pi}\big[\arctan k\big]_{-\infty}^{\infty} = 1$.
- A simulation with 2×10^5 points from three different radially symmetrical distributions (two independent standard normals, uniform on the unit disc, and density proportional to e^{-r}) gave $\mathrm{P}(U/V < 0.7) \approx 0.694$ each time, matching $\frac{1}{2} + \frac{1}{\pi}\arctan 0.7 \approx 0.6944$; the two sides of the identity agreed to within sampling error for several values of k.

Common slips

- Writing $\{U/V < k\} = \{U < kV\}$: multiplying by V reverses the inequality when $V < 0$.
- Taking $\alpha = \arctan(1/k)$ without care: this lies in $(-\frac{\pi}{2}, 0)$ when $k < 0$, whereas the angle we need lies in $(0,\pi)$. Writing $\alpha = \frac{\pi}{2} - \arctan k$ works for every k.

Beyond STEP

The density $\frac{1}{\pi(1+k^2)}$ is the standard *Cauchy distribution*. It is famous for having no mean (the integral of $|k|\,\mathrm{g}(k)$ diverges), so the law of large numbers fails for it: the average of n independent Cauchy variables has the same Cauchy distribution as one of them. The polar-coordinate argument here is a first taste of transforming joint densities with a Jacobian, a standard first-year technique. Grimmett and Welsh, *Probability: An Introduction*, covers both.

WORKED SOLUTION

STEP III 1995, Question 13

Probability and Statistics • Noisy Binary Channel

STEP III 1995, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

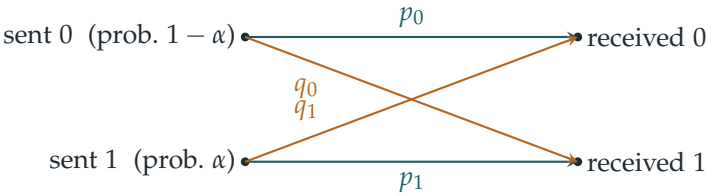
step.maths.org/questions/95-s3-q13

The key idea

Everything follows from looking at one digit at a time. A digit travels through a "binary channel": it is sent as 0 or 1, then either survives or is flipped. Once we know what happens to a single digit, the whole message is just many independent copies, so probabilities multiply. For the booster, the trick is to notice that a digit arrives correctly after two legs if it is flipped an *even* number of times.

Solution

Throughout, write $N = 10^k$ for the length of the message. We assume, as the question intends, that the digits of the original message are independent (each is a 1 with probability α), and that the cable treats each digit independently of all the others. The single-digit picture is below.



The whole message received correctly

Let C be the event that a particular digit is received correctly. The digit is either a 0 that survives or a 1 that survives, so

$$\mathrm{P}(C) = (1-\alpha)p_0 + \alpha p_1.$$

By independence across the N digits,

$$\mathrm{P}(\text{entire message correct}) = \left[(1-\alpha)p_0 + \alpha p_1\right]^{10^k}.$$

The probability β

A digit is received as a 1 either if it was a 1 and survived, or if it was a 0 and was flipped:

$$\beta = \alpha p_1 + (1-\alpha)q_0.$$

Now

$$\begin{aligned}\beta = \alpha &\iff \alpha p_1 + (1-\alpha)q_0 = \alpha \iff (1-\alpha)q_0 = \alpha(1-p_1) = \alpha q_1 \\ &\iff q_0 = \alpha(q_0+q_1) \iff \alpha = \frac{q_0}{q_0+q_1},\end{aligned}$$

where the last step divides by q_0+q_1, which is positive unless the cable is perfect ($q_0 = q_1 = 0$, in which case $\beta = \alpha$ trivially). Every step is reversible, so this is the required "if and only if". In words: the proportion of 1s is unchanged by the cable exactly when the flow $0 \to 1$ (that is, $(1-\alpha)q_0$) balances the flow $1 \to 0$ (that is, αq_1).

A message of all zeros

Suppose the condition holds and every received digit is 0. Because the pairs (sent digit, received digit) are independent across positions, the conditional probability that the whole message is correct is the product of the single-digit probabilities $\mathrm{P}(\text{sent } 0 \mid \text{received } 0)$.

> **Ask yourself** How do we turn "received 0" into a statement about what was sent?

By Bayes' theorem. The probability that a digit is received as 0 is $1-\beta$, and the condition says $\beta = \alpha$, so $1-\beta = 1-\alpha$. Hence

$$\mathrm{P}(\text{sent } 0 \mid \text{received } 0) = \frac{(1-\alpha)p_0}{1-\beta} = \frac{(1-\alpha)p_0}{1-\alpha} = p_0,$$

and therefore

$$\mathrm{P}(\text{message correct} \mid \text{all zeros received}) = p_0^{10^k}.$$

A fifty-fifty chance

With $q_0 = q_1 = q$ and $\alpha = \frac{1}{2}$, each digit is correct with probability $\frac{1}{2}(1-q) + \frac{1}{2}(1-q) = 1-q$. For a message of 10^6 digits we need

$$(1-q)^{10^6} = \tfrac{1}{2} \quad\Longrightarrow\quad 10^6 \ln(1-q) = -\ln 2.$$

Since q is tiny, $\ln(1-q) \approx -q$, giving

$$q \approx \frac{\ln 2}{10^6} \approx 6.93 \times 10^{-7}.$$

Adding a booster station

Halving the length divides q by $2^2 = 4$, since q is proportional to the square of the length. So each half of the cable flips a digit with probability $q/4$, independently of the other half (the booster just re-sends whatever it receives).

> **Ask yourself** Is "both halves transmit correctly" the only way for a digit to arrive correctly?

No: a digit that is flipped in *both* halves is flipped back. So

$$\mathrm{P}(\text{digit correct}) = \left(1-\frac{q}{4}\right)^2 + \left(\frac{q}{4}\right)^2 = 1 - \frac{q}{2} + \frac{q^2}{8} \approx 1 - \frac{q}{2},$$

ignoring terms of order q^2. (Here the double flip is one of those terms, so it makes no difference to the answer, but it is worth seeing why.) The probability that the whole message is correct is now

$$\left(1-\frac{q}{2}\right)^{10^6}.$$

To the same order, $(1-\frac{q}{2})^2 \approx 1-q$, so

$$\left(1-\frac{q}{2}\right)^{10^6} \approx (1-q)^{10^6/2} = \left[(1-q)^{10^6}\right]^{1/2} = \left(\tfrac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}} \approx 0.707.$$

So the booster raises the chance of a perfect message from $\frac{1}{2}$ to about 0.71. Note that we never needed the numerical value of q here: the original condition $(1-q)^{10^6} = \frac{1}{2}$ did all the work. The same idea with $e^{-10^6 q} = \frac{1}{2}$ gives $e^{-10^6 q/2} = 1/\sqrt{2}$.

Checks

- Enumerating every sent and received message of length 4 exactly (with $p_0 = 0.9$, $p_1 = 0.8$) reproduces both $[(1-\alpha)p_0 + \alpha p_1]^4$ and, under the condition on α, the conditional probability $p_0^4 = 0.6561$.
- Solving $(1-q)^{10^6} = \frac{1}{2}$ exactly gives $q = 6.9315 \times 10^{-7}$, agreeing with $\ln 2/10^6$ to five significant figures.
- With this q, the exact booster probability $\left[(1-\frac{q}{4})^2 + (\frac{q}{4})^2\right]^{10^6}$ is 0.707107, matching $1/\sqrt{2}$.
- Sense check: if $q_0 = q_1$ the condition gives $\alpha = \frac{1}{2}$, as symmetry demands.

Common slips

- Taking the error in each half to be $q/2$. The error is proportional to the *square* of the length, so it is $q/4$.

- Answering the "all zeros" part with $(1-\alpha)^N$ or p_0 alone: we need the conditional probability for each digit, raised to the power 10^k.

Beyond STEP

This is the *binary channel* of information theory. When $q_0 = q_1$ it is the binary symmetric channel, the starting point for Shannon's theory of how much information a noisy channel can carry, and for error-correcting codes, which add redundancy so that occasional flips can be detected and repaired rather than merely made rarer. The calculation with the booster is also a two-step Markov chain: the flip probabilities compose by multiplying transition matrices. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and Markov chains at the right level.

WORKED SOLUTION

STEP III 1995, Question 14

Probability and Statistics • Moment Generating Functions

STEP III 1995, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q14

The key idea

The moment generating function turns sums of independent random variables into products: if T_1 and T_2 are independent then $\mathrm{E}\left[\mathrm{e}^{\theta(T_1+T_2)}\right] = \mathrm{E}\left[\mathrm{e}^{\theta T_1}\right]\mathrm{E}\left[\mathrm{e}^{\theta T_2}\right]$. So we find one MGF, raise it to a power, and then recognise the result as the MGF of a density of the same family. Because an MGF determines its distribution, recognising it is enough.

Solution

The MGF of T

For a real number θ,

$$M_T(\theta) = \mathrm{E}\left[\mathrm{e}^{\theta T}\right] = \int_0^\infty t\,\mathrm{e}^{-(1-\theta)t}\,\mathrm{d}t.$$

This converges only when $\theta < 1$ (otherwise the integrand does not tend to 0), so we assume $\theta < 1$ from now on. The integral we need repeatedly is

$$\int_0^\infty t^k \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \frac{k!}{\lambda^{k+1}} \qquad (\lambda > 0,\ k = 0, 1, 2, \ldots),$$

which follows by integrating by parts k times (or by substituting $s = \lambda t$ and using $\int_0^\infty s^k \mathrm{e}^{-s}\,\mathrm{d}s = k!$). With $k = 1$ and $\lambda = 1 - \theta > 0$,

$$\boxed{M_T(\theta) = \frac{1}{(1-\theta)^2}}, \qquad \theta < 1.$$

As a quick sanity check, $M_T(0) = 1$, as every MGF must satisfy.

The MGF of U

Let T_1 and T_2 be the times for the two questions, so $U = T_1 + T_2$, where T_1 and T_2 each have density f and are independent (as the question tells us). Then $\mathrm{e}^{\theta T_1}$ and $\mathrm{e}^{\theta T_2}$ are

independent too, so the expectation of their product is the product of their expectations:

$$M_U(\theta) = \mathrm{E}\left[\mathrm{e}^{\theta T_1}\mathrm{e}^{\theta T_2}\right] = M_T(\theta)^2 = \frac{1}{(1-\theta)^4}, \qquad \theta < 1.$$

The density of U

Now we compute the MGF of the proposed density $\mathrm{g}(u) = \frac{1}{6}u^3\mathrm{e}^{-u}$. Using the integral above with $k = 3$ and $\lambda = 1-\theta$,

$$\int_0^\infty \mathrm{e}^{\theta u}\,\tfrac{1}{6}u^3\mathrm{e}^{-u}\,\mathrm{d}u = \frac{1}{6}\cdot\frac{3!}{(1-\theta)^4} = \frac{1}{(1-\theta)^4}.$$

(Setting $\theta = 0$ confirms that g integrates to 1, so it is a genuine density.)

Ask yourself We have shown that g has the same MGF as U. Why does that mean U has density g?

This uses the uniqueness theorem for MGFs: if two random variables have MGFs that exist and agree on an open interval around $\theta = 0$, they have the same distribution. Here the two agree for every $\theta < 1$, which includes the interval $(-1, 1)$. So U has density

$$\mathrm{g}(u) = \frac{1}{6}u^3\mathrm{e}^{-u}, \qquad u \geq 0,$$

as required.

If you would rather not use the uniqueness theorem, the convolution formula gives the same result directly. Since $T_1, T_2 \geq 0$, for $u \geq 0$,

$$\begin{aligned}\mathrm{g}(u) &= \int_0^u \mathrm{f}(s)\,\mathrm{f}(u-s)\,\mathrm{d}s = \mathrm{e}^{-u}\int_0^u s(u-s)\,\mathrm{d}s \\ &= \mathrm{e}^{-u}\left[\tfrac{1}{2}us^2 - \tfrac{1}{3}s^3\right]_0^u = \frac{1}{6}u^3\mathrm{e}^{-u}.\end{aligned}$$

The total time for n questions

Let $U_n = T_1 + \cdots + T_n$, where the T_i are independent with density f. The same product argument gives

$$M_{U_n}(\theta) = M_T(\theta)^n = \frac{1}{(1-\theta)^{2n}}, \qquad \theta < 1.$$

The pattern from $n = 2$ suggests the density $\mathrm{h}_n(u) = c\,u^{2n-1}\mathrm{e}^{-u}$. Its MGF is $c\,(2n-1)!/(1-\theta)^{2n}$, by the integral with $k = 2n-1$. This equals M_{U_n} exactly when $c = 1/(2n-1)!$. By uniqueness again, the total time for n questions has density

$$\boxed{\mathrm{h}_n(u) = \frac{u^{2n-1}\mathrm{e}^{-u}}{(2n-1)!}, \qquad u \geq 0.}$$

For $n = 1$ this gives f, and for $n = 2$ it gives g, as it should.

Checks

- $\mathrm{E}[T] = M_T'(0) = 2$, which matches $\int_0^\infty t^2 \mathrm{e}^{-t}\,\mathrm{d}t = 2$. So $\mathrm{E}[U_n] = 2n$. The same value comes from h_n: $\int_0^\infty u^{2n}\mathrm{e}^{-u}\,\mathrm{d}u/(2n-1)! = (2n)!/(2n-1)! = 2n$.
- Numerically, Simpson's rule gives $\int_0^\infty \mathrm{e}^{0.5t}\,t\mathrm{e}^{-t}\,\mathrm{d}t = 4.000$, matching $1/(1-0.5)^2$. A simulation of 2×10^5 sums of three times drawn from f gave $\mathrm{P}(U_3 \le 5) \approx 0.386$, against $\int_0^5 u^5\mathrm{e}^{-u}/120\,\mathrm{d}u \approx 0.384$ from h_3.

Common slips

- Forgetting the restriction $\theta < 1$: the MGF does not exist for $\theta \ge 1$.
- Adding MGFs instead of multiplying them. The MGF of U is $M_T(\theta)^2$, not $2M_T(\theta)$. (The MGF of $2T$ would be $M_T(2\theta)$, which is different again.)
- Getting the factorial in the normalising constant wrong. The power of u is $2n-1$, so the constant is $1/(2n-1)!$, not $1/(2n)!$.

Beyond STEP

The density $t\mathrm{e}^{-t}$ is the gamma distribution with shape 2 and rate 1, and h_n is the gamma distribution with shape $2n$. This question shows that independent gammas with the same rate add by adding their shapes. In particular, each T behaves like the sum of two independent exponential times, which is how the gamma distribution arises as the waiting time for the kth event in a Poisson process. At university you will meet the characteristic function $\mathrm{E}[\mathrm{e}^{\mathrm{i}\theta X}]$, which always exists, and use it to prove the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions and these ideas clearly.

Beyond STEP

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